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Developing a Spectral and Colorimetric Database of Artist Paint Materials

by

Yoshio Okumura

B.E. Doshisha University, Kyoto, JAPAN (1997)

A thesis submitted in partial fulfillment of the requirement
for the degree of Master of Science
in the Chester F. Carlson Center
for Color Science of the College of Science
Rochester Institute of Technology

September, 2005

Signature of the Author

Accepted by Dr. Roy S. Berns
Coordinator, M.S. Degree Program

CENTER FOR IMAGING SCIENCE
ROCHESTER INSTITUTE OF TECHNOLOGY
ROCHESTER, NEW YORK

CERTIFICATE OF APPROVAL

M.S. DEGREE THESIS

The M.S. Degree Thesis of Yoshio Okumura
has been examined and approved by two member of the
Color science faculty as satisfactory of the thesis
requirement for the Master of Science degree

Dr. Roy S. Berns. Thesis advisor

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i. DEDICATION

THIS THESIS IS MY THANKS TO MY PARENTS.

For from the multitude of pigments colouring has suffered much. Every pigment has its peculiar nature as regards its effect on the eye; besides this it has its peculiar quality, requireing a corresponding technical method in its application. The former circumstance is a reason why harmony is more difficult of attainment with many materials than with few, the latter, why chemical action and re-action may take place among the colouring substances.

- Johann Wolfgang Van Goethe

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CHAPTER 1. INTRODUCTION

As the project of the author's Master's thesis, the development of a spectral and colorimetric database of artist paint materials for acrylic paints was started. The goal of this research project was to:

- *provide the academic resource of colorant spectral characteristics*
- *give scientific explanations on various paint-particular phenomena (paint mixing, gloss effects and color gamut expansion by varnishing)*

These tasks were planned to satisfy possible interests on paint research from not only conservators in museums but also color educators in schools and color reproduction engineers in imaging companies.

First of all, the coverage of this research was narrowed down to matte acrylic paints that are made from traditional organic and synthetic pigments. That is, the paints of relatively brand-new colorants, fluorescent, metallic, and pearlescent pigments, were not considered herein.

The first mission of the database development was to build up the optical properties of artist paint colors based on spectrophotometric measurement. The contents of the artist paint database were represented by spectral light characteristics, reflectance, absorption, and scattering, so as to simulate specific paint mixtures with the dataset. Since it is derived by a different concept from some encyclopaedic databases that have been introduced to show basic physical and optical parameters such as pigments' refractive indices [Eastaugh 2004] and light reflectivities for pigment-binder combinations [Picollo 2004], our colorant database will be useful for color science researchers as an academic resource.

The advantage of holding the spectral dataset in our database can help spectral im-

aging researchers in a couple of ways. According to the recent study on the spectral-imaging camera system by Mohammadi and Berns [Mohammadi 2004], for example, it clarified that appropriate spectral-reflectance curves of a calibration target should be chosen rather than increasing the number of target paints for spectrally accurate calibration. For such a case, a colorant formulation using the spectral dataset will lead to determining the optimal recipes of acrylic paints providing appropriate reflectance curves to the development of a calibration paint target. Another possible usage of the spectral dataset is for pigment identification, the spectral matching technique of which is to find the recipes of paint mixtures, used in art, from a large amount of colorant spectral information.

The second mission was to simulate paint mixtures with mathematical calculations and verify the accuracy of the simulation performance. The simulation approach, based on the Kubelka-Munk theory, can be considered to have larger possible choices of paints and give better simulation results than gained by a traditional approach based on a look-up table method that characterizes a paint gradient in the $L^*a^*b^*$ coordinates [Carabott 2002]. The spectral-based technique of paint mixing simulation is expected to reveal the contradiction of advocated paint-mixing theory, “yellow, red, and blue can make any colors because these colors are the primaries of the paint system”, to color educators because it is actually not. With regard to paint mixing theory, a professional artist, Michael Wilcox, experientially showed the contradiction by making a large amount of paint mixing palettes [Wilcox 2001].

The third mission was to render paint color gamuts with selected primary paint mixtures and multiple paint mixtures based on the spectral database. Rendering a paint color gamut is an orthodox approach to visualize the possible color coverage of certain paint mixtures. In terms of color reproduction, additionally, it is interesting to

compare color gamut volumes in the $L^*a^*b^*$ space among various coloration systems including the paint system. A general target for calibrating cameras such as the GretagMacbeth ColorChecker is expected to cover a wide gamut far beyond a paint system. Quantifying the color gamut volumes of the systems would be helpful not only estimate color reproduction accuracy but also to develop a paint-color rendering chart for spectral imaging camera.

The last mission was to formulate the gloss effects by varnishing. In conservation works in museums, applying, retouching, and removing a varnish coat are carried out with verifying the appearance of re-painted colors over or without a varnish coat by operators. In order to reproduce the original paint colors, it is necessary for conservators to understand how a new varnish coat makes paint colors changed and visually wet in advance. A scientific explanation on the mechanism of gloss by varnishing will guide conservators to take an appropriate step of choosing a varnish substance as well as an application method to produce their expected surface finishings.

CHAPTER 2. BACKGROUND

2.1. Overview

This chapter summarizes the result of literature reviews on several optical principles of paint systems. To understand the whole range of the listed sections below is of assistance not only to build up such a colorant database but also to solve some enigmas on the visual appearances of artist paintings that we are going to clarify in this research.

2.2 Color Mixing Theory for Artist Paints

2.3 Physical Theories on Gloss Effects

2.4 Physics and Color Science on Varnishing

2.5 Techniques of Paint Color and Glossiness Measurements

The first section is about color mixing theory in the paint system. This section introduces about how to characterize a large number of colorants mathematically with modern computational approaches. The keywords of this section are “*Kubelka-Munk theory*”, “*Saunderson correction*”, and “*non-linear optimization*”.

The second section discusses about gloss modeling. Not only appearance but also measurement values by spectrophotometric and colorimetric instruments are changed by the gloss effects. On “*Fresnel’s law of reflection*” and “*dichromatic reflection model*” are to be mentioned that helps paint researchers to recognize how spectral curves of reflectance by gloss.

The third section is to think about why varnishing make artist paintings glossy and more saturated. Some optical theories on gloss explains changes in color appearance

by varnishing.

The last section is to look through mensuration about spectrophotometric and gloss measurements. The mechanisms of these instruments are summarized at the end of this chapter.

2.2. Color Mixing Theory for Artist Paints

2.2.1. Kubelka-Munk Theory

The Kubelka-Munk theory is a mathematical model characterizing the internal reflectance of a paint medium. The model hypothesizes that light flux absorption and scattering occur in upward and downward directions perpendicular to the medium [Allen 1980; Haase 1992]. Figure 1 is the illustration explaining the optical phenomenon, where i and j are the identifiers of light flowing in two directions. While i represents a light flow moving towards the substrate, j indicates a light flow going up to the top. And then, in this figure, the upper direction from 0 to X is defined as positive in distance. At the specific depth, x , of a paint layer, when the incident light passes through the sub-layer of a small thickness dx , the amount of light penetrating downwards decreases in proportion to absorption and scattering properties. The attenuation is expressed by Equation (2.1), consisting of two Kubelka-Munk coefficients, where K is an absorption coefficient and S is a scattering coefficient. Both of the coefficients are assumed to be constant within the depth of a paint layer.

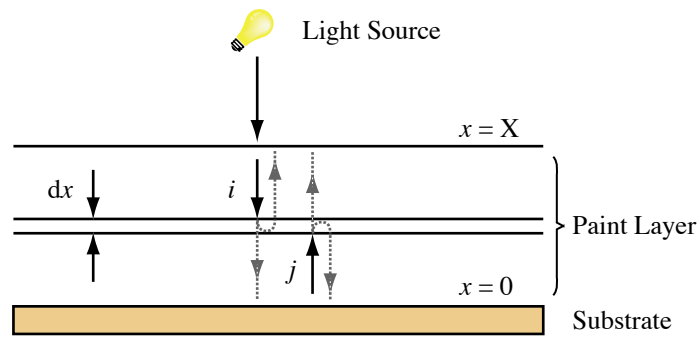


Figure 1. Absorption and scattering components based on two-flux lights in a paint layer.

$$(K + S) i dx \quad (2.1)$$

A light flux reaching the substrate (e.g. canvas) is bounced off and changes its direction upwards. When the reflection light passes through at the sub-layer, the flux also decreases as the following:

$$(K + S) j dx \quad (2.2)$$

By integrating these equations, the total amount of attenuation in each of the up-down directions can be calculated. For example, the light i is not only attenuated by the absorption and scattering of a colorant but also imposed by the light some part of the reflected light j because of scattering. Allen (1980) explained the formulation of such an optical interaction and solved the equations in his reference. By focusing on the two-flux directions, we obtain a pair of the differential equations:

$$\begin{aligned} di &= (K + S) i dx - S j dx \\ -dj &= (K + S) j dx - S i dx \end{aligned} \quad (2.3)$$

These equations can be solved by replacing the ratio (i/j) with a single coefficient, ρ , and those are to be united into a single equation:

$$\frac{d\rho}{dx} = \frac{d(j/i)}{dx} = \frac{i(dj/dx) - j(di/dx)}{i^2} \quad (2.4)$$

The ratios, (di/dx) and (dj/dx) , are already given by transforming Equation (2.3).

Now, therefore, we get a first-order equation consisting of K and S :

$$\frac{d\rho}{dx} = S - 2(K + S)\rho + S\rho^2 \quad (2.5)$$

As the second step, we are required to think of ρ or the ratio (j/i), which are changed by corresponding with the depth x . Just remember, ρ represents the ratio of the returning light flux to the incident light. That is, ρ is equivalent to reflectance at a corresponding depth. When a depth x is equal to X (at the surface), it can be interpreted that ρ is equal to the reflectance of a paint film, R . On the other hand, when x is equal to 0 (at the substrate), ρ must be equivalent to the reflectance of the substrate, R_g , itself. With taking that into account, the equation for R can be solved and obtained as the following:

$$R = \frac{1 - R_g (a - b \coth bSX)}{a - R_g + b \coth bSX} \quad (2.6)$$

where

$$a = 1 + (K/S), \quad b = (a^2 - 1)^{1/2} \quad (2.7)$$

This comprehensive equation is rarely used because paint researchers sometimes meet difficulty in measuring the thickness of an actual paint layer. Therefore, some simplified equation are introduced in response to specific conditions or manners. One situation is, if a specimen does not show any scattering property like a plastic film, Equation (2.6) is transformed into:

$$R = R_g e^{-2KX} \quad (S \rightarrow 0) \quad (2.8)$$

The situation, in which Equation (2.8) is satisfied, is referred to as Bouguer-Beer law [Berns 2000]. However, this equation is not appropriate for paints because they sometimes contain scattering agents like Titanium White.

Another situation is the case of analyzing a opaque sample represented as a ceramic tile. According to the Kubelka-Munk theory, opaque intends the situation that the incident light cannot reach the substrate because of the strong opacity of a paint layer. That is, it is reasonable to consider that the thickness of a paint layer is infinite in this case. Thus, Equation (2.6) is simplified by giving X an infinite number and then can be changed into:

$$R_{\infty} = 1 + (K/S) - [(K/S)^2 + 2(K/S)]^{1/2} \quad (X \rightarrow \infty) \quad (2.9)$$

where R_{∞} is the reflectance of an opaque specimen (or the reflectance at an infinite film thickness). This equation is also solved for K/S into a function of R_{∞} :

$$K/S = (1 - R_{\infty})^2 / 2R_{\infty} \quad (2.10)$$

The concept which we have mentioned about so far is applicable to the case of determining a single K and a single S for an individual colorant. When allowing for the mixture of various colorants with different absorption and scattering properties ($[k_1, k_2, \dots, k_n]$ and $[s_1, s_2, \dots, s_n]$; n is the number of colorants), we need to sum up all those properties somehow leading to being behaved as a unit colorant. Considering of such a paint mixing, it is necessary to correlate internal reflection with absorption coefficients and scattering coefficients.

One of the approaches is to assume the additivity of each of absorption and scattering capabilities individually. The mathematical expression for the additivity appears in Equation (2.11):

$$\begin{aligned} K &= (1 - p)k_t + c_1k_1 + c_2k_2 + \dots + c_nk_n \\ S &= (1 - p)s_t + c_1s_1 + c_2s_2 + \dots + c_ns_n \end{aligned} \quad (2.11)$$

where t represents a tint, k_t and s_t are the absorption and scattering coefficients of a tint respectively. And, c_i is a theoretical concentration by weight for i th colorant, p is the sum of colorant concentrations, and n is the number of colorants mixed:

$$p = c_1 + c_2 + \cdots + c_n \quad (0 \leq p \leq 1) \quad (2.12)$$

where p must be less than 1 because the sum of all the paint concentrations is 1.

The formulation of supporting this hypothesis is called the two-constant Kubelka-Munk theory. In this colorant formulation, we are required to evaluate an absorption coefficient and a scattering coefficient for each substance forming a paint layer. Therefore, K/S is driven by Equation (2.13):

$$\frac{K}{S} = \frac{(1-p)k_t + c_1k_1 + c_2k_2 + \cdots + c_nk_n}{(1-p)s_t + c_1s_1 + c_2s_2 + \cdots + c_ns_n} \quad (2.13)$$

In general, the two-constant Kubelka-Munk theory predicts the (K/S) of a paint precisely.

Another approach is to assume the constancy of a absorption-scattering ratio (K/S) for each colorant. That is, a colorant is characterized by unit (k/s) instead of separate unit k and unit s . If we attempt to suppose that scattering coefficients, $s_1, s_2, \dots, s_n$, are equal to s_t , Equation (2.13) can be simplified:

$$\frac{K}{S} = \frac{(1-p)k_t + c_1k_1 + c_2k_2 + \cdots + c_nk_n}{s_t} \quad (2.14)$$

This approach mentions that the scattering properties of a tint and colorants are independent of their concentrations. However, we assume that scattering effects are not changed regardless of different concentrations for some coloration systems. The final

equation of this approach is written:

$$\frac{K}{S} = (1 - p) \left(\frac{k_t}{s_t} \right) + c_1 \left(\frac{k_1}{s_t} \right) + c_2 \left(\frac{k_2}{s_t} \right) + \cdots + c_n \left(\frac{k_n}{s_t} \right) \quad (2.15)$$

This equation implies the additivity of every (k/s). The formulation on this hypothesis is called the single-constant Kubelka-Munk theory.

In a pilot study, we figured out that the single-constant Kubelka-Munk theory did not show good estimation performance for our paint targets. The problem is that scattering properties at all the wavelengths were changed in response to the increase of a colorant percentage. Therefore, all the paints in this research were characterized by the two-constant Kubelka-Munk theory.

For the estimation of Kubelka-Munk coefficients, the tint ladder method was used. In this method, mixtures of a colorant with a tint are made at different concentrations. That is, tint ladders prepared are expected to show the wide range of a paint color from Titanium White to a masstone. Since the absorption and scattering of a tint, to which those of a colorant are characterized relatively, are denoted as the standard, a low-absorption high-scattering paint, Titanium White, is commonly used as a tint. The tint ladders are measured in reflectance factor and then converted to (K/S) by Equation (2.10) and the Saunderson Correction (cf. 2.2.2 Saunderson Correction).

Now, we are going to estimate k_l and s_l for a target paint by computation with the tint ladder method. For mixtures of two paints, a target paint and Titanium White, Equation (2.13) is simplified:

$$(K/S)_i = [c_i k_1 + (1 - c_i) k_t] / [c_i s_1 + (1 - c_i) s_t] \quad (2.16)$$

where the subscript i is the identifier of the i th tint ladder. Equation (2.16) consists of two unknown coefficients, k_l and s_l . Equation (2.16) can be transformed into Equation (2.17).

$$c_i k_l - c_i (K/S)_i s_l = (1 - c_i) [(K/S)_i s_l - k_l] \quad (2.17)$$

Furthermore, Equation (2.17) can be rewritten using a matrix equation as the following.

$$\mathbf{d} = \mathbf{C}\mathbf{x} \quad : \text{ general expression} \\ (1 - c_i) [(K/S)_i s_l - k_l] = \begin{bmatrix} c_i & -c_i (K/S)_i \end{bmatrix} \begin{bmatrix} k_l \\ s_l \end{bmatrix} \quad (2.18)$$

If we have two tint ladders except for a tint, Equation (2.18) can be solved simultaneously. If we have more than three different concentration series including a tint, moreover, using least square method is to give optimal k_l and s_l values. In this case, the k_l and s_l of an independent matrix in Equation (2.18), $\mathbf{x}$, shall be non-negative. To satisfy this inequality constraint, we used a Matlab optimizing function, “*lsqnonneg*“, in the computation. The algorithm details of Nonnegative Least Squares (NNLS) are introduced in the reference [Lawson 1995].

As a matter of fact, the Kubelka-Munk theory sometimes does not predict the optical properties of some colorants very well. One possible reason is due to poor scalability between concentrations by weight and (K/S) scalars. In such a case, we are sometimes required to employ effective concentration instead of using theoretical concentration. The reference defines that, “effective concentration is concentration determined from colorant measurements of the colored material” [Berns 2000]. To

fit a relationship between theoretical concentration and effective concentration, least squares is also used. The plots of Figure 2 show the relationship for Carbon Black of GOLDEN matte fluid acrylic paints in the two-constant Kubelka-Munk theory. It can be observed that effective concentrations are underestimated comparing to theoretical concentrations in this case.

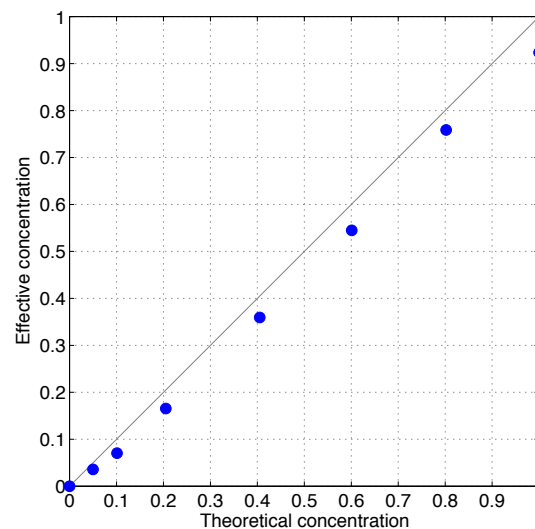


Figure 2. A relationship between theoretical concentration and effective concentration.

After testing the importance of using the effective concentration, it was determined that most of the paints showed linearity for the theoretical concentration under the two-constant Kubelka-Munk theory. Thus, it was determined that the effective concentration was not necessary for this research.

2.2.2. Saunderson Correction

The previous section has mentioned about Kubelka-Munk theory. The internal reflectance of a paint medium is characterized by this theory. The only matter on the model is that it does not take account of surface reflection. As a matter of fact, a

spectrophotometer includes the effects of specular reflection so that it is hard to exclude only internal reflection from the measurement mechanically. We need to extract only internal reflection, R_{∞} , expressed in Equation (2.9), therefore, when using the Kubelka-Munk theory for paint characterization. In order to separate a measured reflectance into the two components, characterizing surface reflection is necessary. Surface reflection is characterized by the Saunderson equation. As mentioned in the previous section of “*Fresnel’s law of reflection*”, first-surface reflection is characterized by the ratio of two different refractive indices like air (1.00) and water (1.33). If a canvas is coated with a paint of 1.50 refractive index, 4% of an incident light will be reflected from the paint surface over all the wavelengths. Now, we can define this reflectance as K_1 . The rest of the light penetrates the surface and then bounces many times between the surface and the substrate. The amount of the penetrating light is defined as $(I-K_1)$. If the light bounced from the top does not scatter at all, the ratio of reflectance on the boundary between an air and a paint layer would be equal to K_1 constantly. However, in the process of a light traveling in the medium, it is scattered in the advancing so that the reflectance at the top boundary characterized as a diffuse reflection ratio, K_2 , shows commonly a higher value than K_1 . The light flow so far is illustrated in Figure 3, that is the first-cycle stage including (1), (2), and (3). This multiple reflection is continued into the second-cycle stage of (4), (5) and (6), the third, and so on.

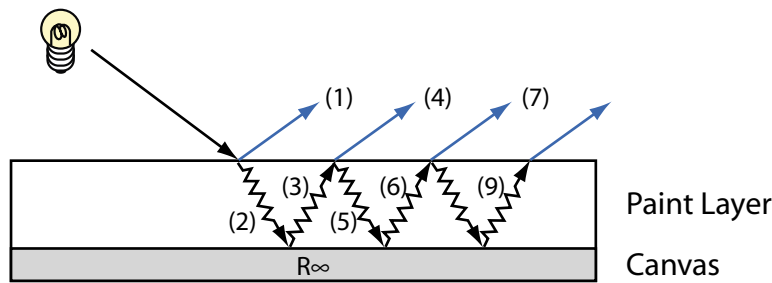


Figure 3. Deviation of Saunderson Correction.

Table II summarizes the derivation of the Saunderson correction [Allen 1980]. The blue terms in the second column represent the components of specular and diffuse reflections, the sum of which are detected by a spectrophotometer.

Table I. Deviation of the Saunderson correction.

Cycle	Light leaving from the surface		Light penetrating into the substrate		Light bouncing from the substrate	
1	(1)	K_1	(2)	$1 - K_1$	(3)	$(1 - K_1) R_\infty$
2	(4)	$(1 - K_1)(1 - K_2) R_\infty$	(5)	$(1 - K_1) K_2 R_\infty$	(6)	$(1 - K_1) K_2 R_\infty^2$
3	(7)	$(1 - K_1)(1 - K_2) R_\infty^2$	(8)	$(1 - K_1) K_2^2 R_\infty^2$	(9)	$(1 - K_1) K_2^2 R_\infty^3$
...		...		...		...

It depends on measurement geometry, however, whether or not a specular component (1) is included in measurement values. By summing up all the components in the second column that is equivalent to reflectance by the specular component included (SPIN) measurement with an integrating sphere instrument, we can describe the measured reflectance, R_m , as:

$$\begin{aligned}
 R_m &= K_1 + (1 - K_1)(1 - K_2) R_\infty (1 + K_2 R_\infty + K_2^2 R_\infty^2 + \dots) \\
 &= K_1 + \frac{(1 - K_1)(1 - K_2) R_\infty}{1 - K_2 R_\infty}
 \end{aligned} \tag{2.19}$$

On the other hand, if we do spectrophotometric measurements in the specular component excluded (SPEX) mode, first-surface reflectance is to be excluded. Hence, the final reflectance will come to:

$$R_m = \frac{(1 - K_1)(1 - K_2) R_\infty}{1 - K_2 R_\infty} \tag{2.20}$$

where R_m indicates a measured reflectance.

As a matter of fact, even though the SPEX mode is selected, it is hard to eliminate only the effects of specular reflection completely because a specular trap may exclude only some portion of a diffused specular reflection trapped with a specular port [Wyble 2003]. If specular exclusion in spectral measurements is necessary as much as possible, using the 45/0 geometry and employing Equation (2.20) will be the best way for the Kubelka-Munk solution.

The explanation so far is only valid for the simple paint coat that consists of three layers, air, a paint layer, and a substrate. However, the more complex structure of a paint coat should be discussed when characterizing the optical properties of variety of artist paint materials. One of the good examples is the varnish coat.

Berns derived the general mathematical equation of an optical model for the varnish coat from the Fresnel law of reflection [Berns 2003]. The model (*c.f.* “2.4.2 *Surface Roughness*”) allows a larger number of the optical properties than the Saunderson equation does:

- The illumination angle of incidence
- Refractive indices of the varnish resin and paint-layer surface
- Spectral transmittance of the varnish
- Internal spectral reflectance of the paint layer

The advantage of using the general model is that gloss effects by varnishing onto the internal reflectance can be formulated that is used for the characterization of varnished paints by the Kubelka-Munk solution.

2.2.3. Optimization Techniques For The Kubelka-Munk Solution

As mentioned in the previous chapter, least-square method is a powerful technique to estimate the Kubelka-Munk coefficients. The only problem of using this method is, however, obtained unit k and unit s for a colorant might not be desired values satisfying an ideal spectral-error criterion. As shown in Equation (2.18), the non-negative least-square solution minimizes the residual, $\|d - Cx\|$, at each wavelength. That is, estimated unit k and unit s for a target paint are expected to minimize root-mean-square (RMS) errors in K/S , not RMS errors in reflectance, R_m . The goal of the coefficients optimization is to make reflectance errors as small as possible so that colorimetric errors are to be minimized simultaneously. The relationship between a R_m error and a K/S error must show some correlation. From Equation (2.9) and Equation (2.17), R_m can be expressed as a function of K/S :

$$R_m = f(K/S) \quad (2.21)$$

By manipulating this mathematical expression, the differential of R_m can be obtained as the following equation:

$$dR_m = \frac{\partial R_m}{\partial (K/S)} d(K/S) \quad (2.22)$$

Equation (2.22) represents how much R_m is changed as K/S increases. The relationship between measures reflectance (in the SPEX mode), R_m , and the partial differential of R_m is shown in Figure 4. One important interpretation from Equation (2.22) and the figure is that, a small change in K/S leads to a large-negative change in R_m at high

R_m . In terms of optimizing an criterion, the blue line in the figure says that, minimizing K/S errors weights R_m errors at high R_m , not at low-middle R_m .

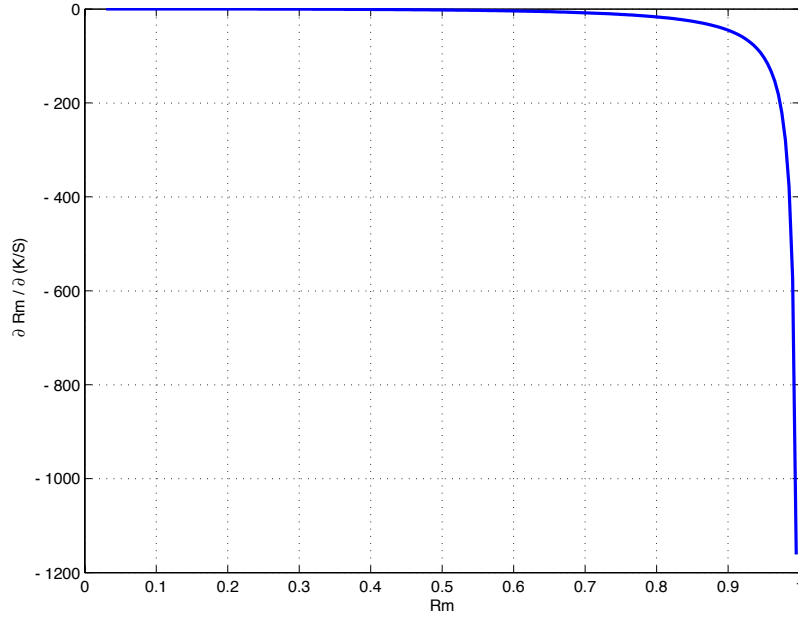


Figure 4. A relationship between the sensitivity of measured reflectance and changes in K/S .

An ideal approach to minimize R_m errors is to use a non-linear optimization technique in the R_m space, instead of using the least-square method in the K/S space that is introduced in Equation (2.18). The basic workflow of a non-linear optimization is illustrated in Figure 5. The flowchart does not apply for the case of incorporating the effective concentration.

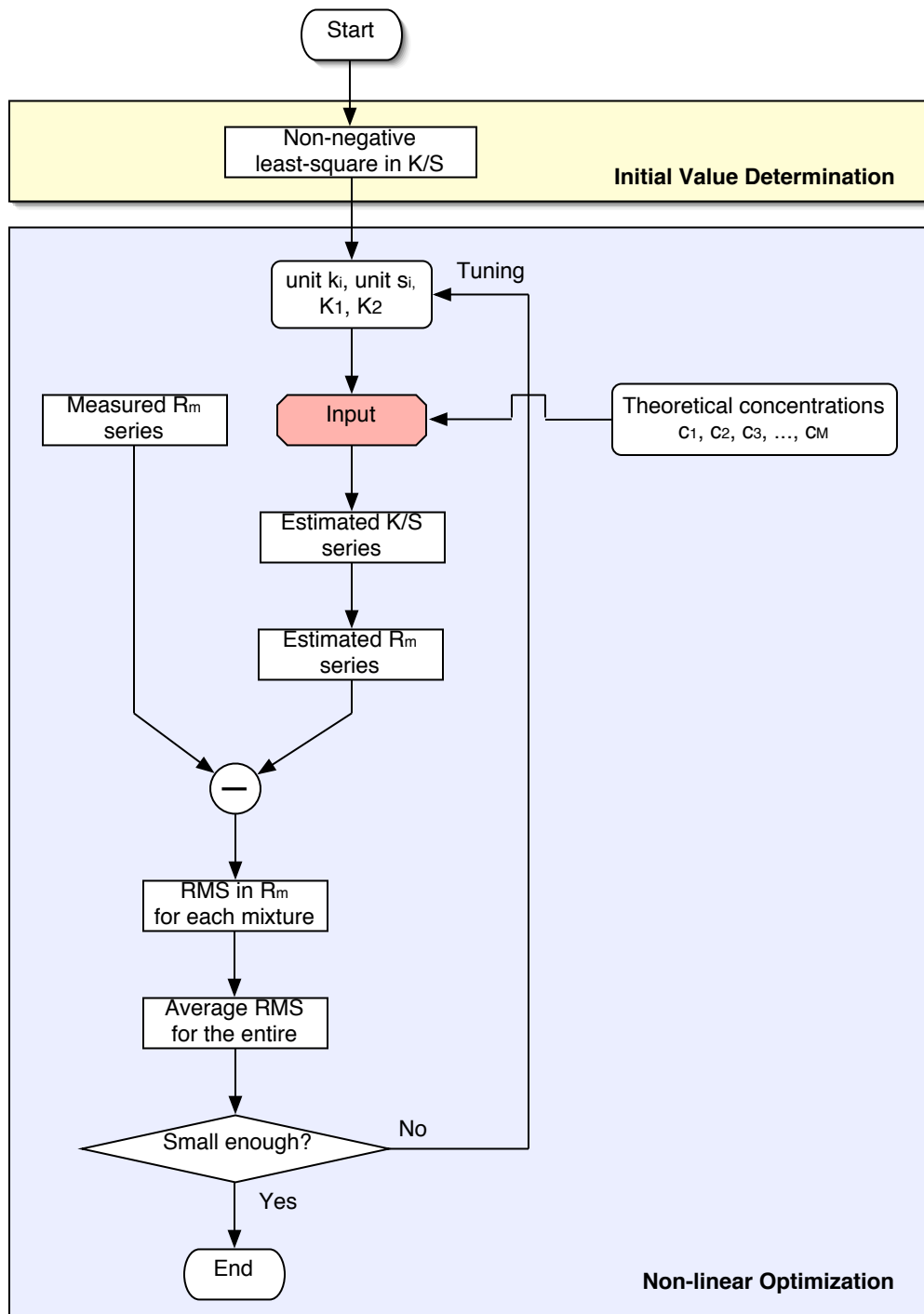


Figure 5. A flowchart of the non-linear optimization process.

This optimization workflow implies that the estimated $R_{m,j}$ (j th tint ladder) is a function of Kubelka-Munk coefficients, Saunderson coefficients, and a theoretical concentration as shown in Equation (2.23):

$$\hat{R}_{m,i} = f(\text{unit } k, \text{unit } s, K_1, K_2, c_i) \quad (2.23)$$

where unit k and unit s are the absorption and scattering coefficients of the two-constant Kubelka-Munk theory, K_1 and K_2 are the Saunderson coefficients, c_i is a theoretical concentration.

Kubelka-Munk coefficients, unit k_j and unit s_j (at j th wavelength), and Saunderson coefficients, K_1 and K_2 , are simultaneously optimized. For the non-linear optimization, our approach is that, the initial values, unit k_i and unit s_i , are pre-assigned the solution from the non-negative least-square method in Equation (2.18). At the same time, while the initial value of K_2 is assigned an empirical value, 0.6, the initial value of K_1 is assigned the minimum reflectance of the tint ladders over all the wavelengths. From experience, the masstone of a colorant have the minimum reflectance in the spectral region because of strong absorption. And then, those initial values are converted into K/S values first, and then into R_m values for each of tint ladders. After this computation, a RMS value between measurement and estimation for i th tint ladder is calculated by using Equation (2.24):

$$\text{RMS}_i = \sqrt{\frac{1}{N} \sum_{j=1}^N \{\hat{R}_{m,i}(\lambda_j) - R_{m,i}(\lambda_j)\}^2} \quad (2.24)$$

where N is the number of wavelengths in spectral measurement. After that, the average RMS for a colorant is calculated:

$$\overline{\text{RMS}} = \frac{1}{M} \sum_{i=1}^M (\text{RMS}_i) \quad (2.25)$$

where M is the number of the tint ladders used.

If the average RMS is under a tolerance, the optimization process ends. If not, the initial values are tuned, and then the process continues from the starting point.

The optimization process explained so far does not employ the effective concentrations even though usually they are estimated first and then curved to the theoretical concentrations. If the paint system shows non-linear scalability in K/S , in addition to unit k , unit s , K_1 , and K_2 , the effective concentrations are also required to be optimized simultaneously. In this case, the theoretical concentrations are expected to be the most appropriate initial values of the optimized effective concentrations.

2.3. Physical Theories on Gloss Effects

In general, gloss is an optical phenomenon in which a surface luster or brightness is observed on a glossy object. Gloss finish (e.g. furniture and floor) gives people the sense of luxury, smoothness, and permanence. An impression from gloss directly affects observers' judgments to select the preferable appearance of colors rather than colors themselves in automotive and home appliance products. A color reproduction system is no exception at this point. A good example of gloss preference is a photo-quality paper designed for inkjet prints that is categorized by three gloss levels: glossy, semi-gloss, and matte. Each gloss level gives you the different final appearance of a reproduced picture. A matte paper is used to emphasize the embossed texture of a paper ground so as to avoid undesirable sheen. On the other hand, a gloss paper produces high contrast, saturated images. Gloss is a familiar topic for photo consumers in daily life.

This section is to discuss about not only how gloss looks like (appearance) but also scientific explanations on gloss (theory).

2.3.1. Gloss Observation

When looking around our living environment, we might notice that there are a lot of objects artificially made glossy by changing their original appearances. One example is a framed poster with a plastic or glass plate on the surface. The flat structure of a glass plate bounces off the projection image of a lighting towards our eyes.

To confirm changes in appearance by the gloss effects, we made the demonstration samples of different gloss-level images. The basic idea of this demonstration is based on a gloss simulation by Berns and de la Rie [Berns 2003b]. A gloss coloration sys-

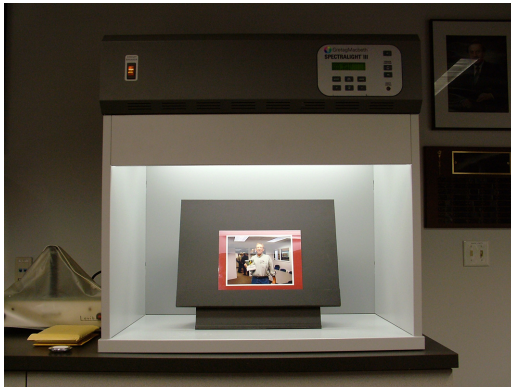
tem was made of an inkjet print and a transparent plastic film. Covering a target photo printed on a semi-gloss paper with an OHP film made the appearance more glossy. A test photo was printed with a Canon S9000 inkjet printer. It is a general explanation on surface reflection that, once the incident light arrives at the film surface, some of the light beam is bounced off directly to the direction of mirror reflection without diffusion. At the same time, the rest of the light penetrates into the coloration medium.

Now, we are going to look at the appearance of a glossy target carefully under a diffuse lighting (Figure 6-(a)). The appearance viewed through a transparent film is more shining because of specular reflection at the film surface as shown in Figure 6-(c). And then, if you tilt it toward or off you without moving a viewing point, the amount of specular reflection will be dramatically changed in a range between gloss and haze. The contrast of perceived specular is getting larger when looking through the transparent film rather than through no film (Figure 6-(b)). It is also interesting that luminance density perceived is higher in dark tone through the film. When looking at the same figure, you might notice that the area where people gathering in the back room is a darkest location in the scene. Corresponding with a darkening effect, colors come to being more saturated. We observed the dark spot through the transparent film that a shade is compressed and tends to be much darker. The reason of darkening is associated with how much specular reflection an observer sees. However, colors in highlight are not significantly changed.

Instead of using a transparent film, a tracing paper covering a photo can apply to a matte finish demonstration as shown in Figure 6-(d). The surface of a tracing paper is almost transparent (not perfectly) and coarse enough for a pencil lead to be transferred to the surface. The surface roughness causes the diffusion of specular reflection. Through a tracing film, the appearance of a photo appears to be whitened over the

surface area because of the homogeneous intrusion of specular reflection. As a result, it can be observed a lower contrast image than the original.

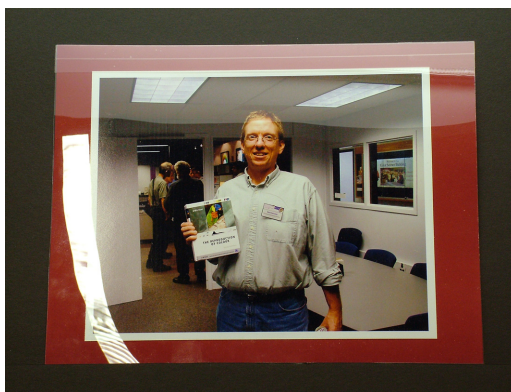
A difference of gloss finish in our demonstration samples comes from the difference in refractive index (RI) between air and a plastic film, or air and print paper. The different optical property prevents a few percentages of the incident light from penetrating the boundary. As a consequence, some portion of the incident light flux is reflected back to our eyes without any color absorption in an ink-dye layer. The details of physical theory on surface reflection, “*Fresnel’s law of reflection*”, is to be explained in the next section.



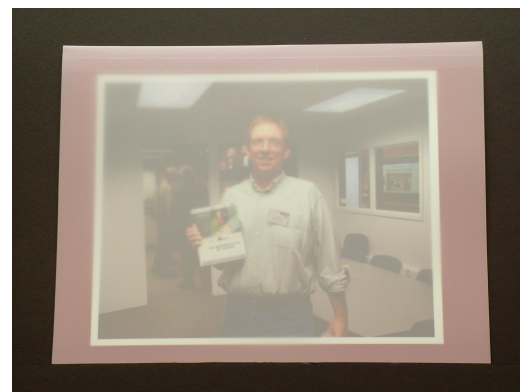
(a) Lighting environment



(b) Without a plastic film



(c) Glossy photo



(d) Matte photo

Figure 6. Glossy and matte demonstrations with photos and plastic films.

2.3.2. Fresnel's Law of Reflection

Fresnel's law of reflection is an optical model of first-surface reflection, which is bounced off at the boundary between two media of different refractive indices. In this theory, normal incident light is simply attenuated by the ratio of reflection to incidence or *reflectance* expressed in Equation (2.18), where n_1 and n_2 represent the refractive indices of two materials like air and a glass.

In addition to for normal incidence, the theory characterizes the total reflectance for multi-angle incidents, ρ_T , as a function of incidence angle, i , and the refractive indices of the media, n_1 and n_2 . Now, we hypothesize a collimated incident light. The amount of first-surface reflection is expressed by the integration of two reflection components: reflectance for a polarized light oriented to the horizon, $\rho_{||}$, and reflectance for a polarized light oriented to the surface, $\rho_{\perp}$. These reflectances are given in Equation (2.19) and (2.20) respectively. The reflectance $\rho_{||}$ increases monotonically corresponding a increase of incident angle i . On the contrary, the reflectance $\rho_{\perp}$ decreases gradually to zero in the range from 0° to the critical angle. And then, it increases exponentially with an increase of incident angle in the rest of the range. Equation (2.21) is given in order to calculate the critical angle f by using n_1 and n_2 . The critical angle is that, a polarized beam to the surface at a incident angle f is not reflected at all but penetrates completely. By taking the average of these reflectances, reflectance ρ_T for unpolarized incident light is finally gained in Equation (2.22). The state of the whole features so far is shown in Figure 7. All the plots are obtained by assuming that (n_2/n_1) is 1.5.

Now, let's move on to looking at the case of a change in relative refractive index (n_2/n_1) . Figure 8 shows relationship between reflectance ρ_T , relative refractive index, and incident angle. As the diagram indicates, a greater refractive-index ratio shows a

higher reflectance. The interesting feature of the figure is also that, whatever relative refractive index is, ρ_T is constant within a range from 0° to 40° in incident angle.

$$\rho = \left(\frac{n_2 - n_1}{n_2 + n_1} \right)^2 \quad (2.18)$$

$$\rho_{\parallel} = \left[\frac{\cos i - \sqrt{(n_2/n_1)^2 - \sin^2 i}}{\cos i + \sqrt{(n_2/n_1)^2 - \sin^2 i}} \right]^2 \quad (2.19)$$

$$\rho_{\perp} = \left[\frac{(n_2/n_1)^2 \cos i - \sqrt{(n_2/n_1)^2 - \sin^2 i}}{(n_2/n_1)^2 \cos i + \sqrt{(n_2/n_1)^2 - \sin^2 i}} \right]^2 \quad (2.20)$$

$$\sin f = \frac{n_2}{n_1} \quad (2.21)$$

$$\rho_T = (\rho_{\parallel} + \rho_{\perp}) / 2 \quad (2.22)$$

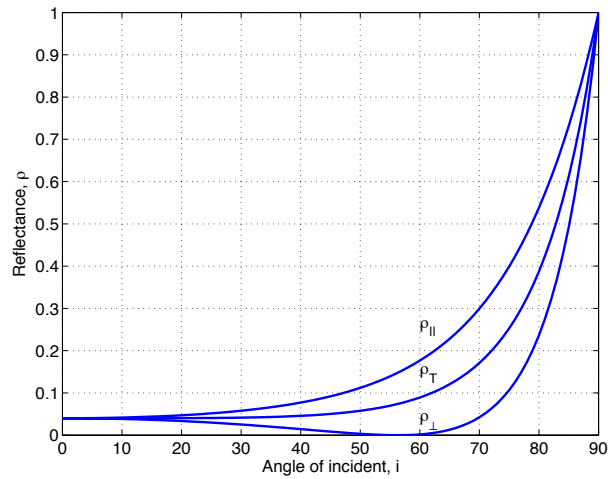


Figure 7. Fresnel reflections based on incident angle, i , and reflectance, ρ .

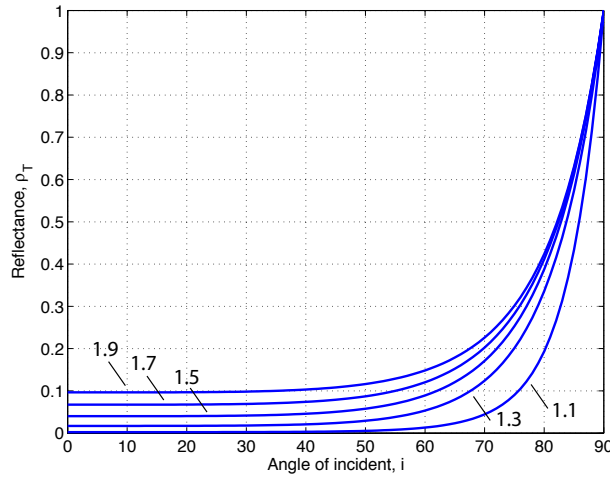


Figure 8. Fresnel reflection for unpolarized light based on incident angle, i , reflectance, ρ_T , refractive index, n (from higher to lower reflectance, $n = 1.9, 1.7, 1.5, 1.3, 1.1$).

As discussed so far, the amount of first-surface reflection under a directional illumination can be explained by Fresnel's law of reflection. That is, the theory can be considered to apply for 45/0 geometry measurements. However, the question is that, what would happen if supposing a diffuse illumination, which is the typical illumination setting for integrating-sphere instruments, instead of a directional illumination. We are interested in the effects of surface reflection onto spectrophotometric measurements. To characterize the optical effects of vanish based on spectrophotometry, understanding the illumination-geometry influence onto spectrophotometric measurements is necessary. Therefore, Fresnel's law of reflection must be extended to apply for a diffuse illumination.

The extension of Fresnel's law of reflection for a diffuse illumination was introduced by Marcus and Pierce [Marcus 1994]. Their approach was to calculate the integration of Equation (2.22) around an incident angle from 0 to π in rad because the incident light is radiated from the entire directions. Now, let Equation (2.22) be

expressed as the following Equation:

$$\rho_T(\mu) = \frac{1}{2} \left[\left(\frac{a - \mu}{a + \mu} \right)^2 + \left(\frac{n^2 \mu - a}{n^2 \mu + a} \right)^2 \right] \quad (2.23)$$

where

$$\mu = \cos i, \quad a^2 = n^2 - \sin^2 i, \quad n = n_2 / n_1 \quad (2.24)$$

To compute the total reflectance in Equation (2.23) for a diffuse illumination, the equation is driven into Equation (2.25). Since $d\mu$ is changed from 0 to 1 that corresponds to a change in angle from 0 to $\pi/2$, the integration factor is multiplied by 2 to cover the entire incident angle from 0 to π .

$$\rho_{T, \text{diffuse}} = 2 \int_0^1 \rho_T(\mu) \mu d\mu \quad (2.25)$$

Figure 9 shows the comparison in Fresnel reflectance for normal incident light and for completely diffuse illumination. Both the blue line (normal) and the red line (diffuse) are drawn by plotting Equation (2.18) and Equation (2.25), respectively. The difference between the lines indicates that the amount of first-surface reflection is expected to be larger for an integrating sphere instrument (diff/8) than for a 45/0 instrument. It reminds in our varnish research that, regardless of whether a varnish is applied or not, a Saunderson coefficient, K_I , must have a larger figure whenever using an integrating sphere instrument. There is no doubt that the determination of a K_I value is affected by measurement geometry. In other words, it can be theoretic-

cally predictable from Figure 9 how many percentages of specular reflection are to be included in integrating-sphere measurement.

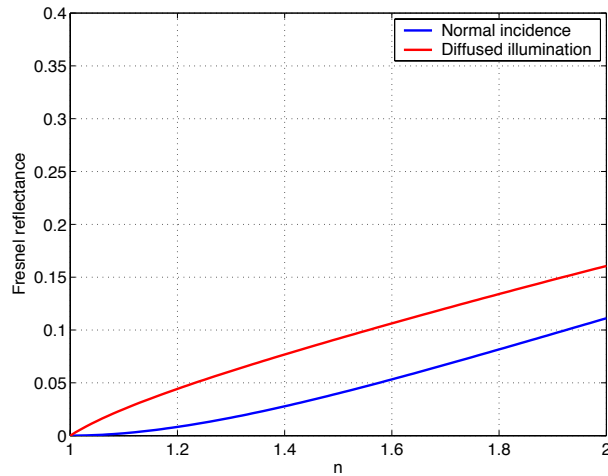


Figure 9. Fresnel reflectances for normal incident light and diffuse illumination.

2.3.3. Dichromatic Reflection Model

The dichromatic reflection model is a model describing two separable states of light reflection in a colored material. The reflection model has been well discussed in the Computer Graphic communities by 3D rendering engineers. According to one explanation, these reflection properties differ in geometry and photometry [Shafer 1985]. The first state of reflection is first-surface reflection, which is described by Fresnel's law of reflection as introduced in the previous section. This type of reflection is induced by the difference in refractive index (RI) the media like air and a paint. Therefore, the spectral radiation curve of surface reflection is identical to that of the illumination. The second state is body reflection, which occurs in the material along with scattering and absorption when the light is traveling through the body. The light enter-

ing a paint layer bounces off many times among pigment particles and binder resins and then returns to the surface. If the material shows perfect random scattering, the surface is called as Lambertian surface. A polytetrafluoroethylene (PTFE) tablet used for spectrophotometer calibration is one example of the Lambertian-like reflectors. The color of body reflection is determined by the absorption of pigments. The mathematical equation of the model is expressed:

$$L(\lambda, i, e, g) = L_{surface}(\lambda, i, e, g) + L_{body}(\lambda, i, e, g) \quad (2.21)$$

L is the light flux reflected from the material whereas $L_{surface}$ and L_{body} are the light fluxes of first-surface reflection and body reflection respectively. Figure 10 explains each of geometry angles (i , e , g) of the equation where a bold arrow indicates normal direction to the surface. λ is the wavelength of a lighting spectra.

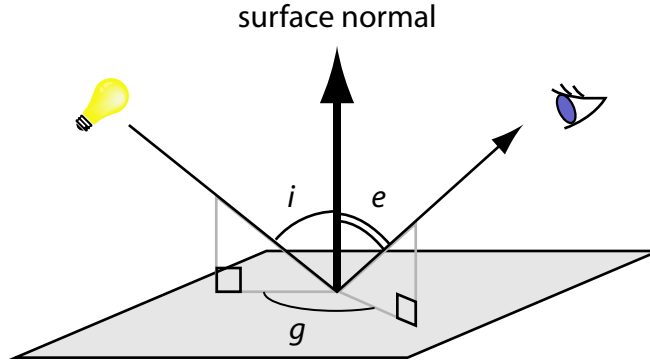


Figure 10. Scheme of explaining a viewing geometry where i is an incident angle, e is a detection angle, g is the phase between incident and detection directions.

This equation can be simplified by assuming that the magnitude of a reflection light is not dependent on wavelength but on geometry [Shafer 1985]. According

to this assumption, each of unique reflection factors is decomposed into a factor of wavelength and that of geometry.

$$L(\lambda, i, e, g) = m_i(i, e, g) c_i(\lambda) + m_b(i, e, g) c_b(\lambda) \quad (2.22)$$

In this simplified equation, $m_i(i, e, g)$ and $m_b(i, e, g)$ are the geometric scalars of first-surface reflection and body reflection ($0 \leq m_i, m_b \leq 1$), $c_i(\lambda)$ and $c_b(\lambda)$ are the spectrum of surface reflection and body reflection, respectively.

The dichromatic reflection model can be transformed to a simplified model when supposing spectrophotometric measurements with a integrating sphere instrument. Since the geometry is diffuse geometry in our experiment, all the geometries are fixed somewhat in Equation (2.22), and the terms of geometry functions, $m_i(i, e, g)$ and $m_b(i, e, g)$, are constants and then the equation can be transformed into:

$$L(\lambda) = m_i c_i(\lambda) + m_b c_b(\lambda) \quad (2.23)$$

where $L(\lambda)$ is the spectral power distribution (SPD) of the reflected light, and m_i and m_b are scalars. Equation (2.23) implies that m_i is determined by following Fresnel's law of reflection because $m_i c_i(\lambda)$ represents the radiation from first-surface reflection. The dichromatic reflection model implies the additivity of specular reflection regardless of what ever geometry used for a spectrophotometer. The basic idea of modeling the gloss effects that is introduced later follows the dichromatic reflection model described by Equation (2.24). According to the model, at most (100 m_i) % of specular reflection can be integrated in spectrophotometric measurement. This specular intrusion in a measurement is what we need to take into account.

2.3.4. Glossiness

The state of gloss is treated differently based on the appearance called “*glossiness*.” The concept of glossiness is related to the specifications of spectrophotometric measurement. A goniophotometer gives a full amount of reflectivity based on measurement geometries. The problem of using such a gonio instrument is the enormous amount of spectral reflectances at an innumerable combination of incident and detection angles. In terms of quality control, therefore, a gloss index of lower dimension has been developed for paint manufacturing. Judd and Wyszecki in their book [Judd 1975] introduce five categories of different glossiness; (1) specular, (2) sheen, (3) contrast, (4) distinctness of image, and (5) absence of bloom. To categorize each gloss state, incident angle, viewing angle, and the beam size of illumination have to be determined. All of them are quantified by the ratio of luminance between standard and a trial specimen at a specific geometry. Among various gloss definitions available, the degree of specular or specular gloss is commonly used as a gloss index in the paint industry.

“Specular” is defined as a glossiness characterized by the relative luminous reflectance factor of a specimen in the mirror direction [ASTM D523, 1999]. Both of incident and viewing angles are 60° and -60° from the normal in most cases. Unless measuring metallic and interference paints with a glossmeter, specular gloss will be confined within a range from 0 to 100. Metallic and pearlescent paints might indicate over 100 specular gloss because of oriented metallic flakes with high reflectances. If, however, the gloss indication of a glossmeter for a mirror reflection might be too small or out of the range because of extremely matte or gloss, using other geometries is recommended (e.g. 20 degree for over detection and 85 degree for poor detection, both of which in 60 degree).

2.4. Physics and Color Science on Varnishing

2.4.1. Refractive Indices of Artist Materials

A couple of optical phenomena that are associated with different refractive indices of artist materials can be considered.

In general, after solvent evaporation, the structure of a dried paint is condensed tightly enough to stabilize pigments and binder resins, both of which have identical or different RIs. As far as several types of varnishes are concerned, it will be noticed that varnish RIs vary because of different chemical compositions. We recognize from these facts that the appearance of a paint or a varnished paint is composed by complicated optical systems.

For example, the difference in refractive index between pigments and binder resins causes scattering in the medium. This case is applicable to the combination of Titanium White pigments and an acrylic emulsion. Comparing to the RI of Titanium Oxide (2.71), acrylic emulsion has obviously smaller RI (1.48) so that the large difference in RI increases scattering capability. Some portion of the incident light bounces around in the vehicle of pigments and binder resins back and forth in this manner (Figure.11). At the same time, the other portions are absorbed and captured within the fine crystals of pigments. The encapsulation of a light flux within a pigment crystal would be happened when pigment's RI is extremely higher than the binder's surrounding RI.

This phenomenon is explained as total internal reflection as shown in Figure 12. The illustration mentions about the critical angle, f , between air (1.00 RI) and a acrylic emulsion (1.48 RI) is 42.5° . If an incident angle is above 42.5° as shown in Figure 12, a light flux will be reflected to a mirror direction. As a whole, the amount of an incident light is reduced in the process of reflection so that such an optical mechanism is usually described as “subtractive.”

Additionally, the difference in refractive indices of air and a binder (or a varnish resin) causes surface reflection. An optimal varnish material used for picture varnish is expected to have the same RI as the binder. The basic requirement of varnish is to protect paintings from humidity, dust, and the other chemical pollutions by a thin film coat without changing the original appearance. Therefore, a pre-investigation on a binder's RI is required to be successful as a picture varnish.

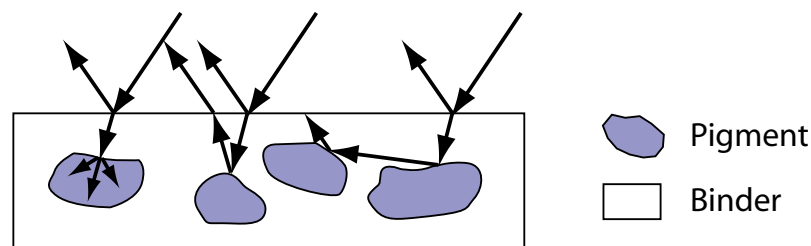


Figure 11. Light interactions in the medium.

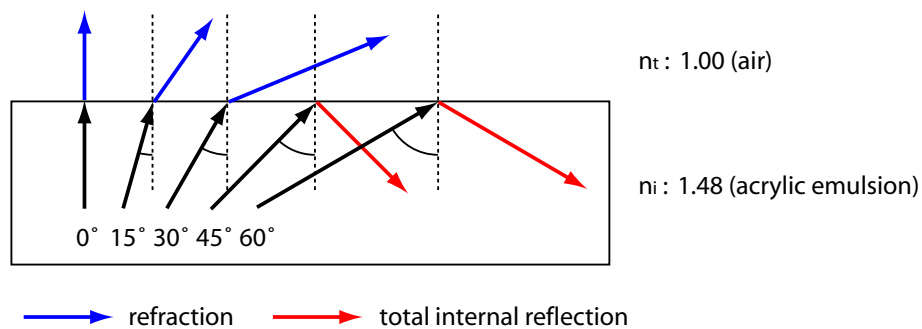


Figure 12. Light refraction and total internal reflection at above a critical angle.

2.4.2. Surface Roughness

With the increase of the RI difference between air and a varnish layer (or a paint layer), the amount of a light reflected from the surface to a viewer increases. However, the influence of surface roughness upon the appearance of a painting is more significant than that of RI differences. Not only the Fresnel reflection from the surface but also body reflection from a paint media that is diffused according to micro surface structures. We assume that, whether a surface is rough or smooth determines whether a specimen is gloss or matte respectively.

Berns and de la Rie analyzed the effects of RI and surface roughness on color appearance to prove the assumption [Berns 2003a]. According to their research, a varnish RI affects high reflectance rather than low reflectance. The effects of varnishing to diffuse reflectance factor had been formulated with the Ryde transmission equation that is introduced in their paper.

$$\Phi_{\text{total}} = \Phi_{\text{in air}} + \Phi_{\text{first surface}} \quad (2.24)$$

where

$$\Phi_{\text{in air}} = \frac{(1 - \rho_{\text{d,va}}) \Phi_{\text{in varnish}} T}{(1 - \rho_{\text{d,vp}} R)(1 - \rho_{\text{d,va}} R) - \rho_{\text{d,vp}} \rho_{\text{d,va}} T^2} \quad (2.25)$$

By using these equations, Berns and de la Rie simulated varnishing effects for a yellow paint in four cases: a 1.57 RI varnish, a 1.54 RI varnish, a 1.47 RI varnish, and no varnish. Their computational approach did not analyze the effects of first surface reflection in this case. In addition, it was hypothesized that the surface is

smooth enough to yield mirror reflection because an isotropic surface makes surface reflection diffused to some extent. The result implied that, even though the large difference in spectral curve was observed in high reflectance region, color difference caused by the spectral difference was less than 2 in L^* under the worst case (1.47 RI varnish). The reason of the small difference in lightness was due to perceptive contrast compression represented by a non-linear relationship of lightness perception to luminance. That is, the color appearance change of a paint sample by varnishing was almost indistinguishable.

On the contrary, to simulate two extremely different surface states, completely leveling roughness and completely replicating roughness, the researchers excluded the component of first surface reflection associating with the leveling from and added it associating with the replicating into the total reflection. That approach is equivalent to the simulation of different geometry measurements: SPEX for the leveling and SPIN for the replicating. A spectral curve of the complete replicating was equally higher at all wavelengths than that of complete leveling. Changes of colorimetric data based on the spectral curve differences appeared in L^* and C^*_{ab} . With leveling of the surface, lightness decreased as well as chroma increased. However, if replicating the surface, even if applying different RI varnishes, no significant difference appeared in colorimetric data.

2.4.3. Color Gamut Expansion by Varnishing

The research on the effects of a picture varnish to color gamut has been done by Berns and de la Rie [Berns 2003b]. Their experimental approach was using a glass sandwich to simulate the optical effects of varnishing mechanically. One inner side of the sandwich had the smooth surface of a glass plate. Another inner side was blasted

with sands on a glass plate so as to make the surface uneven. And then, the uneven surface was coated with different RI varnishes or nothing with different thicknesses. A positive photo film was inserted between the glasses as a coloration material. For this specimen, a light source illuminated the specimen from the backside, and a digital camera, which was calibrated to work as a colorimeter, captured the light passed through on the other side.

The result of colorimetric analysis showed that the varnish made the colors darken and lively in chroma (based on specular excluded measurement). That is, a color gamut expanded by varnishing was observed according to this simulation. As the authors mentioned, however, there is the unreasoning phenomenon that an increase in lightness by varnishing should not have occurred for bright colors. If a varnish coat on a paint film is actually exemplified instead of using such a transmission system, some percentage of a reflection flux from a paint layer shall be bounced off at the boundary because of different RIs. In other words, lightness for each of all the varnished paints is expected to be reduced. A ground glass without sand blast shows high scattering because of significant difference in RI between air and a varnish coat. It is reasonable to suppose that, on the other hand, difference in RI between sand particles and a varnish coating them is too slight to induce scattering. As a result, the amount of an incident light passing comes to being larger through a varnished ground glass than through a non-varnished ground glass.

To comprehend a mechanism of color gamut expansion, as a consequence, a reflection sample rather than such a transmission media needs to be exemplified. Therefore, we prepared actual varnished paint samples and analyzed them to clarify the gloss mechanism in a subtractive media.

2.5. Techniques of Paint Color and Glossiness Measurements

2.5.1. Spectrophotometer

Spectrophotometric measurement is the fundamental process to quantify the reflectivity of a paint medium. Of the variety of geometry reflectometers available, an integrating sphere spectrophotometer (diff/8°) with a specular port was used for this research. The main reason of this choice was our aim to collect all portions of diffuse surface reflection from a replicated or leveled surface structure formed by varnishing. In SPIN measurements, all the surface reflection is detected regardless of the different degree of light diffusion caused by surface roughness. On the other hand, SPEX geometry eliminates not all but most surface reflection with a specular trap in measurement. Considering of dichromatic reflection model, difference in spectral reflectance between SPIN and SPEX is expected to be a constant value at all wavelengths. For a glossy surface, at most 5% of incident light, which is calculated by Fresnel's law of reflection, are added in SPIN measurements [Hunter 1987]. In other words, the offset amount can be considered to represent the surface roughness of a paint specimen.

Johnston-Feller exemplified the effects of surface differences with colored plastic-gel coats [Johnston-Feller 2001]. In her experiment, red and gray plastic-gel coats were prepared and molded to replicate different gloss levels: matte (rough), semi-gloss (textured), and gloss (smooth). And then, spectral reflectance curves were measured with an integrating sphere instrument by switching the SPIN mode and the SPEX mode. The instrument was calibrated with a pressed BaSO₄ reference. In the SPEX mode, measured reflectance curves indicated that two glossy samples appeared darker than semi-gloss and matte samples. The offset amounts in her figures were 4 to 6 % reflectance. On the other hand, the spectral curves of those in the SPIN mode were indistinguishable at all wavelengths. Besides, while the reflectance curves of

the matte samples were almost identical regardless of whether SPIN or SPEX, those curves of glossy samples varied.

What is important is, no matter what specular mode is used, an integrating sphere cannot separate a spectrophotometric curve into two components, first-surface reflection and body reflection, when analyzing a rough surface material. Some operators might confuse the terminologies, first-surface reflection and body reflection with specular reflection and diffuse reflection, the later of which is actually separated in spectrophotometric measurements.

2.5.2. Glossmeter

To quantify the gloss level of a paint surface, a glossmeter, which is commonly used for quality control, is required to be prepared. The basic structure of a glossmeter is standardized in ASTM 523. According to the standard, glossmeter geometries are 20°, 60°, and 85°, each of which is selected by an instrument operator corresponding to the gloss level of a test specimen: paint, coating, and panel. Table II is the summary of gloss levels and their applicable geometries.

Table II. Specular gloss levels and appropriate gloss geometries.

Gloss level	Appropriate geometry
Ordinary	60°
60° gloss > 70	20°
60° gloss < 10	85°

Glossmeters give specular gloss or relative luminous reflectance factor as a gloss index. The ASTM standard defines that “*relative luminous reflectance factor is the ratio of the luminous flux reflected from a specimen to the luminous flux reflected from*

a standard surface under the same geometric conditions” [ASTM D523, 1999].

Figure 13 shows the structure of a parallel-beam glossmeter. From the left, an incandescent light source, the beam of which is focused by condenser lens attached in front of the source, is used to illuminate the surface of a test specimen. A source field aperture controls the amount of the light passing through. A source lens next to the aperture collimates the beam onto the surface with an appropriate beam diameter for projection. At the surface boundary, the light reflected from the surface travels into the direction of mirror reflection. A receptor lens focuses the reflected beam into a receptor field aperture so as to build up a source mirror image. And then, the image is focused by a collector lens through a spectral correction filter onto a photo detector at the end.

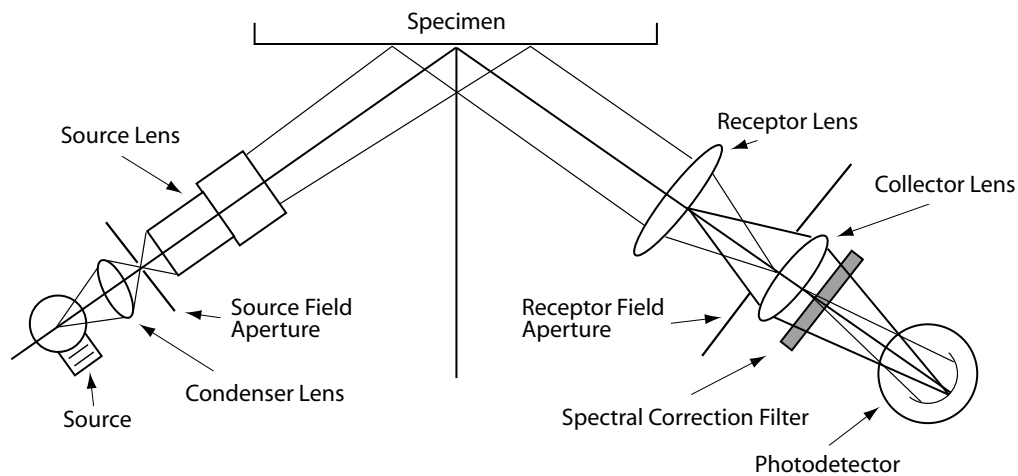


Figure 13. Structure of a parallel-beam glossmeter.

The important point of the glossmeter structure from color science’s aspect is its receptor design. A receptor consisting of a spectral correction filter and a photo detector is designed to simulate CIE luminous efficiency under a CIE standard illuminant

C. Therefore, it can be easily assumed that there is high correlation between specular gloss in a glossmeter and luminance factor in a colorimeter.

According to the ASTM standard, primary standards used for glossmeter calibration shall be made of highly polished, plane, black glass with a refractive index of 1.567 for the sodium D line (589 nm). The color of a standard tile, black, is expected to avoid an unexpected light absorption of colorants possibly happened as a result of body reflection. The reflection amount from the standard is normalized into 100, which is the standard value relative to the specular gloss of any test specimen. However, a commercial glossmeter produced by BYK Gardner is calibrated with a working standard (tile), which is produced for practical purposes to satisfy mass-production and low cost materials. For example, a BYK Gardner tri-micro-gloss meter in our laboratory must be calibrated with a black glass tile that indicate 92, 95, and 99 specular glosses at 20°, 60°, and 85° geometries respectively.

Before actual gloss measurement, the preparing size of a specimen need to be carefully determined. If the size is too small, a beam spot projecting on the surface might go beyond it at a large specular angle. Equation (2.26) indicates the relationship between spot long axis and measurement geometry, shown in the cosine equation:

$$r = \frac{d}{\cos \theta} \quad (2.26)$$

where d is a diameter of the normal incident light, θ is an incident angle, and r is the length of long axis for a projected light spot at some incident angle onto the surface. Figure 14 shows changes in lighting spot length based on measurement geometries following the cosine function.

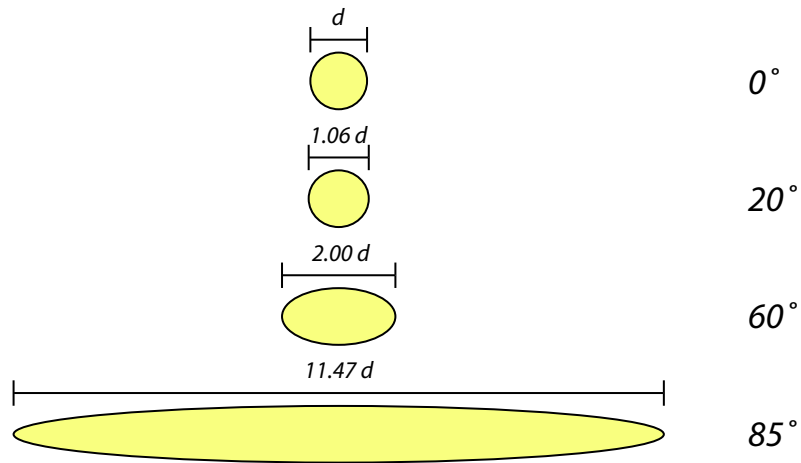


Figure 14. Comparison of the beam spot projection.

An advanced knowledge of light beam diffusion is useful when determining the size of a paint sample for gloss measurement. Since the size of each patch is large enough, for example, a GretagMacbeth ColorChecker is measurable with a BYK Gardner glossmeter. However, a GretagMacbeth ColorChecker DC is not. Therefore, our sample size introduced in the next chapter was determined to be at least a 3.1 cm by 3.1 cm square.

CHAPTER 3. SAMPLE PREPARATION

This chapter introduces the basic information on artist paint materials and drawdown preparation in this research: GOLDEN Matte Fluid Acrylic Paints, GOLDEN MSA Gloss Varnish, and paint drawdowns.

3.1. GOLDEN Matte Fluid Acrylic Paints

The target materials for the artist paint database are matte acrylic paints. A typical acrylic paint consists of pigments, acrylic emulsion (binder), water (solvent), and several coating additives. The entire set of the GOLDEN acrylic paints, which were chosen for this research, has 26 color variations and a white. While some of the paints were made of single pigments, the other paints contain two or three pigments. Table III shows paint names and their containing pigments with Colour Index Name (C.I. Name).

Table III. List of GOLDEN paint names, C.I. names, and pigment names.

Paint Name	C.I. Name	Pigment Name
Hansa Yellow Opaque	PY 74	Arylide Yellow 5GX
Diarylide Yellow	PY 83	Diarylide Yellow HR-70
Pyrrole Red	PR 254	Dipyrrolopyrrol
Naphthol Red Medium	PR 5	Naphtol ITR
Quinacridone Crimson	PR 206 / PR 202	Quinacridone
Quinacridone Magenta	PR 122	Quinacridone
Dioxazine Purple	PV 23	Carbazole Dioxazine
Anthraquinon Blue	PB 60	Indanthrone Blue

Ultramarine Blue	PB 29	Polysulfide of Sodium-Alumino-Silicate
Cobalt Blue	PB 26	Oxides of Cobalt and Aluminum
Cerulean Blue, Chromium	PB 36:1	Oxides of Cobalt and Chromium
Phthalo Blue (Green Shade)	PB 15:4	Copper Phthalocyanine
Phthalo Green (Blue Shade)	PG 17	Chlorinated Copper Phthalocyanine
Jenkins Green	PBk 9 / PY 150 / PG 36	Amorphous Carbon Nickel Complex Azo Brominated & Chlorinated-Copper Phthalocyanine
Permanent Green Light	PY 3 / PG 7	Arylide Yellow Chlorinated Copper Phthalocyanine
Chromium Oxide Green	PG 17	Anhydrous Chromium Sesquioxide
Green Gold	PY 150 / PG 36 / PY 3	Nickel Complex Azo Brominated & Chlorinated-Copper Phthalocyanine
Yellow Ochre	PY 43	Natural Hydrated Iron Oxide
Raw Sienna	PY 43	Natural Iron Oxide
Burnt Sienna	PBr 7	Calcined Natural Iron Oxide
Red Oxide	PR 101	Synthetic Red Iron Oxide
Burnt Umber	PBr 7	Calcined Natural Iron Oxide containing Manganese
Raw Umber	PBr 7	Natural Iron Oxide containing Manganese
Paynes Gray	PB 29 / PBk 7	Ultramarine Blue Carbon Black
Carbon Black	PBk 7	Nearly Pure Amorphous-Carbon
Titan Buff	PW 6	Titanium Dioxide Rutile
Zinc White	PW 4	Zinc Oxide
Titanium White	PW 6	Titanium Dioxide Rutile

We expect that some of the acrylic paints contain the same pigments as the line-up of Gamblin Conservation Colors. The reason is that the Gamblin paint line-up is a popular oil-paint set used for painting conservation in museums. The research on acrylic paints is expected to contribute to a future related-paint research on the pigment identifications of cultural heritage oil paintings (e.g. Van Gogh's, Henri Matisse) with a spectral imaging system. Table IV shows the table of paint names using the same pigments between GOLDEN products and Gamblin products.

Table IV. GOLDEN acrylic paints and Gamblin oil paints having the identical pigments.

GOLDEN Matte Fluid	C.I. Name	Gamblin Conservation
Hansa Yellow Opaque	PY 74	Hansa Yellow Medium
Diarylide Yellow	PY 83	Indian Yellow Permanent
Dioxazine Purple	PV 23	Dioxazine Purple
Ultramarine Blue	PB 29	Ultramarine Blue
Phthalo Green (Blue Shade)	PB 15:4	Manganese Blue Hue
Chromium Oxide Green	PG 7	Chromium Oxide Green
Red Oxide	PR 101	Transparent Earth Red
Burnt Sienna	PBr 7	Burnt Sienna
Raw Umber	PBr 7	Raw Umber
Burnt Umber	PBr 7	Burnt Umber
Yellow Ochre	PY 43	Yellow Ochre
Titanium White	PW 6	Titanium White

The spectral and colorimetric data shows the unique properties of opaque color of each acrylic paint. Figure 15 shows the spectral reflectance curves of masstones for all the paint set. The spectral reflectances were measured with an integrating sphere instrument, GretagMacbeth ColorEye XTH, with a small aperture in the SPIN mode. The spectra range in measurement is from 360 nm to 750 nm. One of the interesting features is that, Cobalt Blue and Cerulean Blue have the similar reflectance curves of

two broad peaks. Both of them show higher reflectances in the red spectral region. The peak wavelengths of the blue paints at the visual short wavelengths are slightly different each other. While Cobalt Blue has a peak at 430 nm, Cerulean Blue has a peak at 460 nm. Actually we can observe the difference that Cerulean Blue appears more greenish by comparing with Cobalt Blue. Another interesting feature of the reflectances appears in Titanium White. Titanium White shows strong absorption at the short wavelengths from 360 nm to 420 nm. The reflectance of the white in that spectral region is much smaller than those of Cobalt Blue and Cerulean Blue. This result implies that the scattering of the blue paints is larger than that of the white in coloration systems because colors showing higher scattering properties have higher reflectances.

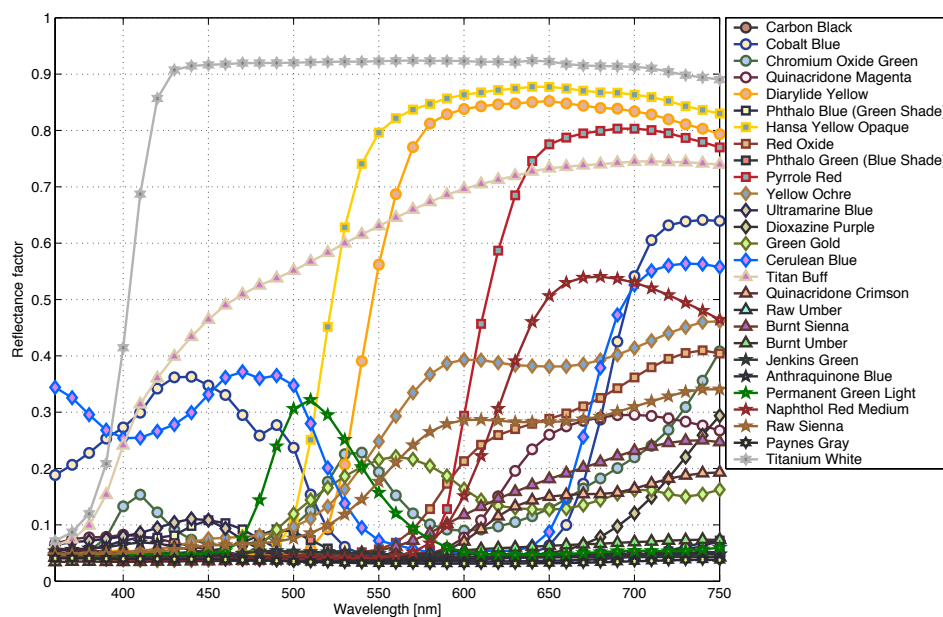


Figure 15. Spectral reflectance curves for 27 paints of the acrylic paint set.

Figure 16 shows spectral K/S curves of masstone for 27 paints in log scale. The K/S curve for each paint was computed with the certain Saunderson coefficientss, $K_1 = 0.003$ and $K_2 = 0.061$, from the measured reflectance curves shown in Figure 15. Tha paint having an interesting feature, which we can observe in Figure 16, is Payness Gray. Payness Gray shows extremely strong absorption so that a K/S value is significantly different by six order from Titanium White at 570 nm.

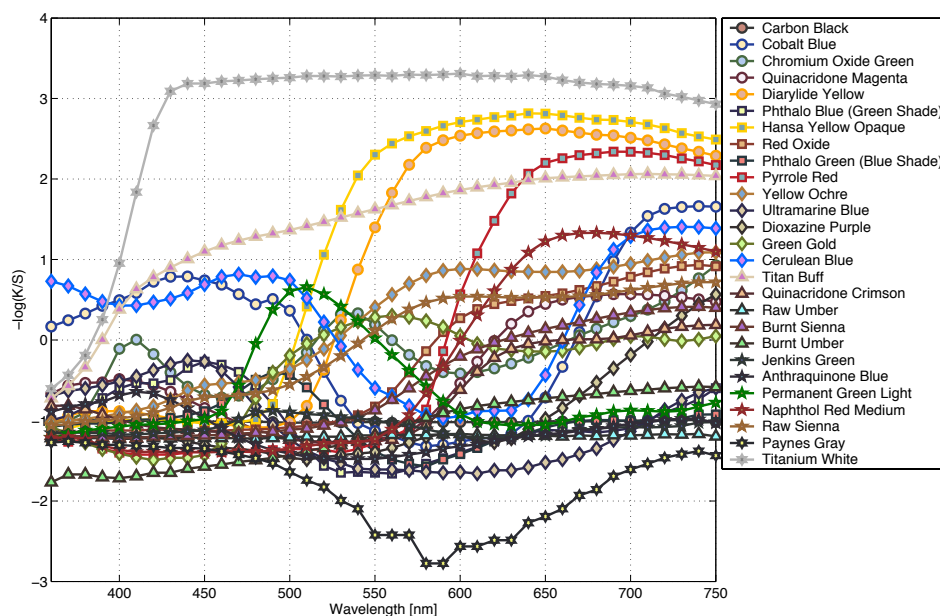


Figure 16. Spectral K/S curves for 27 paints of the acrylic paint set.

Figure 17 shows changes in chroma with a increase of colorant percentage for the paint set. Lightness and chroma for some paints represented as Hansa Yellow Opaque and Chromium Oxide Green linearly decrease in the transition from light tone to masstone. On the other hand, the other paints including Green Gold and Quinacridone Magenta show the peaks of chroma at the middle tones. To understand the presence

of maximum chroma is intrinsic when rendering color gamut. Mohammadi and Berns (2004) used maximum chroma for reducing the required number of tint ladders for a colorant formation based on the Kubelka-Munk solution [Mohammadi 2004]. Their analyzing result emphasized that using only three tint ladders, tint, maximum chroma, and masstone, are necessary and then achieve a good estimation performance of absorption and scattering properties. Their approach is a good example of trying to render a paint color gamut accurately and with a minimum number of samples.

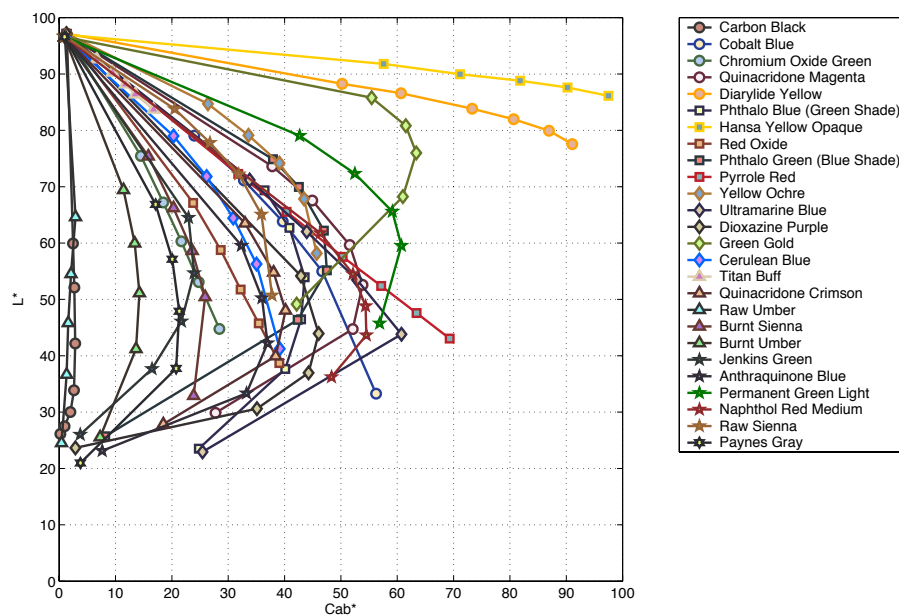


Figure 17. Colorimetric plots for 27 paints of the acrylic paint set.

3.2. GOLDEN MSA Gloss Varnish

To analyze the gloss effects by varnishing, we applied a GOLDEN Mineral Spirits Acrylic (MSA) Gloss varnish for dried matte-acrylic paints. The MSA varnish is soluble in a strong solvent, (e.g. pure mineral spirits). When thinning with a mineral spirits, it is desirable for the solvent to have a high concentration of xylene in terms of solubility. We used Stoddard mineral spirits giving off a strong odor in this research. Since the irritating odor of a strong solvent gives damages to your lung and kidney, any operation on varnishing must be done in a room with a ventilation system. For the details of paint material safety, we would like paint researchers to read the Material Safety Data of the Products.

The requisite process for varnishing is to find an appropriate recipe of mixture of a varnish and a solvent. Viscosity controlled by the ratio of a varnish and a solvent determines the surface construction of a varnish coat as shown in Figure 18. In order to disperse varnish resins uniformly over a paint surface, a high viscous solution has to be used because strong adhesive force stabilizes varnish resins dispersed in liquid. The drawback in using a thick varnish solution is difficulty in application caused by strong friction force. A high viscous solution may give mechanical surface defects like scratches onto a coating surface when applying a varnish with a metal bar. On the other hand, a thin varnish solution is applicable for making a thin coat and a low-gloss finish. However, a low-viscous solvent might sometimes cause too much fluid and a flocculation of varnish resins in a wet coat. Pros and cons both are available for each recipe. By following an instruction manual guided by GOLDEN, therefore, we attempted several recipes from 2:1 (varnish:solvent) at the starting point to varnish only (no solvent). With the thought of replicating a surface roughness and economy, we determined that our recipe of a varnish solution was 2:1 (varnish:solvent) leading

to a thin varnish coat. This case is equivalent to the middle illustration of Figure 18.

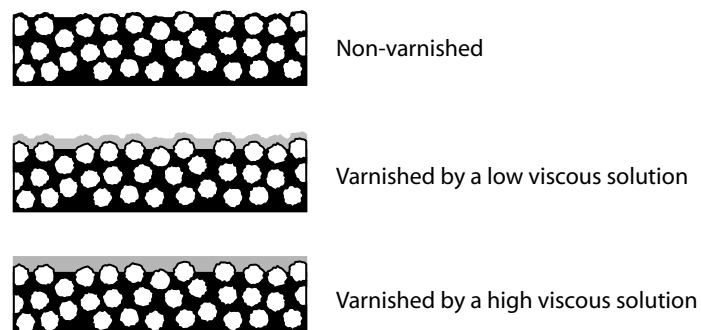


Figure 18. Surface constructions based on different viscous varnishes.

3.3. Drawdowns

Making paint drawdowns is a requirement for paint characterization. As a drawdown substrate, a LENETA opacity chart was used. The chart has a wide black stripe between two white stripes, the design of which is used for evaluating paint opacity in contrast. For each of the target paints, in addition to the drawdown of a tint (Titanium White), 5 or 6 different concentration series (20%, 40%, ..., 100%) of a colorant must be prepared. Determining the appropriate steps of concentration is dependent on the tinting strength of a colorant. For example, Phthalo Green and Phthalo Blue show strong tinting strength. A colored paint of strong tinting strength is hard to be tinted with Titanium White. As a result, the tint ladders made at the even concentration steps have fewer bright colors perceptually. In such a case, we need to prepare additional paint samples at light tone (e.g. 5%) in order to improve the performance of the paint

characterization.

For paint application, a BYK Gardner wide applicator with a 10mil gap was used. A target paint is rubbed on an opacity chart by moving an applicator at 6 cm/s from the top to the bottom as shown in Figure 19-(a) [ASTM D2805, 1996]. At this thickness, however, some of the GOLDEN acrylic paints showed translucency. The trend was observed especially for yellow paints (e.g. Hansa Yellow Opaque and Diarylide Yellow) and phthalo-based paints. In such a case, we needed to make a more thick paint film than 10mil somehow. Hence, we attempted two approaches for a thick coating: (1) apply a paint multiple times crossways (2) making the more gap of an applicator by adding papers into between the bar edges and a substrate. The first approach sometimes causes an uneven paint surface. For instance, the first-made layer of a paint might trap some air bubbles so as to be an uneven paint surface. It is true that the surface structure of an upper layer given in the next trial would depend on the smoothness of the underlying layer.

If the large area of a paint coat is necessary, the second approach was more appropriate. However, the uncertainty of a film thickness by taking the second approach would be a problem if we need to control a film thickness. Fortunately, we did not need to take care of an exact film thickness because paint drawdowns were expected to be opaque in this research. So we used the second approach to produce more thick films.

The approximate time required for perfect drying was several hours at room temperature. During the drying process, wet paint drawdowns were protected from dust and other air pollutions.

Regarding varnish application, a BYK Gardner narrow applicator with a 10mil gap was used. Before varnish application, we confirmed if a paint film to be varnished

was dried completely. The portion of a varnish solution was dripped near a paint film, and then it was smoothed over the film surface crossways as shown in Figure 19-(b).

According to an instruction manual by GOLDEN, recommendable drying time for varnish is 3-6 hours. However, we observed that the surface of a varnish was still sticky even 1-2 days after varnish application. Of course, it depends on drying room environments, our recommending drying time for varnish is at least one week.

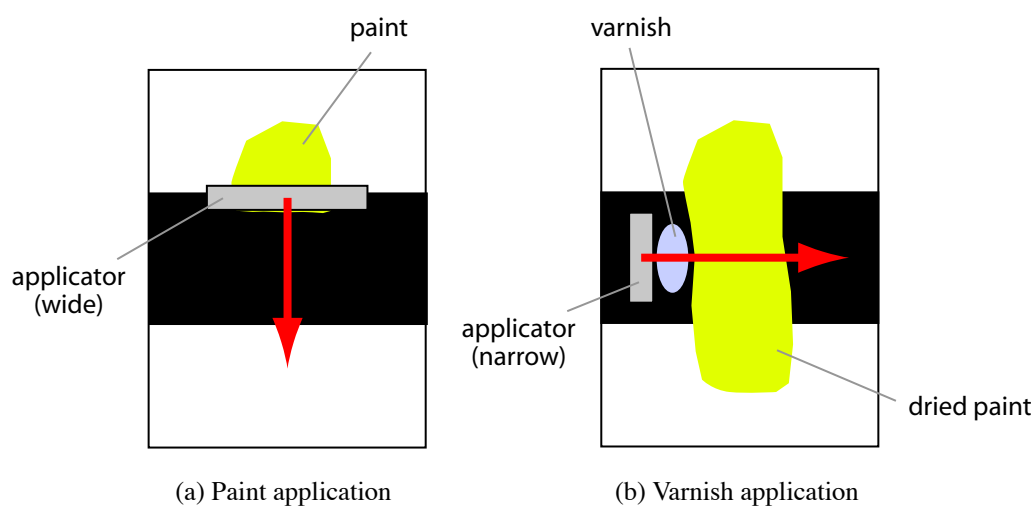


Figure 19. Paint and varnish applications.

CHAPTER 4. OPTIMIZING PAINT MIXING MODEL COEFFICIENTS

This chapter discusses the optimization approaches used to determine the paint mixing model coefficients: unit k , unit s (in the Kubelka-Munk theory), K_1 and K_2 (in Saunderson correction), and optimization results for the acrylic paint set.

4.1. Validity of the two-constant Kubelka-Munk theory

We used the two-constant Kubelka-Munk theory to build up the artist paint database. The reason of choosing the two-constant theory, not the single-constant theory, was due to better characterization performance.

In order to confirm the advantage of using the two-constant theory, we chose 13 paints of the acrylic paint set and compared characterization performances for them by the two theories. The paint characterization was executed for each individual paint by using each of the two theories and the least square method. For only the single-constant theory, we used effective concentrations, which were calculated by the summation of K/S curves, instead of theoretical concentrations. With regard to the characterization of an individual paint, the Saunderson coefficients, K_1 and K_2 , were assigned empirical values.

Table V shows the summary of characterization performances in CIEDE2000. The table apparently indicates that the characterization performances by the two-constant theory is much better than those by the single-constant theory. While those in the single-constant theory are located around between $1.0 \Delta E_{00}$ and $5.0 \Delta E_{00}$, all the mean CIEDE2000 values in the two-constant theory are lower than $1.0 \Delta E_{00}$.

Table V. The two-constant theory vs. the single-constant theory in characterization performance for 13 acrylic paints in CIEDE2000.

Paint name	K1	K2	Two-constant		Single-constant	
			<i>mean</i>	<i>max</i>	<i>mean</i>	<i>max</i>
Carbon Black	0	0.4	0.66	2.00	0.81	1.38
Cobalt Blue	0.03	0.6	0.24	0.41	1.18	3.47
Chromium Oxide Green	0.04	0.6	0.22	0.65	2.05	5.38
Quinacridone Magenta	0.04	0.6	0.41	1.31	1.55	7.78
Diarylide Yellow	0.04	0.6	0.23	0.87	1.83	4.48
Phthalo Blue (Green Shade)	0.03	0.6	0.37	1.12	4.83	14.72
Hansa Yellow Opaque	0.04	0.6	0.12	0.26	1.99	6.15
Red Oxide	0.04	0.6	0.21	0.71	3.86	12.55
Phthalo Green (Blue Shade)	0.03	0.6	0.64	2.12	4.34	20.87
Pyrrole Red	0.03	0.6	0.21	0.52	2.33	6.14
Yellow Ochre	0.04	0.6	0.18	0.44	1.26	4.04
Ultramarine Blue	0.03	0.6	0.19	0.59	0.52	2.17
Dioxazine Purple	0.03	0.6	0.22	0.80	2.70	11.96

The reason of poor characterization for the single-constant theory is due to the poor scalability of K/S curves. Figure 20 shows the theoretical-effective concentration curve and the K/S curves of tint ladders for Phthalo Green (Blue Shade) as an example. The relationship between theoretical concentration and effective concentration is non-linear or exponential. In addition, the peak of a K/S curve at each colorant concentration varies. This result implies the phenomenon that scattering properties are changed with corresponding to colorant concentrations. The single-constant theory did not compensate such a spectral change in scattering so that characterization performance might be worse for this green paint. In addition, the same trends were confirmed in some color paints of the paint set.

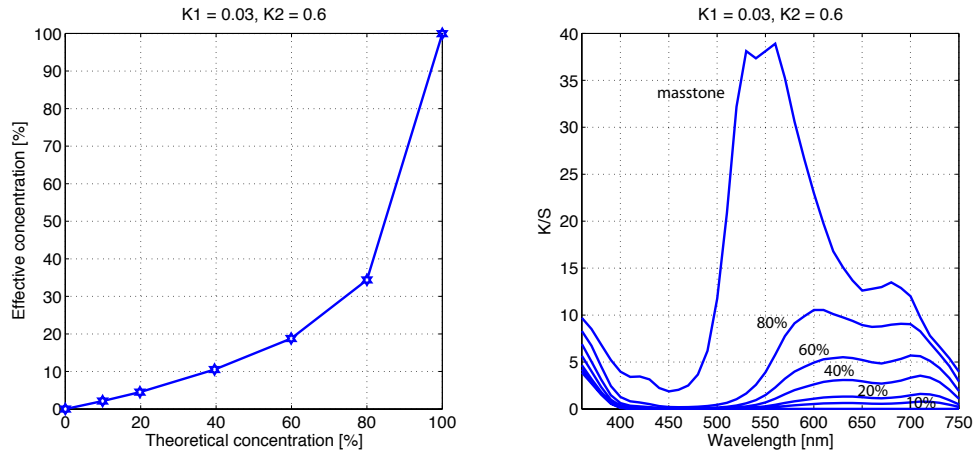


Figure 20. Poor scalability of K/S curves based on the single-constant theory for Phthalo Green (Blue Shade).

From these results, we decided to use the two-constant Kubelka-Munk theory to build up the artist paint database.

4.2. Metrics

In the two-constant Kubelka-Munk solution, we employed three different metrics in spectral comparison that is to be minimized when optimizing the paint mixing model coefficients. Each of the metrics has unique features designed for specific purposes.

The first metric is to use root-mean-square (RMS) in reflectance as shown in Equation (2.24). Minimizing a RMS value is statistically meaningful because this approach is equivalent to minimizing the variance of a spectral error distribution.

The second metric was to use weighted RMS (wRMS) based on the diagonal of Matrix R that is developed by Berns. The concept of using this metric is to minimize spectral estimation errors and XYZ estimation errors simultaneously. The spectral

curve of the diagonal of Matrix R is almost equal to the summation of color matching functions, $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, and $\bar{z}(\lambda)$. Figure 21 shows the spectral curve of the diagonal of Matrix R. The mathematical expression of wRMS (Matrix R) is exhibited in Equation (2.27) where i is the i th colorant of the paint set, j indicates the j th wavelength, N is the number of wavelengths in spectral measurement, w_i is the diagonal of Matrix R at the i th wavelength, R_m and $\hat{R}_m$ are measured reflectance and estimated measured reflectance respectively.

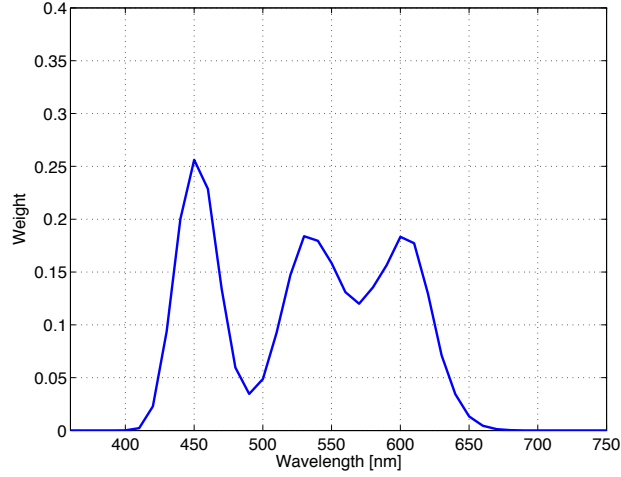


Figure 21. The diagonal of Matrix R for weighted RMS.

$$\text{wRMS}_i = \sqrt{\frac{1}{N} \sum_{j=1}^N \sqrt{w_j} \{\hat{R}_{m,i}(\lambda_j) - R_{m,i}(\lambda_j)\}^2} \quad (2.27)$$

The last metric to be introduced is to use Viggiano's Spectral Comparison Index (SCI) [Viggiano 2001]. The concept of the SCI is to quantify a spectral difference and a color matching error simultaneously with a single index. The SCI is calculated by

using Equation (2.28):

$$M_v = \sum_{j=1}^N w(\lambda_j) \cdot \|\Delta \beta(\lambda_j)\| \quad (2.28)$$

where $w(\lambda)$ is Viggiano's spectral weight based on ΔE_{ab}^* as described in Equation (2.29), and $\beta(\lambda)$ is the difference between R_m and $\hat{R}_m$. The Viggiano's spectral weight is computed by calculating the derivatives of L^* , a^* , and b^* . The detailed explanation of mathematical equations is written in his paper.

$$w(\lambda) = \sqrt{\left(\frac{dL^*}{d\beta(\lambda)}\right)^2 + \left(\frac{da^*}{d\beta(\lambda)}\right)^2 + \left(\frac{db^*}{d\beta(\lambda)}\right)^2} \quad (2.29)$$

The interesting feature of the SCI is a flexible changing of spectral weight in response to tristimulus values of a target color. Figure 22 shows spectral weight curves changed by the reflectances of tint ladders for Hansa Yellow Opaque as an example. The calculation is based on D65 illuminant and the 1931 standard observer. The reflectance curves of the yellow are different in the blue spectral region because of absorption by yellow pigments. Therefore, it can be observed that the spectral weights in Figure 22 are dramatically changed in that spectral region. While the highest peak in a yellow line indicates the Hansa Yellow Opaque masstone, the lowest flat curve expresses a Titanium White. That is, as SCI's concept on color difference is introduced, it is actually confirmed that a lower reflectance color has a higher spectral weight and vice versa.

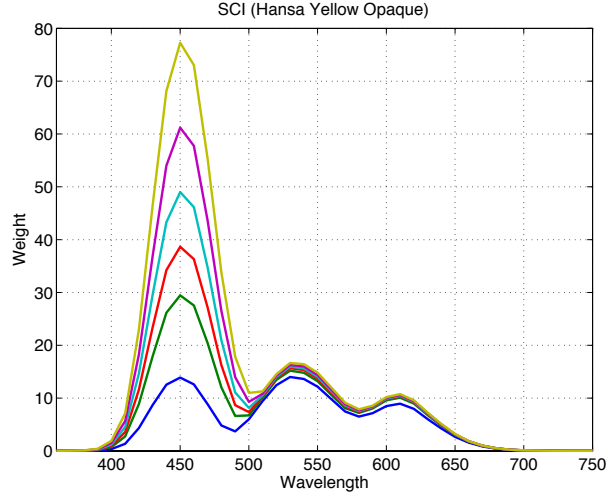


Figure 22. Spectral weights in the SCI for Hansa Yellow Opaque.

The drawback of the SCI for optimization is that the equation calculates a residual instead of root-mean-square. In other word, minimizing the SCI does not lead to minimizing the variance of a spectral error distribution in optimization. This index architecture was an issue in a precedence experiment when using SCI as an objective function in an optimization processing. To solve this problem, we designed a new weighted RMS criteria, which is similar to Equation (2.27), based on Viggiano's spectral weight:

$$\text{wRMS}_i = \sqrt{\frac{1}{N} \sum_{j=1}^N w(\lambda_j) \{\hat{R}_{m,i}(\lambda_j) - R_{m,i}(\lambda_j)\}^2} \quad (2.30)$$

where w_i is Viggiano's spectral weight as a function of R_m .

Finally, to discriminate two types of weighted RMS, we call wRMS based on the diagonal of Matrix R “wRMS1” and wRMS based on Viggiano's spectral weight “wRMS2” in this thesis.

4.3. Local and Global Optimization Procedures

To obtain the Kubelka-Munk coefficients and Saunderson Coefficients, we tested two optimization approaches, “*local optimization*” and “*global optimization*”. By using each of the optimization approaches, we characterized 27 acrylic paints including Titanium White.

Local optimization:

The first approach is called “*local optimization*”. This optimization process is to accomplish two computation tasks simultaneously for each colorant: (1) solving the two-constant Kubelka-Munk equation (Equation (2.18)), and (2) optimizing the Saunderson coefficients. In these procedures, the average of R_m RMS errors for the entire paint set are to be minimized. In local optimization, as a result, different paints have different Saunderson coefficients. This approach is expected to give the best characterization performance for each colorant.

Global optimization:

The second approach is called “*global optimization*”. The purpose of using global optimization is to eliminate the mismatch of the Saunderson coefficients among all the colorants that would occur in local optimization. When using global optimization, it is important to consider of how to assign the initial values of the coefficients.

In global optimization, a single K_1 and a single K_2 , are determined for the entire paint set. That is, an optimal K_1 and an optimal K_2 are obtained that minimized the average of R_m RMS errors for the entire paint set. Unlike local optimization, the initial value of K_1 is assigned the least reflectance of the paint set (not each individual

paint) over all the wavelengths because assigning a larger K_I value than the least reflectance causes the over-reduction of first-surface reflection by the Saunderson correction so that the internal reflectances of some colorants will be negative.

In addition to optimizing the Saunderson coefficients, each of two following computations must be executed simultaneously to obtain optimal unit k and unit s for each colorant.

The first computation is to process the least square method for each colorant as introduced in Equation (2.18). The mathematical computation is not complex, however, it depends on paint researcher's decision to figure out how many tint ladders are to be prepared to achieve an accurate characterization. We attempted two cases: (1) using 5 to 6 tint ladders in a regular approach, and (2) using only three tint ladders (tint, maximum chroma, and masstone), proposed by Mohammadi and Berns [Mohammadi 2004]. Table VI exhibits the list of paints having maximum chroma and their concentrations. The concentrations of maximum chroma in the table can be read from Figure 17. For the paints having no maximum chroma (e.g. Carbon Black and Diarylide Yellow), additionally, we chose tint ladders with middle chroma (for chromatic paints) or middle lightness (for achromatic paints) in the gradations instead of maximum-chroma tint ladders.

Table VI. List of paints having maximum chroma and their colorant concentrations.

Paint name	Concentration [%] (maximum chroma)
Quinacridone Magenta	60.3
Phthalo Blue (Green Shade)	39.6
Phthalo Green (Blue Shade)	60.1
Ultramarine Blue	79.7
Dioxazine Purple	40.4
Green Gold	59.8
Quinacridone Crimson	60.4
Burnt Sienna	80.0
Burnt Umber	59.9
Jenkins Green	39.8
Anthraquinone Blue	59.9
Permanent Green Light	78.4
Naphthol Red Medium	78.4
Paynes Gray	60.2

The second computation is to process non-linear optimization. The process of the non-linear optimization is already introduced in Figure 5 (2.2.3 *Optimization Techniques for the Kubelka-Munk Solution*). The criteria, the average of which is to be minimized for the tint ladders of all the colorants, are three metrics: (1) R_m RMS, (2) R_m RMS1, and (3) R_m wRMS2.

Table VII summarizes the details of several optimization approaches mentioned above, each of which was tested to evaluate characterization performance.

Table VII. Summary of the optimization approaches and their details.

#	Approach	Method	Optimizing	Characterizing
1	Local	least square & minimizing R_m RMS	$K_{1,j}, K_{2,j}$ ($i = 1, \dots, 26$)	for each individual paint
2	Global	least square & minimizing R_m RMS	K_1, K_2	for the entire paints simultaneously
3	Global	least square & minimizing R_m RMS (maximum chroma)	K_1, K_2	for the entire paints simultaneously
4	Global	minimizing R_m RMS	unit k_i , unit s_i ($i = 1, \dots, 26$) K_1, K_2	for the entire paints simultaneously
5	Global	minimizing R_m wRMS1	unit k_i , unit s_i ($i = 1, \dots, 26$) K_1, K_2	for the entire paints simultaneously
6	Global	minimizing R_m wRMS2	unit k_i , unit s_i ($i = 1, \dots, 26$) K_1, K_2	for the entire paints simultaneously

4.4. Characterization Performance

Table VIII summarizes the characterization performance of the artist paint database based on each optimization method. The characterization errors in R_m RMS and CIEDE2000 are the average values for the entire tint ladders. These results indicate that, there are no significant difference in characterization performance even though taking different optimization approaches.

Table VIII. Characterization performances for the paint database based on optimization approaches.

Optimization approach	R_m RMS			CIEDE2000		
	mean	max	90th*	mean	max	90th*
Local optimization (least square method)	0.004	0.028	0.001	0.29	2.05	0.78
Global optimization (least square method)	0.005	0.058	0.012	0.29	3.24	0.70
Global optimization (least square method, using tint, max chroma, and masstone)	0.006	0.041	0.021	0.39	3.60	1.27
Global optimization (minimizing R_m RMS)	0.004	0.029	0.013	0.31	3.11	0.74
Global optimization (minimizing R_m weighted RMS by the diagonal of Matrix R)	0.006	0.065	0.015	0.33	3.15	0.68
Global optimization (minimizing R_m weighted RMS by Viggiano's weight)	0.006	0.058	0.015	0.29	3.19	0.71

* 90th percentile

The analyzed results of characterization performance for each optimization method in more details are listed in Chapter10. Appendices, “*E. Spreadsheets of Paint Database performances*”.

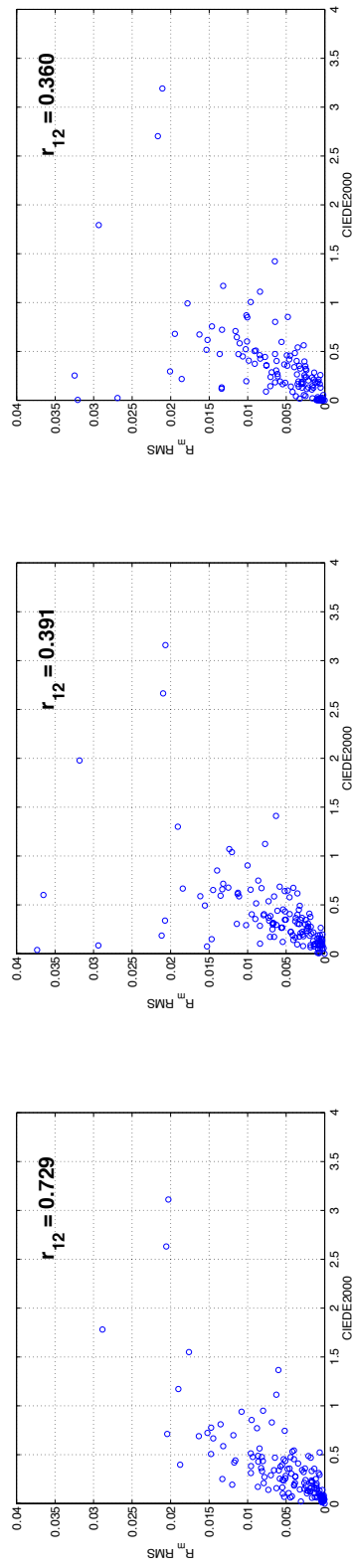
4.5. Discussion

We attempted several optimization approaches to find the best characterization method for GOLDEN matte fluid acrylic paints. On the whole, fortunately, the characterization performance by each method shown in Table VIII is not significantly

different. This result can be interpreted that our target paints were well characterized by the two-constant Kubelka-Munk theory regardless of different minimizing criteria and not using effective concentration. Therefore, we could not find any advantage of using weighted RMS error metrics ($wRMS1$ and $wRMS2$).

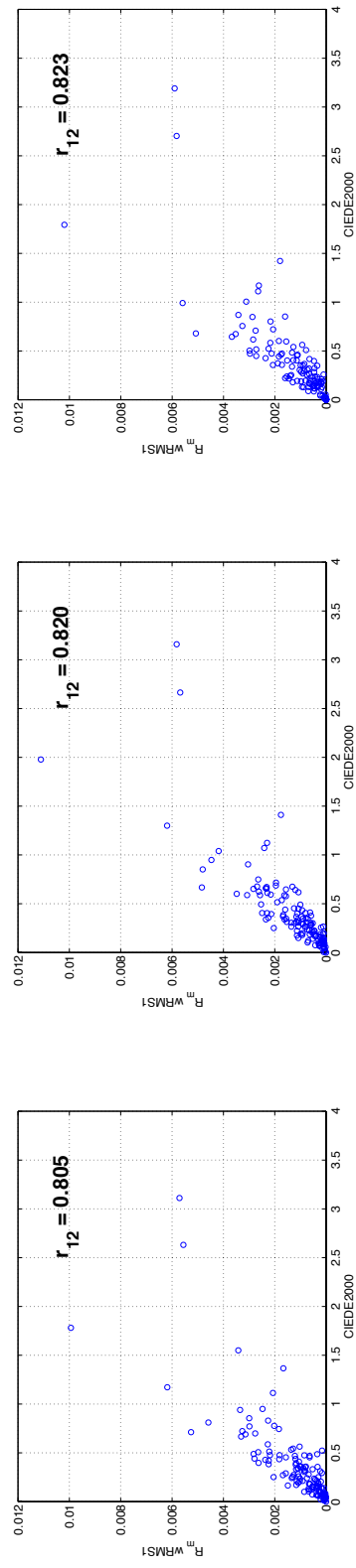
However, we would like to rate these optimization methods somehow. Therefore, we kept our eyes on the versatility of a spectral error metric. In terms of spectral matching, a spectral error metric (e.g. R_m RMS) rather than CIEDE2000 can be considered to be the best error criterion for spectral imaging systems when concerning metamerism. However, it is also true that we still need to evaluate spectral imaging systems in terms of colorimetry. Therefore, we expect that an ideal spectral-error metric should quantify not only spectral errors but also colorimetric errors. If some spectral error metric is the best, it must show a high correlation between spectral and colorimetric errors.

There are differences in correlation between CIEDE2000 and spectral error metrics (RMS, $wRMS1$, and $wRMS2$) in reflectance that resulted from global non-linear optimizations for the paint set. Figure 23 shows the plot of correlation between CIEDE2000 and R_m RMS for each optimization method. The optimization by minimizing R_m RMS shows high correlation that is not observed in the cases of the optimization by minimizing R_m $wRMS1$ and the optimization by minimizing R_m $wRMS2$. Figure 24 and Figure 25 show the plots of correlation between CIEDE2000 and R_m $wRMS1$, CIEDE2000 and R_m $wRMS2$ respectively for each optimization method. Since all the optimization method show high correlation, there is no significant difference in these figures.



(a) Optimization by minimizing R_m RMS (b) Optimization by minimizing R_m wRMS1 (c) Optimization by minimizing R_m wRMS2

Figure 23. Correlation of characterization errors between CIEDE2000 and RMS.



(a) Optimization by minimizing R_m RMS (b) Optimization by minimizing R_m wRMS1 (c) Optimization by minimizing R_m wRMS2

Figure 24. Correlation of characterization errors between CIEDE2000 and wRMS1.

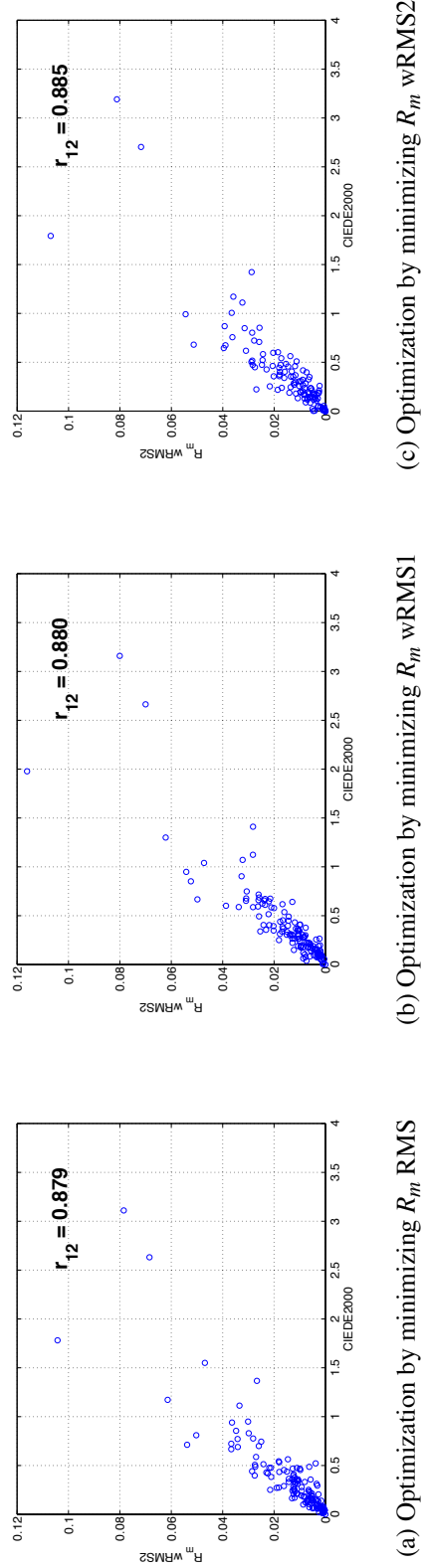


Figure 25. Correlation of characterization errors between CIEDE2000 and RMS2.

Table IX shows the summary of the correlation assessments based on correlation coefficient (r_{12}). From this table, it can be read that, even though it depends on what metric is used for spectral error evaluation, minimizing R_m RMS in optimization covers all of the spectral comparison metrics for the paint set.

Table IX. Correlation ratings between CIEDE2000 and spectral error metrics for global optimization methods. (⊙: excellent*, ○: good*, △ : poor*)

Correlation	Optimization metric		
	R_m RMS	R_m wRMS1	R_m wRMS2
CIEDE2000 vs. R_m RMS	○	△	△
CIEDE2000 vs. R_m wRMS1	⊙	⊙	⊙
CIEDE2000 vs. R_m wRMS2	⊙	⊙	⊙

* $r_{12} > 0.80$

** $r_{12} > 0.70$

*** $r_{12} < 0.50$

From this result, we concluded that the optimization method by minimizing R_m RMS is the best approach for GOLDEN matte fluid acrylic paints.

CHAPTER 5. PAINT MIXING MODEL PERFORMANCE

This chapter discusses the computational performance of the paint mixing estimation. Since the variation of paint mixing recipes is enormous (e.g. the number of paint choices, concentrations), this research verified a limited number of paint mixtures. Secondary mixtures of two paints of three selected primaries, Hansa Yellow Opaque, Pyrrole Red, and Ultramarine Blue, and with or without a tint, Titanium White, were targeted in this experiment.

5.1. Test Samples

To evaluate the accuracy of the paint mixing simulation, we prepared actual paint drawdowns that were made by the combinations of certain primary paints. The recipes of 27 drawdowns are illustrated in Figure 26. Nine drawdowns were secondary colors without a tint, which were made by the combinations of two paints of the primaries. The rest of the mixing paints were the secondary mixtures tinted with Titanium White.

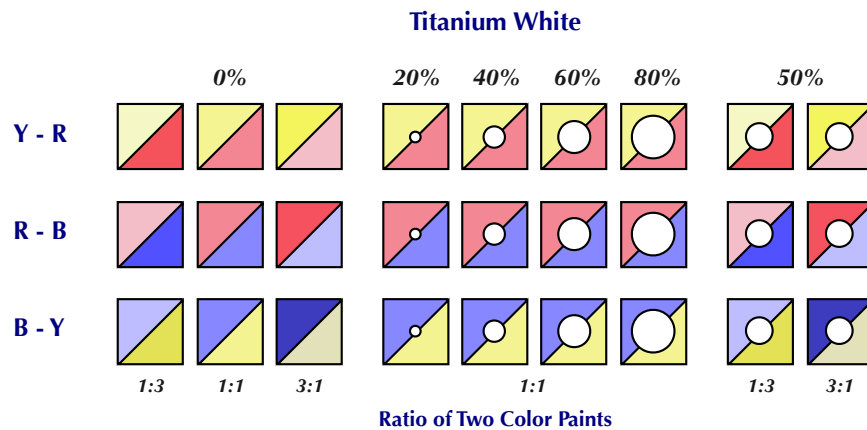


Figure 26. The recipes of 27 paint mixtures for verifying simulation performance.

5.2. Simulation Performance

In order to evaluate simulation performances for secondary paint mixtures, we estimated the spectral reflectances based on the recipes illustrated in Figure 26. And then, we calculated the CIE LAB values of the corresponding reflectances between measurement and estimation under D65 illuminant and the 1931 standard observer with specular reflection (3%). Finally, we compared in CIEDE2000 the measurements of the paint drawdowns with the estimations.

Figure 27 shows the colorimetric vector plots between measurements (arrow bottoms) and their estimations (arrow heads). It can be confirmed from this figure that large errors appears in the green region between the hues of Hansa Yellow Opaque and Ultramarine Blue. And then, the figure reminds us that the measured CIE LAB values of mixed greens are located outside of the color gamut that has been estimated by the Kubelka-Munk approach (the white area). That is, the simulation performance for mixed greens were worse than for the other mixed colors including orange and purple.

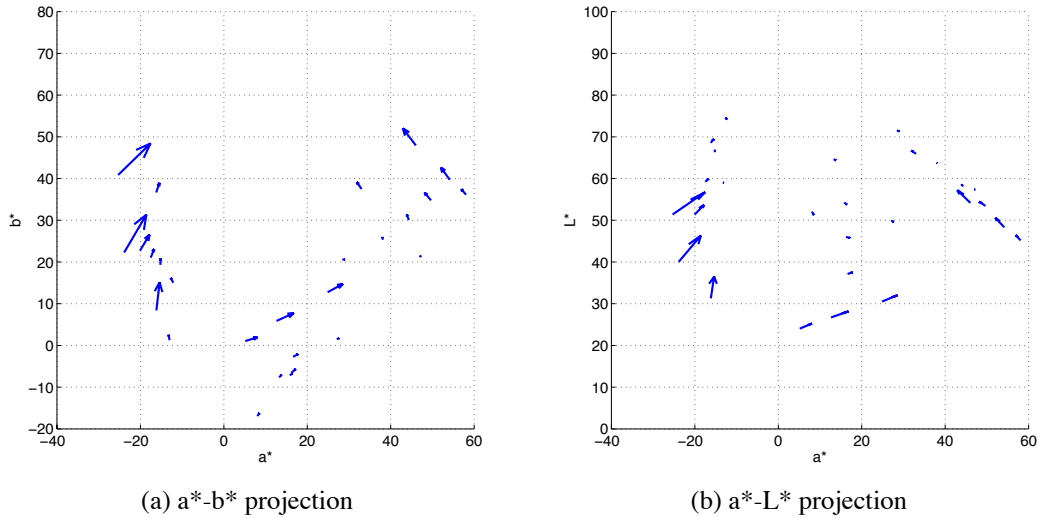


Figure 27. Colorimetric plots for secondary mixtures between measurements and their estimations (arrow bottoms: measurements, arrow heads: estimations).

Table X shows the statistical details of the estimation errors. The error index used in the tables is CIEDE2000 under D65 illuminant and the 1931 standard observer based on SPIN measurements. The left of the table indicates the estimation errors for secondary mixtures. The right of the table shows the estimation errors for individual colorants that resulted from analyzing the tint ladders for each paint.

Table X. The statistical data of the estimation errors in CIEDE2000 for secondary mixtures and individual paints.

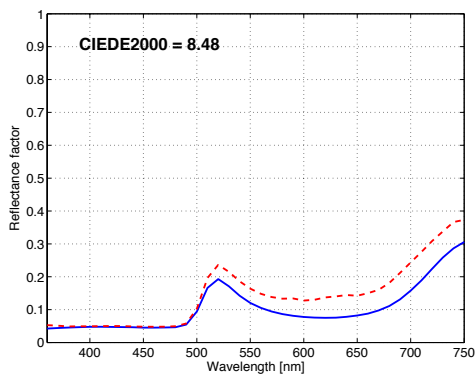
(a) Secondary mixtures		(b) Tint ladders for each colorant			
	Secondary		Y*	R**	B***
mean	2.21	mean	0.11	0.21	0.19
maximum	8.48	maximum	0.19	0.38	0.51
standard deviation	2.20	standard deviation	0.07	0.14	0.19
90 th percentile	4.19	90 th percentile	0.18	0.32	0.30

* Hansa Yellow Opaque
 ** Pyrrole Red
 *** Ultramarine Blue

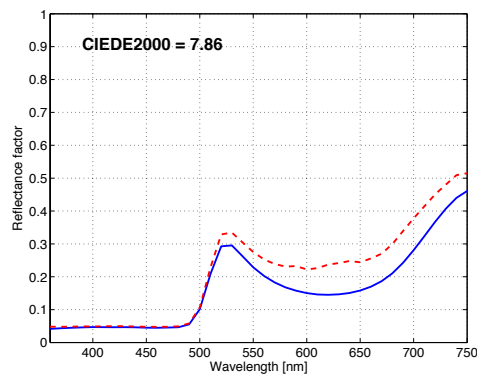
These tables clearly show that mixing simulation performance for secondary mixtures was worse than for individual colorants. According to Table VI-(b), for three primary paints, Hansa Yellow Opaque, Pyrrole Red, and Ultramarine Blue, the absorption and scattering properties were characterized individually accurately so that the net color difference was under $0.3 \Delta E_{00}$. However, this result does not come into the improvement of estimation performance of secondary mixtures.

Figure 28 exhibits the worst six of the spectral estimation results. The worst result is the case of mixture of one blue and three yellow. The colorimetric error was $8.48 \Delta E_{00}$. The second worst was mixture of 1 blue and 1 yellow with $7.86 \Delta E_{00}$. The common trends, in which the worse estimation results occurred, were related to the yellow paint. Yellow mixtures (*five of the worst six were yellow mixtures*) as well as their

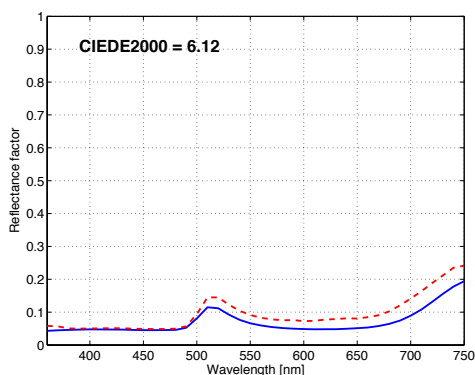
high-concentrations are applicable for such a worse case. Moreover, most of them were not tinted with Titanium White.



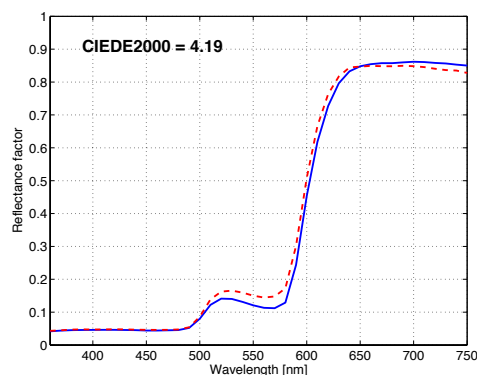
[Worst] Blue : Yellow = 1 : 1 (no tint)



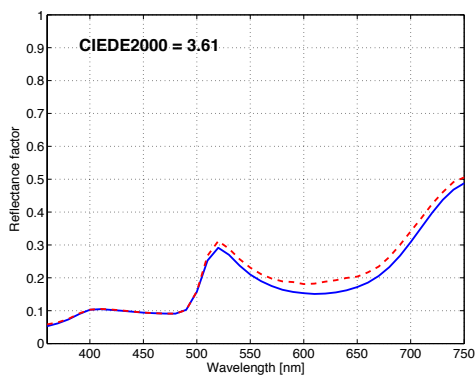
[2nd] Blue : Yellow = 1 : 3 (no tint)



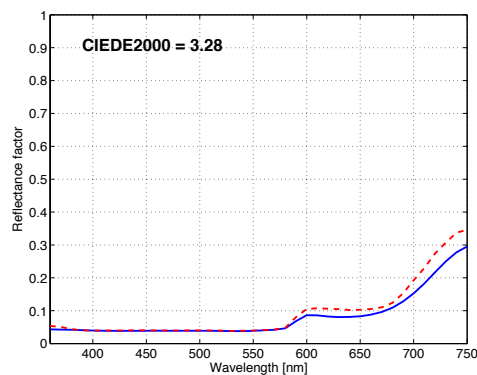
[3rd] Blue : Yellow = 3 : 1 (no tint)



[4th] Yellow : Red = 3 : 1 (no tint)



[5th] Blue : Yellow = 1 : 1 (20% white)



[6th] Red : Blue = 1 : 1 (no tint)

Figure 28. Worst six of the spectral errors for secondary mixtures (blue line: measurement, red line: estimation).

5.3. Discussion

We assume from these results of the worst six that the opacity imperfection of the tint ladders for Hansa Yellow Opaque might cause poor estimation performances. The opacity of a typical yellow paint is slightly weak compared to that of the other colored paints. A yellow paint shows high reflectance over most of the wavelengths or the green-to-red spectral region. The high reflectance results from weak absorption, not high scattering. The weak scattering property results in translucency for a paint. One good explanation on weak scattering is to discuss about difference in the refractive index (RI) of a pigment. According to the reference listing pigment RIs [Eastaugh 2004], while titanium white, which also has high reflectance over all the wavelengths, has a 2.71 RI, hansa yellow opaque has a 1.66 RI. The binder surrounding pigments in the liquid of an acrylic paint has a 1.48 RI, the value of which is cited from the technical information from GOLDEN Artist Colors, Inc. A slight difference in RI between a yellow pigment and the binder makes a yellow paint more translucent.

We observed that the masstone of the tint ladders for Hansa Yellow Opaque still showed slight translucency even after improving opacity by making a thick paint layer. Figure 29 shows a reflectance decrease by thickening a paint film layer for Hansa Yellow Opaque. The spectral curves were measured on the white background of opacity charts and by the SPIN mode. With an increase of the film thickness or opacity, a reflectance factor of the yellow in the red spectral region will be lower.

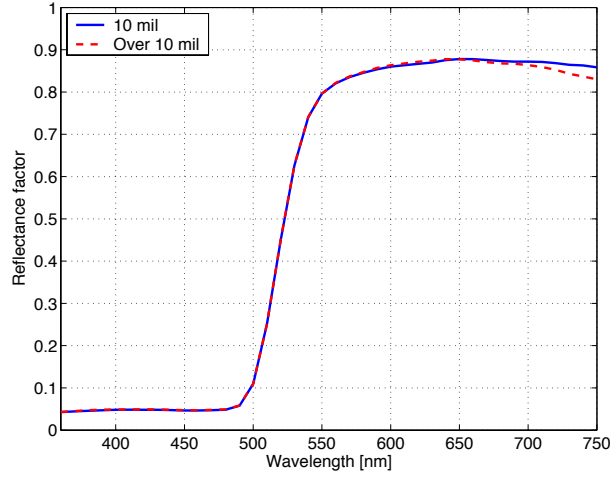


Figure 29. Reflectance change by a paint film thickness (SPIN measurement and on the white background).

Let us now return to the discussion on mixing simulation performance. The estimated curves (red lines) in Figure 28 are higher in the green-to-red spectral region than the measurement curves (blue lines). What if we prepared complete opaque yellow drawdowns? We expect that using an opaque drawdown leads to being better estimation performances (lower and then closer to measurement) for yellow mixtures.

In order to confirm how opacity affects the paint mixing simulation, we attempted to reduce the unit k of Hansa Yellow Opaque in the paint database. Figure 30-(a) shows the unit k (blue line) and the reducing amount of unit k (green line) in this trial. The original unit k was spectrally reduced by 0.05 from 560 nm to 750nm. This amount was arbitrarily determined for a demonstration of opacity effects. By this reduction, the spectral curve of masstone was got lower as shown in Figure 30-(b), where red line and green line indicate the spectral reflectance curves of masstone of an over 10 mil paint layer (measurement) and an improved opacity paint layer (simulation) for Hansa Yellow Opaque. That trend of a reflectance change by simulation

seems the same as by thickening an actual paint layer as shown in Figure 29.

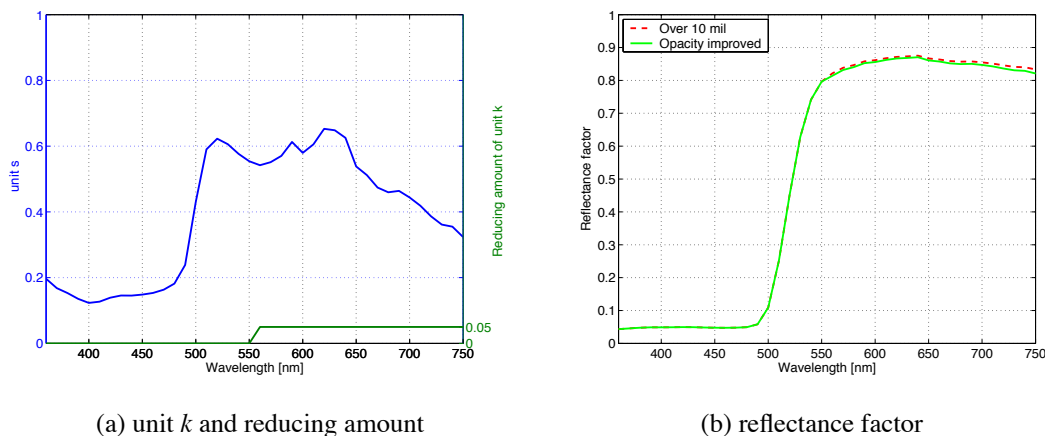
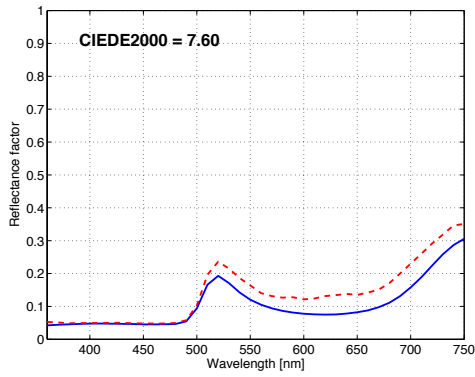
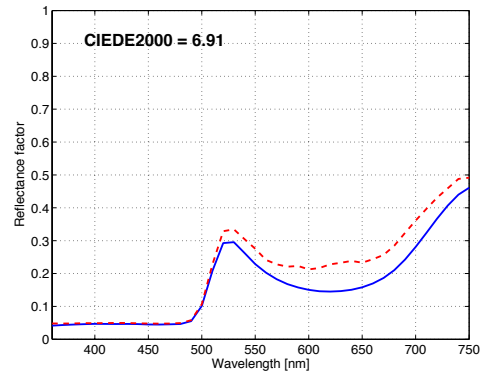


Figure 30. Simulation of opacity effects onto the spectral reflectance curve of masstone by reducing the unit s of Hansa Yellow Opaque.

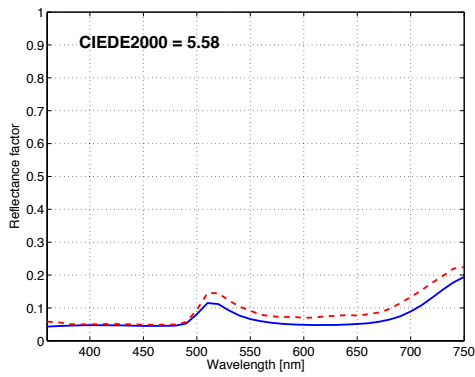
By using the new unit k of Hansa Yellow Opaque. We estimated the spectral reflectance curves for the worst five of the secondary paint mixtures, which was shown in Figure 28. These five mixtures were mixtures of Hansa Yellow Opaque. The result is exhibited in Figure 31. For each mixture, we can observe that an estimated reflectance curve (red line) decreases so as to approach to a measurement reflectance curve (blue line). As a result, the estimation errors in CIEDE2000 were improved to a $0.33 \Delta E_{00}$ to $0.95 \Delta E_{00}$ decrease. From these figures, it can be said that making an opaque drawdown for Hansa Yellow Opaque would improve the simulation performance of the secondary paint mixtures. However, the estimation errors are still larger than the result of an individual paint (Table X). One possible reason for poor estimation might be bad compatibility of pigment mixtures with different chemical properties.



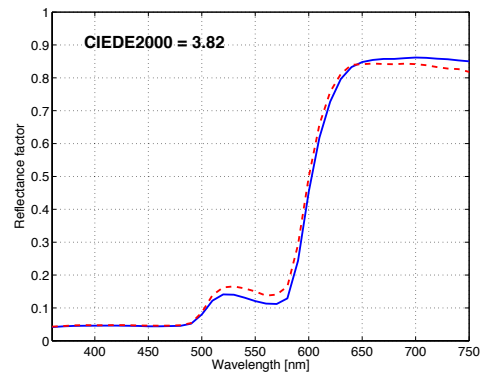
[Worst] Blue : Yellow = 1 : 1 (no tint)



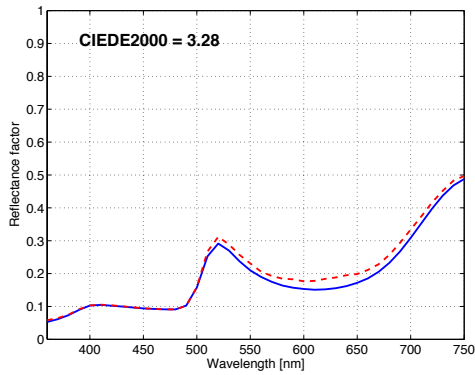
[2nd] Blue : Yellow = 1 : 3 (no tint)



[3rd] Blue : Yellow = 3 : 1 (no tint)



[4th] Yellow : Red = 3 : 1 (no tint)



[5th] Blue : Yellow = 1 : 1 (20% white)

Figure 31. Improvement of the secondary mixture estimation by a opacity change for the previous worst five mixtures (blue line: measurement, red line: estimation).

Unfortunately, we gave up thickening a yellow paint film to improve opacity because the substrate of a thick film was curved by shrinking force during drying. As a future research, therefore, we need to deal with several processes to confirm if our assumption is correct.

- (1) Clarify how to make a thick paint film for Hansa Yellow Opaque without shrinkage.*
- (2) Characterize both of translucent and opaque yellow paints with the Kubelka-Munk solution.*
(In this case, we need to measure an exact film thickness and transparency.)
- (3) Simulate the mixtures of primary paints as translucent colors.*
- (4) Formulate the effects of translucency onto mixing estimation performance.*

CHAPTER 6. RENDERING ARTISTS' PAINT COLOR GAMUT

This chapter discusses color gamut rendering based on the various number of paints used and paint combinations assumed. As background, there is the traditional paint mixing theory that primary paint mixtures of yellow, red, and blue, can make any possible colors. To justify the belief, for example, some color educators say that a green is made of the combination of a yellow paint and a blue paint. The first interest come to our minds is in the scientific validity of paint mixing phenomenon. If the phenomenon is correct anytime, we will be convinced that a color gamut for the combination of three primary paints must cover the broad area of a color space.

We are also expecting that a scientific interpretation of paint mixing theory will be helpful for paint education. Another interest of paint color gamut is from color science aspect; *how large is the color gamut going to be when mixing all the paints of a GOLDEN acrylic lineup*. Topics on what is the optimal number of primary colors are well discussed in the printer industry. For example, the reason of well-discussion is due to the recent trend of using multiple inks (e.g. CMYKRG) more than traditional CMYK in an inkjet printer that leads to dramatically expanded color gamut. From this point, we believe that comparing color gamut volume between acrylic paints and other coloration systems is meaningful, particularly when taking a step to reproduce acrylic paintings with color accuracy by using a spectral imaging system that includes digital color printers.

6.1. Primary Paint Mixture

First of all, we already know the absorption and scattering properties for all of the 27

acrylic paints in the previous experiment. That is, any mixed colors can be mathematically produced by using this database.

The second step, which we need to think about, is to generate a color gamut by presuming some paint set of primaries from the database. For this purpose, we are required to determine lots of concentration series that represent mixtures of three individual primaries and a white in advance. A concentration series of paints chosen must satisfy the following restriction that, *“the sum of all the paint concentrations is equal to 1”*, as shown in Equation (2.12). In other words, if we determine the values of concentration series for each of the colorants, the concentration of a tint is automatically determined in response to the colorant concentrations. The restriction makes it more complex to design the algorithm of color gamut rendering for a paint system rather than for a printer system. The procedures we took to determine concentration values for primary paint mixtures of yellow (Y), red (R), blue (B), and white (W), called as *“a primary paint set”* in this thesis, were:

[Step 1] Determining intervals which separates evenly the ranges of concentration between 0 and 1 for each color paints (e.g. 0, 0.2, ..., 1 at 0.2 intervals).

[Step 2] Listing all the combinations made by the equally discrete concentrations at the determined intervals for three colorants.

[Step 3] Filtering the only concentration series in which the sum of the colorant concentrations is less than 1.

In Step 1, it must be noticed that, the number of all the combinations is enormous with an increase of the number of paints and intervals (e.g. 6^{26} combinations for the mixtures of 26 paints at 0.2 intervals). That implies, the fineness of intervals depends on the indexing number of a data array that a computer system is allowed to handle.

Figure 32 shows an example of the concentration series, in which the data manipulation process from Step 2 through Step 3 is observed. “Sum” in the left table indicates the sum of yellow, red, and blue concentrations that is identical to “ p ” in Equation (2.12).

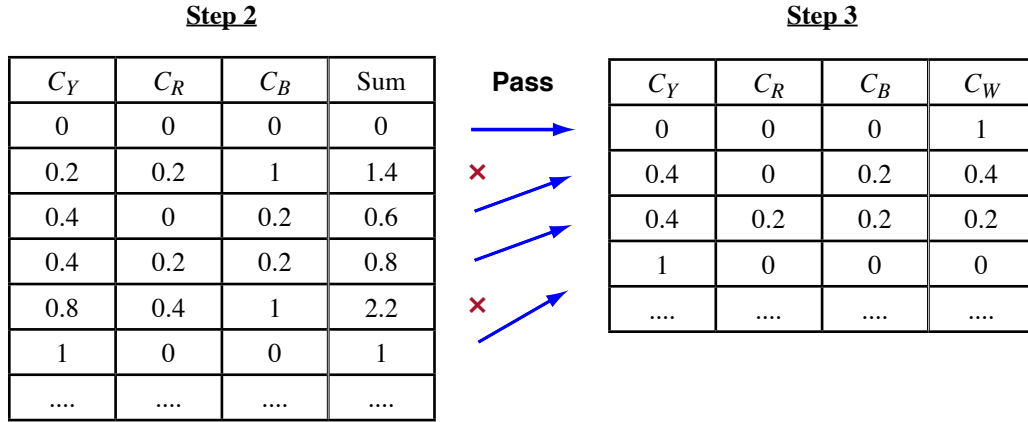


Figure 32. Data filtering process from Step 2 through Step 3.

As the figure shows, a concentration series in which the sum of yellow, red, and blue paint concentrations less than 1 can be picked up. The concentrations of a tint in the right table is calculated in Equation (2.31):

$$c_t = 1 - (c_Y + c_R + c_B) \quad (2.31)$$

where C_t , C_Y , C_R , and C_B are the concentrations of white, yellow, red, and blue paints respectively. Since the whole process introduced so far are easily integrated in a gamut rendering program, we used this procedure for color gamut rendering.

Another possible approach might be to use the Monte Carlo method to generate concentration series. The values of C_Y , C_R , and C_B are randomly generated between 0 and 1 under the following constraint based on Equation (2.12):

$$c_Y + c_R + c_B \leq 1 \quad (2.32)$$

Even though this method is mathematically simple, it can be hard to downsize the total number of concentration series leading to covering color gamut uniformly. Of course, the algorithm can be easily integrated in to a gamut simulation program. However, we did not use the method for this experiment.

The third step of color gamut rendering is to convert the concentration series to CIE LAB values. First of all, each concentration series of a specific primary paint set can be converted into a spectral reflectance curve estimated by the Kubelka-Munk equation and the Saunderson equation (for SPIN), Equation (2.13) and Equation (2.19). And then, for the estimated spectral reflectances, the CIELAB values are to be computed under D65 illuminant and the 1931 standard observer.

The final step is to render a color gamut from the estimated CIELAB values. The surface of a color gamut in CIELAB space is computed in Matlab with two Matlab functions, “*convhulln*” and “*trisurf*”. While the function “*convhulln*” computes the convex hull from a 3D dataset, the function “*trisurf*” plots triangular facets covering the gamut surface based on the convex hull. The details of the functions are explained in the Matlab help menu.

In our experiment, we computed color gamuts based on our database for two possible sets of the primary paints individually.

Case I

*Hansa Yellow Opaque, Pyrrole Red, Ultramarine Blue, Titanium White**

Case II

*Hansa Yellow Opaque, Pyrrole Red, Phthalo Blue (Green Shade),
Titanium White**

** Titanium White is necessary for both cases to tint colors.*

Figure 33 shows the color gamuts of two primary paint sets that are viewed from the top of the a^*-b^* plane. The colored area indicates the paint color gamut, the colors of which were reproduced from the estimated spectral reflectances under D65 illuminant and the 1931 standard observer to sRGB digits. If a simulated color was out of the sRGB space, its digital count (or counts) below 0 and beyond 255 was assigned 0 and 255, respectively. The gray shadow areas in the plots represent the MacAdam limits. The MacAdam limits were calculated by adding an offset of 3% specular reflection into the original zero-unity reflectances at all the reflectances [MacAdam 1935]. The purpose of specular addition is to make the MacAdam limits more realistic in terms of spectrophotometry. Imposing specular reflection results from taking account of difference in refractive index between air and a paint film. Comparing to the MacAdam limits, we can confirm that both of the paint color gamuts based on primaries are small.

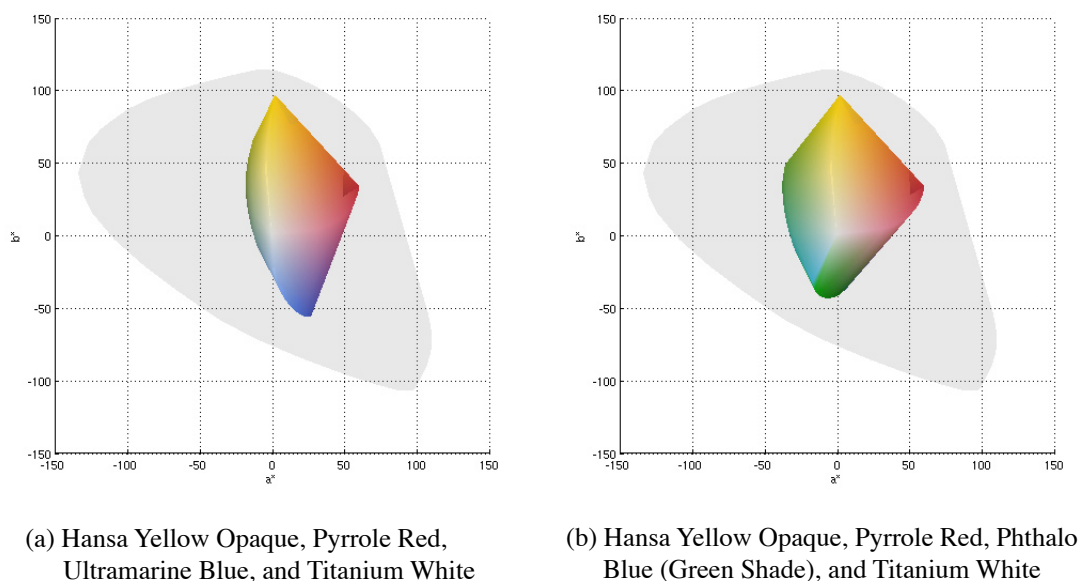
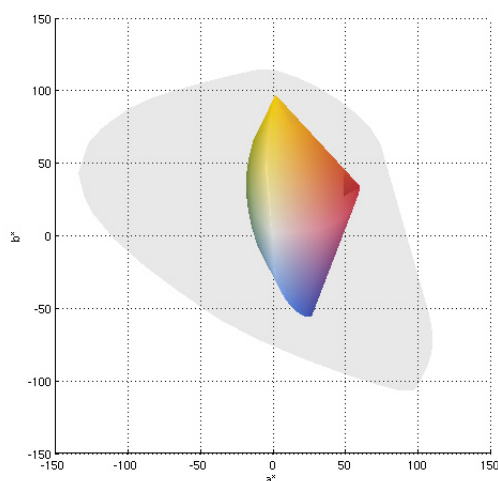


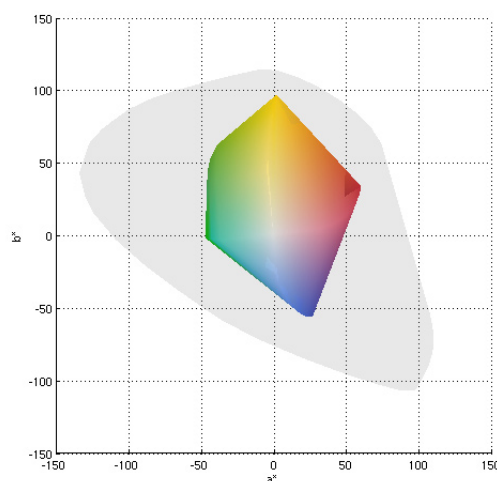
Figure 33. Rendering color gamuts for two three-primary paint sets.

Additionally, it is obvious from the figure that the coverage of a paint color gamut depends on the choice of primary paints. For Case I (Figure 33-(a)), high-chroma colors from yellow through red to blue appears in the plot. However, that is not applicable to green. With regard to Case II (Figure 33-(b)), even though some saturated greens are produced, however, bluish purples are not observed in the color gamut. These simulation results clarify that the choice of a blue paint determined the coverage of greens and purples in these cases.

To enlarge the green coverage, we re-simulated a color gamut by adding an extra green paint, Phthalo Green (Blue Shade) to the recipe of Case I. This decision is realistic when thinking of the situation that painters make color palettes from paint tubes. That is, the mixtures are made of three primaries, white, and green. In this case, we extended the generation method of concentration series in Figure 34 for a four color-paint combination. In this case, the sum of C_Y , C_R , C_B , and C_G (green) has to be equal to or less than 1. Figure 34 shows the simulation result.



(a) Hansa Yellow Opaque, Pyrrole Red, Ultramarine Blue, and Titanium White



(b) Hansa Yellow Opaque, Pyrrole Red, Ultramarine Blue, Titanium White, and Phthalo Green (Blue Shade)

Figure 34. The effects of an extra green paint addition for color gamut expansion for Case I.

It can be observed from the figure that the green coverage is improved dramatically. The result is affected by the addition of a green paint. Now, we compared the volume of each color gamut in CIELAB space. In this comparison, we assigned 100 to the volume of the MacAdam limits as the standard. While the volume of a color gamut for Case I is 17.5, the volume resulted from a green paint addition is 26.5. The volume is expanded 151% relative to the original.

In the same fashion, we also observed a color gamut expansion by adding the same extra green paint for Case II. The results of color gamut rendering for this case are shown in Figure 35. In the figure, the estimated colors in the blue region were drawn as greens because of out of the sRGB gamut. The volume of the primary set was changed from 19.6 to 23.2 relative to the volume of the MacAdam limits (100). That is equivalent to a 118% gamut expansion to the original by adding Phthalo Green (Blue Shade).

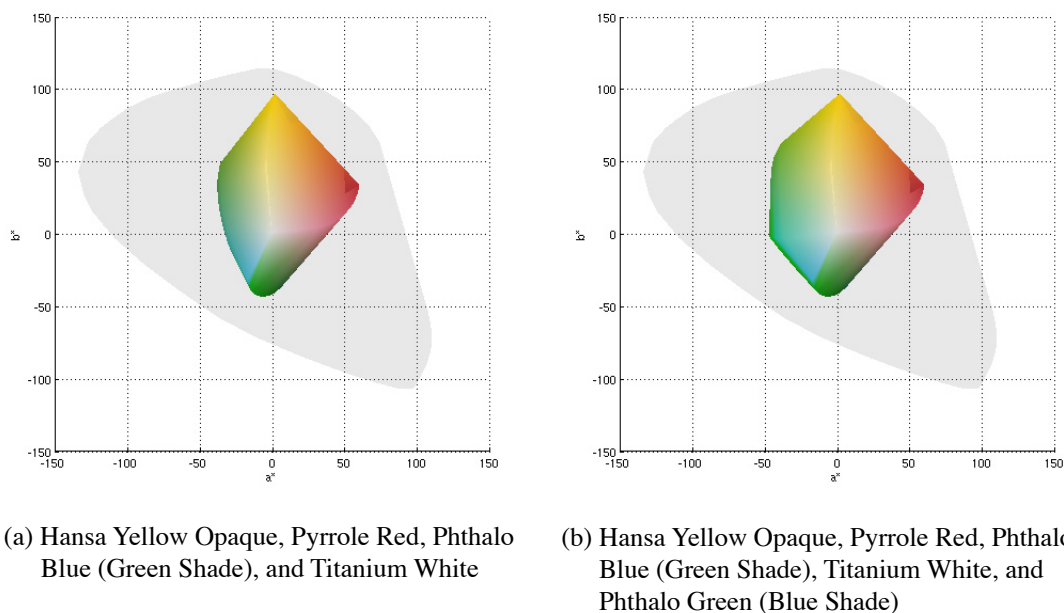


Figure 35. The effects of an extra green paint addition for color gamut expansion for Case II.

These results prove the fact that there are a lot of green paints available in addition to primary paints.

6.2. Multiple Paint Mixture

With an increase of the number of paints used, the process of color gamut rendering gets more complex. For instance, if we have 27 paints including a white and 6 different concentrations are considered for each paint gradation, the total number of all the combinations is 6^{26} . In such a case, our approach of generating concentration series (c.f. “6.1 Primary Paint Mixture”) needs a huge memory size for data storage. However, it is not necessary to use all the combinations because there are several trends of paint color gamut leading to data reduction:

Trend 1: A primary paint set basically generates a large color gamut.

Trend 2: A combination of two high-chroma paints with a tint generates a large color gamut.

Trend 3: Only four paints (two color paints, black, and white) are necessary to generate a local color gamut.

“A local color gamut” in this thesis is defined as a color gamut generated within a specific range of hue angles. The hull of a local color gamut must be part of a global color gamut. The angle range is dependent on a pair of neighboring paint pairs chosen. By taking account of these trends, we attempted to downsize the bulk of concentration series in the following manner:

- [Step 1]** Aiming to generate concentration series needed for rendering a local color gamut, which is made of two chromatic paints, black, and white.
- [Step 2]** Seeking for a pair of neighboring chromatic paints in terms of chroma and hue for each locality in a direction of hue angle (e.g. Diarylide Yellow and Hansa Yellow Opaque).
- [Step 3]** Summing up the picked-up bulks of concentration series for local color gamuts.
- [Step 4]** Adding concentration series for a primary paint set into the collected bulks by Step 3.

Figure 36 exhibits the illustration explaining the processes from Step 1 to Step 4 above. In the figure, the two sets, paint 1 and paint 2, and paint 3 and paint 4, are the pairs of neighbors. The primary paints in the paint set are Hansa Yellow Opaque, Pyrrole Red, and Ultramarine Blue.

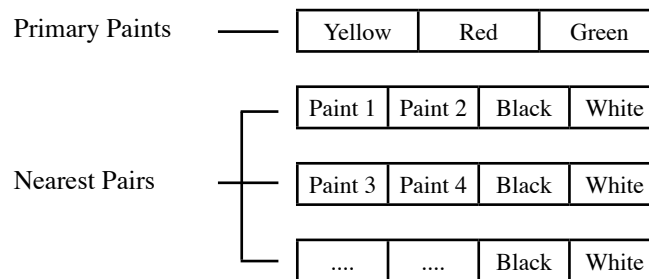


Figure 36. Segmentation for global color gamut.

Figure 37 shows the measured colorimetric plots for 26 acrylic paints that is useful to seek for neighbor paint pairs. For example, Diarylide Yellow and Hansa Yellow Opaque, Permanent Green Light and Phthalo Green (Blue Shade), Naphthol Red Medium and Pyrrole Red, are pairs of the neighbor paints. To find neighbor paints around a hue angle might be easily done by seeking closest hues in computa-

tion. However, we must draw attention to the fact that some nearest paint pairs in hue might be completely different in chroma (e.g. Cobalt Blue and Permanent Green Light). Selecting such a paint pair is not appropriate because a local color gamut resulted from the pair does not render the outer hull of a global color gamut. Therefore, we need to pay attention to how to choose neighbor paints carefully.

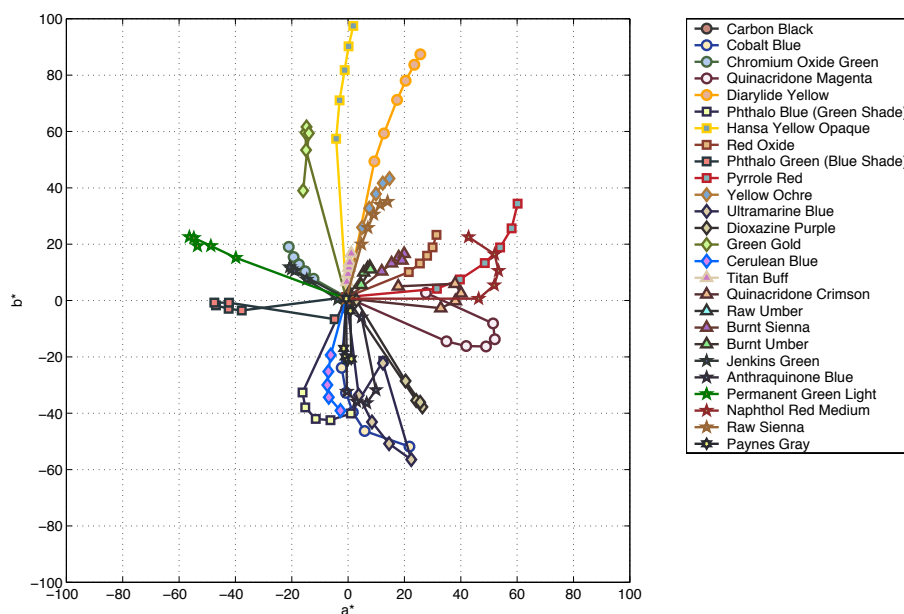


Figure 37. Chromatic plots for 27 acrylic paints.

Figure 38 shows the simulation result of color gamut for the 27 paint mixture. Even though the coverage of the color gamut is still smaller than the MacAdam limits (gray shade), it uniformly covers the whole hue angle and constant high chroma.

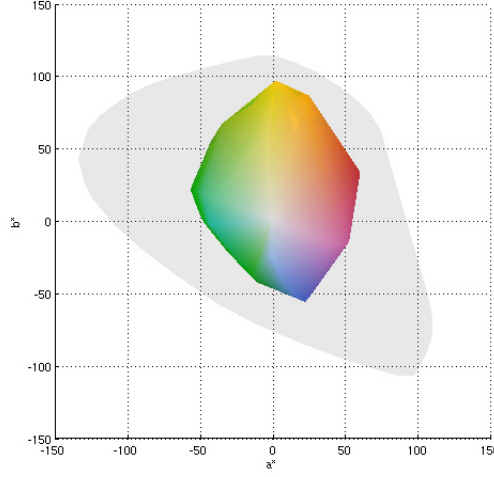


Figure 38. Paint color gamut by using all the 26 colorants and a tint.

6.3. Volume Comparison to Other Coloration Systems

Paint drawdowns, which we made to build up the database, are expected to be used as portions of a color rendering chart in a spectral imaging system. Therefore, comparing color gamut volume among various coloration systems is useful for future research about color reproduction accuracy.

For this purpose, in addition to using the MacAdam limits and Pointer datasets [Pointer 1980], we measured three printer outputs and two color rendering charts. The measurement values were CIELAB coordinates calculated under D65 illuminant and the 1931 standard observer in the SPIN mode.

Table XI shows the summation of the volume comparison results. Each of the volumes is normalized relative to the volume of the MacAdam limits (100). The interesting figure of the table is the volume of a inkjet printer (Canon PIXUS iP8600) because the size is not different from the volume of the Pointer's color gamut. The result

reminds us of an innovative printing technology in which eight dye inks of the printer reproduce almost all the surface real colors introduced by Pointer [Pointer 1980]. The color gamut rendered for each coloration system, which is listed in Table XI, is shown in Chapter 10, “*F. Color Gamut Collection for Various Coloration Systems*”.

Table XI. Volume comparison of color gamut among various coloration systems.

	Coloration System	Volume
Paint	GOLDEN Acrylics ^{*1}	30.9
Photo Printer	Photo ^{*2}	16.1
	Laser Printer ^{*3}	13.7
	Inkjet Printer ^{*4}	42.5
Color Chart	ColorChecker DC	26.4
	ColorChecker SG	23.0
Academic	Pointer	45.7
	MacAdam limits	100

*1 Computational mixtures of 27 acrylic paints

*2 Kodak Q-60 Color Input Target (ANSI IT8.7/2)

*3 Xerox Phaser 6500

*4 Canon PIXMA iP8500

Another interesting point in the figure is about color rendering charts, the Macbeth ColorChecker DC and ColorChecker SG. Since we have used those for calibrating a spectral imaging system, we supposed that the color gamuts of the rendering charts must cover the entire color ranges of artist painting targets. If, however, thinking of an acrylic paint as a target, some colors of a painting might not be captured with color reproduction accuracy due to their out of being gamut. Therefore, it is reasonable to use an actually-painted color target as a camera calibration chart.

6.4. Discussion

Whatever combination of acrylic paints is used, a paint color gamut can be rendered with the database and by following the previous steps that have been introduced so far.

However, there is a possible issue of rendering accuracy on color gamut that can happen when using the Matlab function, “*convhulln*“. “*convhulln*“ is the 3D version of “*convhull*” that is for 2D. Figure 39, which is cited from the Matlab help menu, explains how “*convhull*” works, where a blue “+” indicates a dataset, and red line represents the result of comprising a convex hull. Even if a dataset shows the trend of a concave as shown in the figure, only a farthest outer hull is rendered in a drawing routine. This tendency would cause a problem of color gamut rendering when the transition curve of hue between two paints shows not convex, but concave.

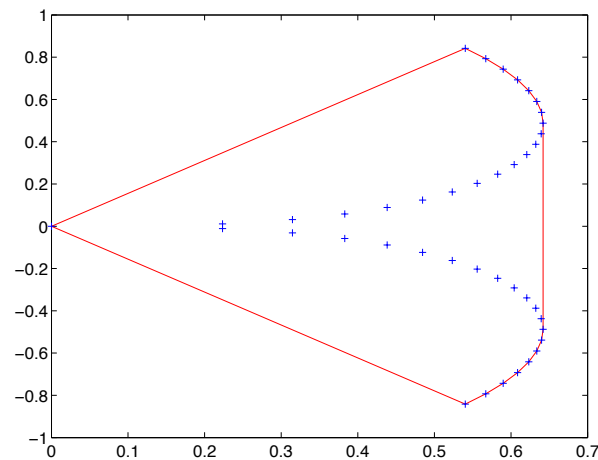


Figure 39. Extrapolation drawing by the Matlab function, “*convhull*”.

Figure 40 shows the simulation result of hue transition of secondary paint mixtures

in the a^*-b^* projection. The recipes for this simulation are mixtures of Hansa Yellow Opaque, Pyrrole Red, and Ultramarine Blue without a tint. The figure clearly presents that the transition in hue from Pyrrole Red to Ultramarine Blue shows a dent to low chroma in the middle. From this point, we need to pay attention to rendering a gamut shell between hues of red and blue.

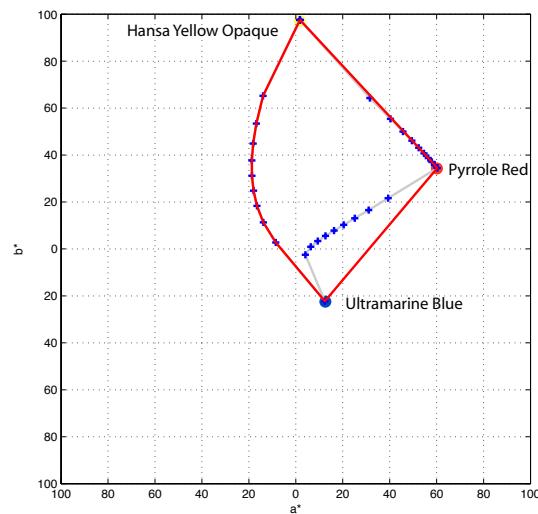


Figure 40. Contour of secondary colors by computer simulation (blue plus signs: the estimated plots of secondary mixtures without a tint, red line: the rendering result by “*convhull*”).

An optimal approach for accurate color-gamut rendering is to develop an algorithm applicable for not only a convex hull but also a concave hull. To solve this technical issue, it is definitely worthwhile to verify the validity of other rendering techniques including Delaunay triangulation and Voronoi diagram (from the Matlab help menu).

Another issue on rendering imperfection for a concave shape is associated with the calculation accuracy of a color gamut volume. As Figure 40 shows, the volume of an actual color gamut would be smaller than that of the result of a computer color-gamut

rendering. If so, the volume sizes of the coloration systems listed in Table V might be imprecise and smaller than the actual volume sizes. Fortunately, it seems that such a concave hull appears only in a paint system. For example, it is different from the paint system that the color gamut of a typical CMYK printer shows the projection of a hexagon (C, M, Y, R, G, B are the peaks), not a concave hull, onto the a^*-b^* space. Therefore, it is also necessary to develop an 3D rendering algorithm to calculate a outer hull containing concave shape portions or dents to estimate a paint color-gamut volume.

CHAPTER 7. MODELING GLOSS EFFECTS BY VARNISHING

This chapter discusses the modeling of the gloss effects by varnishing. As the first step to this topic, we figured out how gloss affects spectrophotometric measurements in terms of specular integration, SPIN and SPEX. And then, we also attempted to correlate the gloss effects based on measurement geometry with a specular gloss index. The result of this challenge led successfully to characterizing changes in spectral reflectance by varnishing with a single metric. Finally, we succeeded formulating the gloss effects onto appearance for acrylic paints by using our developed gloss model.

7.1. SPIN vs. SPEX

The first important thing which we needed to confirm about a varnishing model is if a varnish layer was completely transparent. If a varnish layer shows translucency or is colored, we must formulate the complex phenomenon of light absorption within a varnish layer. Another important thing is to confirm how the gloss effects by varnishing give influence to spectrophotometric measurements. Both of the optical phenomena were quantified by analyzing both of the SPIN measurements and the SPEX measurements for varnished and non-varnished paint specimens.

First of all, we measured spectral reflectances for the tint ladders of all the paints. For the measurement, we used an integrating sphere instrument, a GretagMacbeth ColorEye XTH, with a small aperture. Both of SPIN and SPEX geometries were chosen. And then, we compared reflectance curves based on different measurement geometries.

Figure 41 shows the results of spectrophotometric measurement in the SPIN mode

for the tint ladders of Carbon Black. Figure 41-(a) exhibits the spectral curves, and Figure 41-(b) indicates the residuals in reflectance between the non-varnished specimens and the varnished specimens. As we expect, it can be observed that the spectral curves at each concentration is almost zero in the range from 450 to 750 nm. However, the strong light absorption in the UV region (360-450nm) appears for the varnished specimens. The reason of the absorption is due to Ultra Violet Light Stabilizer (UVLS) contained in this MSA varnish. The medium works to protect acrylic paintings from damages by UV radiation. This result implies that UVLS works very well as a UV cut filter.

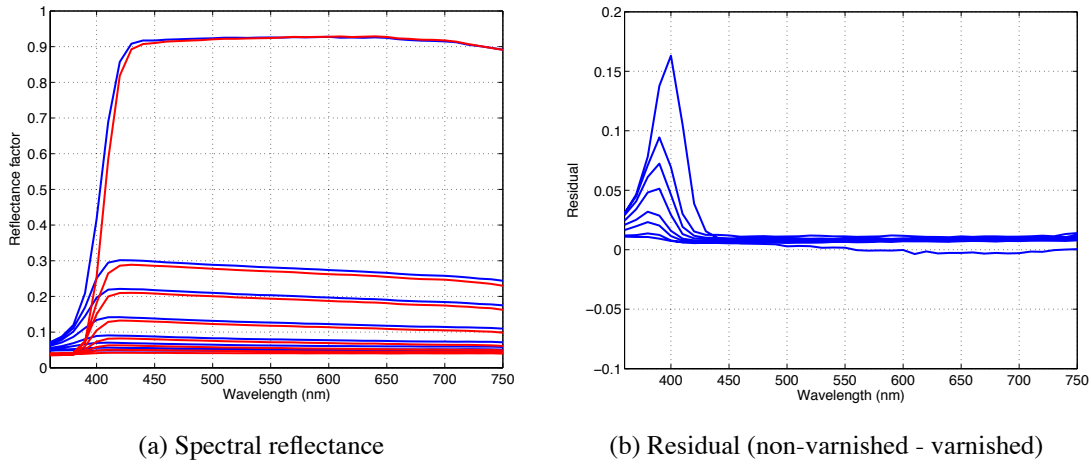


Figure 41. Spectrophotometric measurements for the tint ladders of Carbon Black in the SPIN mode (blue lines: varnish, red lines: non-varnish).

Table XII shows color differences between the non-varnished paint specimens and the varnished specimens in the SPIN measurements. While the second column is the result for Carbon Black, the third column is for all the 26 paint tint ladders including Carbon Black. These results clearly show that the expected performance of our gloss

modeling for the entire paints can be considered to be above 1.0 ΔE_{00} on the average because the model does not take account of the transparency of a varnish coating.

Table XII. The statistical data of CIEDE2000 color difference between non-varnished paint specimens and varnished paint specimens in the SPIN mode.

	Carbon Black	26 paint tint ladders*
mean	1.24	0.73
maximum	1.64	3.55
stdandard deviation	0.30	0.42
90 th percentile	1.46	1.17

* The varnished and non-varnished paints of masstone for Quinacridone Magenta are not included in this figures because of the surface defects occurred in sample preparation.

The next experiment was to measure the same specimens in the SPEX geometry with the same instrument. Since a specular trap of the instrument removes some amount of specular reflection, differences in spectral reflectance between the varnished paints and the non-varnished paints were expected. The amount is determined by the distribution of specular reflection and the size and position of a specular port.

Figure 42 exhibits the result of the SPEX measurements. The common feature of residual curves between SPIN and SPEX is the peaks upraised by UVLS in the short-wavelength region. However, the entire amount of residual curves for the SPEX measurements is constantly higher compared to for the SPIN measurements. Now, let us focus on only the wavelengths from 450 to 750 nm. With regard to the Figure 42-(b), the constant offsets over the range are varied even though those do not appear in the case of the SPIN (Figure 41-(b)). We suppose from this result that the variance in the SPEX measurements is due to surface roughness differences for each tint ladder.

We can also say that the variance does not result from some factors related to refractive index because the same vanish with a single RI was used for all the varnished specimens except for different measurement geometries. It is inferred from the results that the capability of specular exclusion is dependent on surface construction. In other words, if surface construction is characterized with some gloss index, it would be possible to correlate the gloss effects onto spectrophotometric measurement with the gloss index. Unfortunately, we will have no clue to discuss the possibility without considering of both of SPIN and SPEX. Even though the non-varnished specimens are matte, however, they are not Lambertian-like reflectors represented as a PTFE tablet so as to produce specular reflection slightly. On the other hand, as Figure 41 shows, the SPIN geometry is not sensitive to surface roughness at all. Therefore, we need to take into account of both of SPIN and SPEX in order to quantify the gloss effects to spectrophotometric measurement.

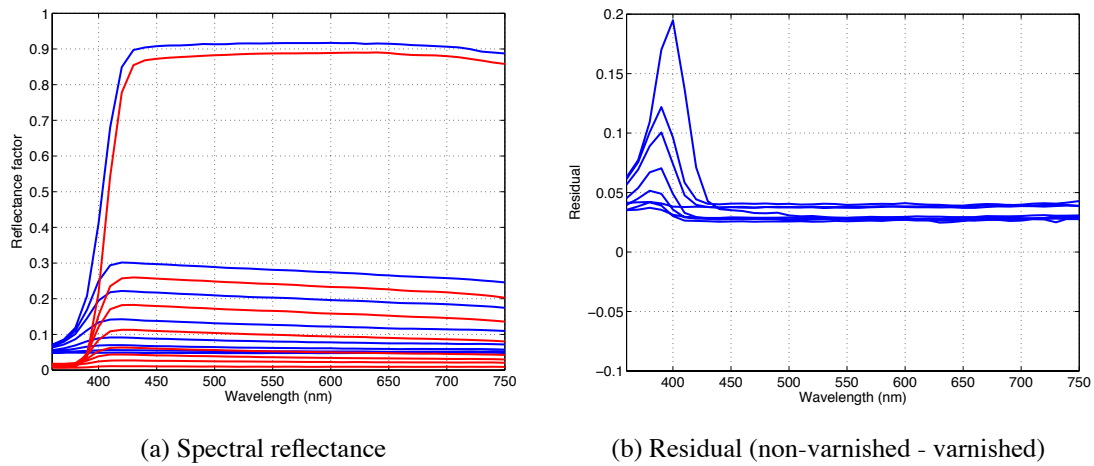


Figure 42. Spectrophotometric measurements for the tint ladders of Carbon Black in the SPEX mode (blue lines: varnish, red lines: non-varnish).

Figure 43 shows the comparison between SPIN and SPEX for the non-varnished

tint ladders of Carbon Black. The slight differences in spectral curve explains that the specimens are matte. Of the reflectance curves, only the reflectances of a tint (at the top) between SPIN and SPEX shows slight difference. The reason might be due to miss white calibration at the beginning of spectral measurement. However, the difference (1.2%) is small compared to the Fresnel reflection (4% in a 1.5 RI paint to air).

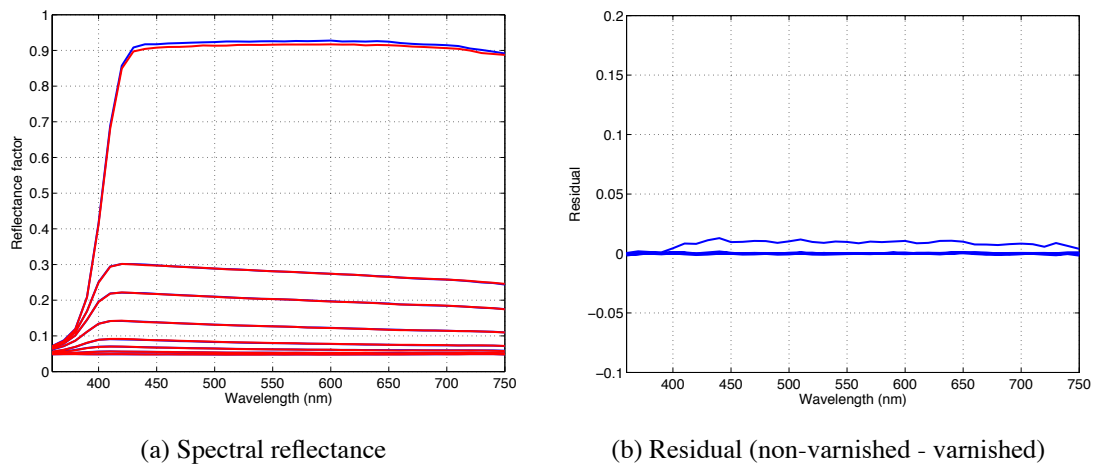


Figure 43. Spectrophotometric measurements for the non-varnished tint ladders of Carbon Black (blue lines: SPIN, red lines: SPEX).

Figure 44 shows the comparison between SPIN and SPEX for the varnished tint ladders of Carbon Black. The figure presents the large differences between SPIN (blue lines) to SPEX (red lines). The difference amounts are not identical but constant for each tint ladder over all the wavelengths. The range of the differences is from 2% to 4% in reflectance factor.

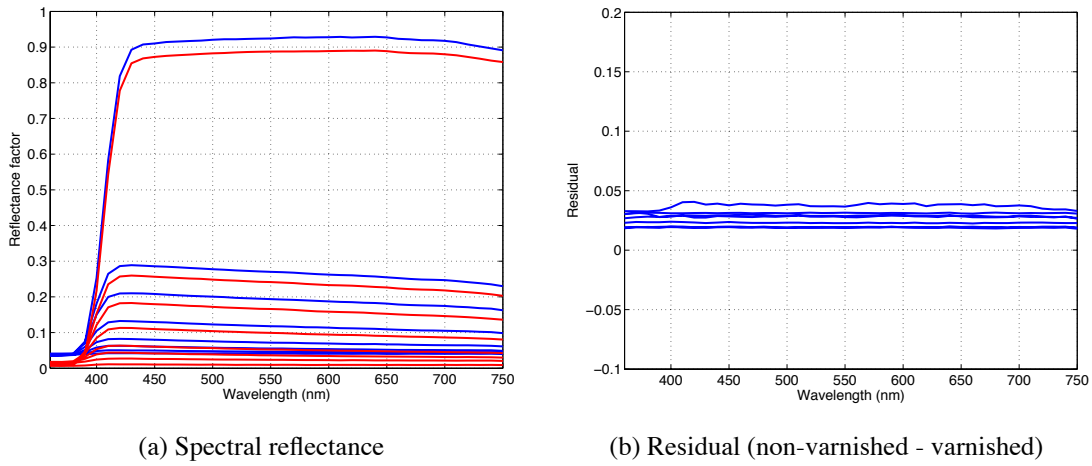


Figure 44. Spectrophotometric measurements for the varnished tint ladders of Carbon Black (blue lines: SPIN, red lines: SPEX).

7.2. Spectrophotometry vs. Specular Gloss

According to the previous experiment, the gloss effects by varnishing can be confirmed so that differences in spectral reflectance between SPIN and SPEX get larger with an increase of gloss. Now, we attempted to figure out a relationship between a gloss index and the amount of the gloss effects.

Table XIII shows the results of specular gloss measurements for both of the varnished and non-varnished tint ladders of Carbon Black. The reason for choosing Carbon Black is to exclude unexpected light absorptions by colored pigments. It is possible that light flux, some of which is absorbed by color pigments once, returns to the same direction as specular reflection's. However, we would like to focus on the gloss effects coming from surface structure, not pigment absorption. The specular gloss values for the non-varnished shows matte, the level of which is close to a GretagMacbeth ColorChecker DC (1.2 specular gloss at 60° gloss geometry).

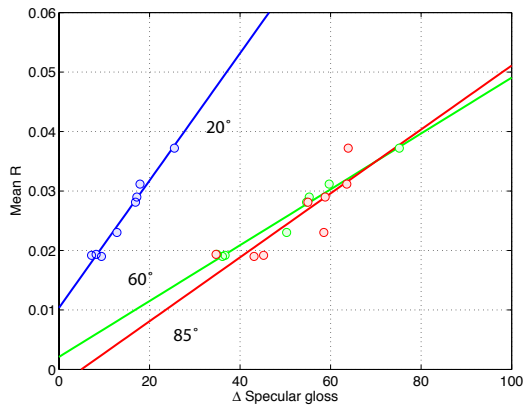
By varnishing, gloss levels jump up to the range from 37.6 to 62.5 at 60° gloss geometry. This result of various gloss levels available was surprising because our first assumption was that the same varnish and the same application produced the same reflective level of varnish coats. Actually, all the varnished paints were made of the same viscosity solution and by the same drawdown bar. The variance in specular gloss might be caused by compatibility between paint media and varnish materials at chemical level leading to different surface roughness.

Table XII. Specular gloss measurement for Carbon Black.

<i>Conc.</i>	Non-varnish			Varnish		
	20°	60°	85°	20°	60°	85°
5%	0.4	2.1	3.2	17.6	57.4	62.0
10%	0.3	2.1	3.2	17.2	56.8	58.2
20%	0.2	2.4	2.5	9.6	38.5	45.6
40%	0.2	2.7	3.2	8.4	37.6	37.9
60%	0.2	2.9	2.8	7.4	39.6	48.0
80%	0.2	2.7	2.3	13.0	53.0	60.8
100%	0.2	2.8	2.0	18.1	62.5	65.6

Now, let's move on to relating specular gloss with the gloss effects onto spectrophotometric measurement. As the first step, we averaged the spectral differences (SPIN - SPEX) that were shown in Figure 44 over all the wavelengths. The reason for using the SPIN measurements of the varnished paints instead of the SPEX measurements of those is to discount the UV absorption in the modeling. As mentioned before, fortunately, spectral reflectance values between non-varnished paints and varnished paints would be identical regardless of using each of SPIN and SPEX except in the blue region (Figure 41). And then, we plotted a relationship between the

differences of specular gloss and the average spectral differences based on the different geometries. Figure 45-(a) shows the result of plotting where blue circles, green circles, and red circles indicate 20°, 60°, and 85° gloss geometries, respectively. The distribution of each geometry plot is fitted with a linear model. The degree of association between specular gloss and the average spectral differences, R^2 , were computed as listed in Figure 45-(b). This result means that the order of goodness-of-fit is 20°, 60°, and 85°. However, the fitting line of 20° gloss geometry has a slight positive offset at 0 Δ specular gloss. This high value is not appropriate for gloss characterization because it can be interpreted that, even a paint after varnished is still matte (no specular gloss difference), a spectrophotometric instrument gives spectral differences in reflectance based on specular manipulation. That is not correct in an actual measurement for matte objects. Therefore, we chose the fitting line of 60° gloss geometry with a high R^2 and a close-to-zero offset at perfectly matte.



(a) Linear modeling

Geometry	Fitting Equation*	R^2
20°	$y = 0.0011 x + 0.0104$	0.976
60°	$y = 0.0005 x + 0.0021$	0.968
85°	$y = 0.0005 x - 0.0027$	0.730

* x is Δ specular gloss, y indicates a constant based on the gloss effects onto spectrophotometry.

(b) Statistical data

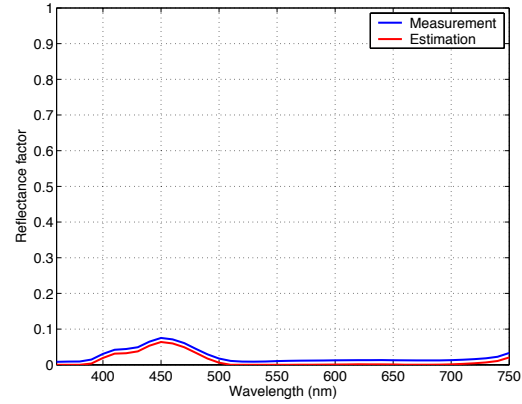
Figure 45. Correlation between the differences of specular gloss and the average spectral reflectance differences for Carbon Black.

7.3. Simulation of SPEX measurement

We already developed the model describing the relationship between the differences of specular gloss between varnished and non-varnished and the average spectral reflectance differences based on specular integration, both of which occur by the gloss effects. This relationship was characterized by using the Carbon Black tint ladders. Now, we attempted to evaluate the SPEX estimation performance by the gloss model for all the varnished paints. An estimated SPEX measurement is calculated by subtracting the gloss effects from the measured SPIN measurement for all the varnished paints. Over-subtraction sometimes happens because a statistical model has noise so that an estimated reflectance will be negative. In such a case, a negative reflectance was assigned 0 in the computational process.

By using the gloss model, we estimated the SPEX measurements from the measured SPIN measurements for all the tint ladder data. And then, the comparison was done between the measured SPEX reflectances and the estimated SPEX reflectances, the latter of which were generated by the gloss model. The estimation performance was evaluated in CIEDE2000 under D65 illuminant and the 1931 standard observer. The result is shown in Figure 46-(a). The average ΔE_{00} is very small so as not to distinguish these differences visually. However, the masstone of Phthalo Blue shows the worst performance with 7.11 ΔE_{00} . Figure 46-(b) exhibits the spectral curves of measurement and estimation based on SPEX. Even though the difference looks spectrally slight (2% in reflectance), however, high sensitivity of our vision to a low reflectance color made the color difference large.

	CIEDE2000
mean	0.41
maximum	7.11
standard deviation	0.70



(a) Estimation performance

(b) Worst - Phthalate Blue (Green Shade) masstone

Figure 46. Worst estimation performance for the masstone of Phthalate Blue (Green Shade).

7.4. Paint Gamut Expansion

Color gamut expanded by varnishing can be generated by the same technique as shown in the previous chapter (*c.f.* “6.2 Multiple Paint Mixture”). The new approach is to use the gloss model leading to generating color gamut for SPEX at any gloss level. The original data used for color gamut rendering in Chapter 6 is based on SPIN. To simulate color gamuts at various gloss levels, we assigned three different gloss levels: (1) non-varnished (0 Δ specular gloss), (2) matte (1 Δ specular gloss), (3) semi-gloss (40 Δ specular gloss), and (4) high-gloss (60 Δ specular gloss), the gloss geometry of which are 60° gloss geometry.

Figure 47 shows the simulation results of color gamut expansion at three gloss levels. These figures say that a color gamut is expanded into the chroma and darkness directions with an increase of gloss level. On the contrary, bright colors do not change

so much. These simulation results fit into the result of gloss observation in gloss demonstrations with plastic films that was discussed in “2.3.1. Gloss Observation” of this paper.

The numbers of Figure 47-(b) exhibits the volumes of paint color gamuts in CIE $L^*a^*b^*$ space corresponding to gloss levels (non-varnished = 100). From this numbers, it can be confirmed that the volume of a color gamut is expanded by varnishing up to 150 % comparing to the color gamut of the non-varnished paints.

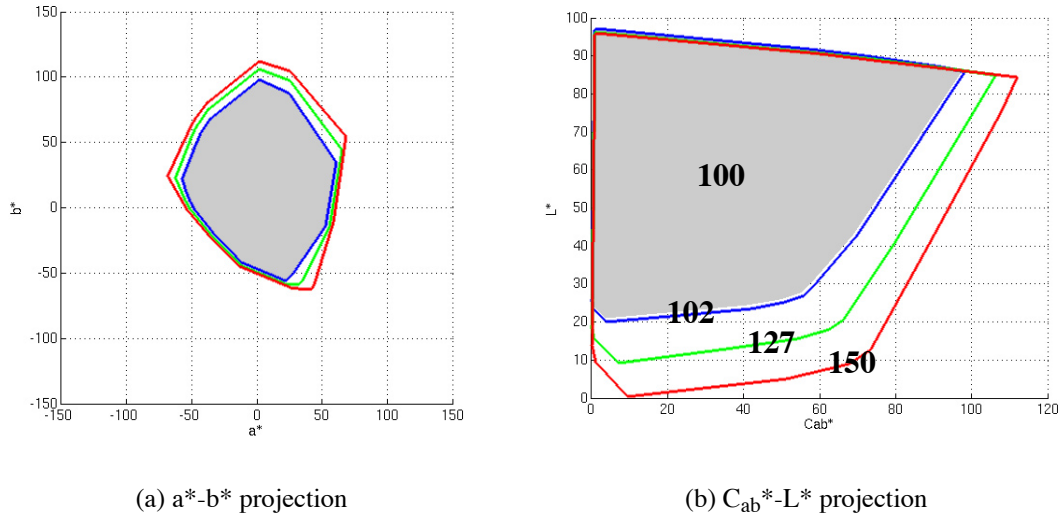


Figure 47. Color gamut expansions by varnishing based on the gloss simulation for 27 acrylic paints (gray area: non-varnished, blue area: matte, green area: semi-gloss, red area: high-gloss).

7.5. Discussion

The simulation of the gloss effects onto appearance matches the result of gloss demonstrations. However, there is still a question on if the SPEX geometry is most cor-

related with gloss perception rather than the other geometries. It seems to be no doubt that we naturally observe not full, but some specular reflection from glossy objects. A good example is a occasion that we look at varnished oil paintings in museums. By moving towards and far from varnished paintings, an audience can exclude first-surface reflection projected by the ceiling light from their sights to some degree. At this point, the SPEX measurement, which integrates a certain amount of specular reflection to colorimetric measurement values (like to our sights), might quantify the state of gloss observation properly. Therefore, it must be a future challenge to figure out what measurement geometry is most associated with visual observation in museum settings.

One of the other challenges is to extend the gloss model applicable for the case of using a different RI varnish from binder's RI. Since varnish's RI controls the replication state of a paint surface structure, painting conservators use a variety of varnish materials with different RIs for picture varnish [de la Rie 1987]. Fortunately, a MSA gloss varnish has 1.47 RI similar to the RI of an acrylic emulsion (1.48 RI) so that our simple gloss model characterized gloss effects precisely. A difference in RI between the two media makes gloss formulation more complex because of internal multiple reflections happened within a paint layer. Additionally, some of the light flux passing through a varnish layer bounces off at the boundary between a varnish layer and a paint layer. It is easily assumed that, since no light absorption makes the amount of first-surface reflection and other reflections at a varnish-paint boundary increased, a varnished paint with a higher RI varnish would lead to being more achromatic. In such a case, it must be hard to characterize Fresnel reflections (e.g. first-surface reflection).

Another challenge is to extend the gloss model for the case of using a colored var-

nish coating. There are a lot of occasions to deal with colored varnish coatings in art conservation. For example, one of the popular varnishes, Darmar varnish, is pale yellow. In addition, if a varnish coating is oxidized for a long time, the color of a coating film will be changed into being yellowish. Unfortunately, our gloss model does not formulate this phenomenon because the amount of a spectral difference between SPIN and SPEX is based on a varnish coating during constant transmittance without scattering. In addition, for example, a gloss meter is not sensitive to a change in the spectral regions other than green.

A further direction of this thesis is to figure out how to characterize the varnish coating that is made by various RI varnishes with translucency. For this research, we would need to develop an evaluation method to surface reflection and the transparency of a varnish film with a more complex gloss model (c.f. Equation (2.25)) and some instrument rather than by using the simple model and a gloss meter. The simulation result of color darkening and color saturation particularly for middle-to-dark paint colors matched with visual observation for the glossy and matte demonstrations with plastic films as well as the result of the past experiment by Berns and de la Rie mostly [Berns 2003b].

CHAPTER 8. SUMMARY AND CONCLUSIONS

8.1. Summary

We developed a spectral and colorimetric database of artist paint materials for 27 paints of the GOLDEN matte fluid acrylic paint lineup. The database includes the paint mixing model coefficients, unit k and unit s for each colorant (the Kubelka-Munk coefficients), and a single K_1 and a single K_2 for the paint set (Saunderson coefficients). These coefficients were optimized by minimizing the spectral errors of the paint characterization. In an optimization process, several optimization approaches, local optimization and global optimization, were tried to find the best optimization technique leading to minimum characterization errors. With regard to local optimization, the least-square method by using the full set of concentration series and the limited number of them (or the maximum chroma dataset). For global optimization, three spectral error metrics, R_m RMS, R_m weighted RMS by the diagonal of Matrix R (wRMS1), and R_m weighted RMS by Viggiano's spectral weight (wRMS2), were used in addition to a traditional approach, the least-square method in the K/S space.

By using the absorption and scattering properties for all the paints stored in the spectral database, we simulated secondary paint mixtures with and without tinting with computation and verified simulation performances in CIEDE2000.

We also rendered paint color gamuts by using the database that were made by the combinations of primary mixtures (yellow, red, blue), 4 paints (three primaries + green), and all the 26 paints. The volume sizes of several coloration systems including paint, printer, and photo, were compared to confirm if the colorimetric coverage of the paint set is large enough to be used as base colorants for a color rendering

chart.

Finally, we developed a gloss model that estimated spectral reflectances in the SPEX measurement geometry from measured spectral reflectances in the SPIN mode and specular gloss (at 60° gloss geometry). We also demonstrated the trend of color gamut expansion by varnishing a matte paint that is associated with visual perception for varnished glossy paintings.

8.2. Conclusions

The characterization performances of the acrylic paint, which resulted from local optimization and global optimization, were satisfactory. The reason of the good performances is due to the validity of the two-constant Kubelka-Munk theory for this particular paint set. With regard to global optimization, as a result, different spectral comparison indices, R_m RMS, R_m weighted RMS by the diagonal of Matrix R, and R_m weighted RMS by Viggiano's spectral weight were not significantly different as objective functions in the optimization. The only difference related to the spectral metrics was seen in correlation between CIEDE2000 and R_m RMS when analyzing spectral and colorimetric characterization errors based on each optimization technique. The optimization by minimizing R_m RMS shows high correlation between two metrics while the optimization by minimizing R_m wRMS1 and the optimization by minimizing R_m wRMS2. This result of the high correlation implies that, minimizing R_m RMS is the best approach because it accelerate the convergences of optimal values spectrally and colorimetrically at the same time in an optimization process.

The simulation performances of secondary paint mixtures were not good particularly for yellow paint mixtures. One of the reasons for the poor performances was re-

lated to the imperfection of opacity for the drawdowns of Hansa Yellow Opaque and difficulties in separating the optical properties of the yellow from the tint, Titanium White. To improve the simulation accuracy, the paint film of a yellow paint mixture must be extremely thick enough to be opaque, or we need to consider about both cases of opaque and translucent paints using the general form of the Kubelka-Munk theory. In this case, the opacity of mixtures of translucent and opaque paints can be considered to be determined by the hiding powers and the balance of them.

Paint color gamuts were rendered by using the spectral database for any number of paint combinations. In addition, as a result of comparison in color gamut, we confirmed that, GretagMacbeth ColorChecker DC and GretagMacbeth Color Checker SG, which are used for spectral image capturing as calibration charts, do not cover the possible maximum volume of this paint system. This is strong evidence that an actual painted-color chart is necessary for accurate color reproduction when targeting acrylic paintings in a spectral imaging system. However, this is very dependent on the method of building the color profile.

The color gamut expansion by varnishing could be simulated with our gloss model. The gloss model was driven by the fact that a specular gloss (at 60° gloss geometry) and the amount of spectral differences between SPIN and SPEX were highly correlated.

8.3. Future Research

For future research, the first priority is to build up the same spectral database for oil paints. This destination is more helpful for research on the digital image archive of cultural heritages particulally on oil paintings. By association with the oil paint data-

base, we are also interested in comparison in spectral property between acrylic paints and oil paints, both of which are made of the identical pigments as shown in Table IV. If colors of both paints with the same CI Name show the same spectral reflectances or some spectral curve trends, that would mean that oil paint colors might be able to be replaced with acrylic paint colors. Replacing traditional paints with modern paints is very helpful for imaging researchers because acrylic paints are significantly easier to use for creating a paint color rendering chart without the difficulties in application and chemical toxicity.

Another of the future direction on this thesis is to improve the simulation performance of multiple paint mixtures. Our target mixtures were limited in secondary paint mixtures so that it is still unclear for the case of simulating mixtures of more than three colorants. The only problem to implement is that it takes a lot of time for sample preparation. However, challenging a time-consuming experiment of multiple colorant mixtures would clarify a physical mechanism of colorant mixtures and the limitation of a Kubelka-Munk based simulation caused by chemical (e.g. pigments' compatibility) and optical factors.

A continuous examination of gloss study is to extend our gloss model for a varnish coat with a different RI from paint's binder and a certain degree of translucency. Success in this challenge will lead to not only disclosing the gloss effects by varnishing from optics' perspective but also might be able to formulate the aging effects of a varnish coat onto appearance by light exposure and oxidization (e.g. yellowing).

Finally, it is also necessary to establish our hypothesis on gloss modeling that the SPEX geometry is most associated with gloss perception. Such a topic has been discussed in terms of spectrophotometry so far. At present, in spectral imaging capture, the topic is discussed as a different form that a glare resulted by a lighting condi-

tion determines the final appearance of spectral images. At this point, it might be a good occasion now to re-consider an association between an instrument (or imaging) geometry and visual perception for not only glossy colors but also universal colors including metallics and pearlescents.

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CHAPTER 10. APPENDICES

A. Spreadsheet of Measured Paint Dataset

A.1. Recipes of Paint Drawdowns

Table A-I shows the recipes of paint drawdowns used for the artist paint material database. To calculate a concentration by weight, we used the following equation.

$$\text{Concentration [\%]} = 100 \times \text{Colorant weight} / (\text{Colorant weight} + \text{Tint weight})$$

Table A-I. Recipes of paint drawdowns used for the artist paint database.

Paint Name	Concentration		Weight	
	Target [%]	Actual [%]	Colorant [g]	Tint [g]
Titanium White	100	100.0	5.00	0.00
Carbon Black	5	5.0	0.25	4.78
	10	10.1	0.51	4.53
	20	20.5	1.03	4.00
	40	40.5	2.04	3.00
	60	60.1	3.21	2.13
	80	80.2	4.01	0.99
	100	100.0	5.00	0.00
Cobalt Blue	20	20.0	1.00	3.99
	40	40.2	2.01	2.99
	60	59.7	2.99	2.02
	80	80.2	4.01	0.99
	100	100.0	5.00	0.00
Chromium Oxide Green	20	19.9	1.00	4.03
	40	40.2	2.02	3.00
	60	59.5	3.00	2.04
	80	80.6	4.04	0.97
	100	100.0	5.00	0.00
Quinacridone Magenta	10	10.1	0.51	4.52
	20	20.5	1.03	4.00
	40	40.1	2.01	3.00
	60	60.3	3.02	1.99

	80	80.4	4.02	0.98
	100	100.0	5.00	0.00
Diarylide Yellow	10	10.3	0.52	4.53
	20	20.5	1.03	4.00
	40	40.2	2.04	3.03
	60	59.7	2.99	2.02
	80	80.2	4.00	0.99
	100	100.0	5.00	0.00
Phthalo Blue (Green Shade)	10	9.9	0.50	4.54
	20	19.8	1.00	4.04
	40	39.6	1.99	3.03
	60	59.9	3.02	2.02
	80	80.0	4.00	1.00
	100	100.0	5.00	0.00
Hansa Yellow Opaque	20	20.1	1.01	4.01
	40	39.9	2.01	3.03
	60	60.4	3.03	1.99
	80	80.5	4.04	0.98
	100	100.0	5.00	0.00
Red Oxide	20	20.3	1.02	4.00
	40	39.8	2.01	3.04
	60	60.5	3.03	1.98
	80	79.6	4.02	1.03
	100	100.0	5.00	0.00
Phthalo Green (Blue Shade)	10	10.2	0.51	4.51
	20	20.5	1.03	4.00
	40	40.2	2.01	2.99
	60	60.1	3.01	2.00
	80	79.7	4.01	1.02
	100	100.0	5.00	0.00
Pyrrole Red	10	9.9	0.50	4.53
	20	19.8	1.00	4.05
	40	40.0	2.01	3.01
	60	60.1	3.03	2.01
	80	80.2	4.01	0.99
	100	100.0	5.00	0.00
Yellow Ochre	20	20.4	1.03	4.03
	40	40.2	2.02	3.00
	60	60.2	3.01	1.99

	80	80.4	4.02	0.98
	100	100.0	5.00	0.00
Ultramarine Blue	20	19.8	1.00	4.05
	40	40.0	2.01	3.01
	60	60.0	3.02	2.01
	80	79.7	4.01	1.02
	100	100.0	5.00	0.00
Dioxazine Purple	20	19.9	1.00	4.02
	40	40.4	2.02	2.98
	60	59.9	3.02	2.02
	80	79.7	4.01	1.02
	100	100.0	5.00	0.00
Green Gold	20	20.7	1.04	3.98
	40	40.2	2.02	3.01
	60	59.8	3.00	2.02
	80	79.9	4.05	1.02
	100	100.0	5.00	0.00
Cerulean Blue	20	20.4	1.03	4.01
	40	40.2	2.03	3.02
	60	60.6	3.03	1.97
	80	80.2	4.02	0.99
	100	100.0	5.00	0.00
Titan Buff	20	19.9	1.08	4.36
	40	40.2	2.01	2.99
	60	60.4	3.02	1.98
	80	79.9	4.05	1.02
	100	100.0	5.00	0.00
Quinacridone Crimson	20	20.2	1.02	4.04
	40	40.1	2.02	3.02
	60	60.4	3.02	1.98
	80	80.5	4.05	0.98
	100	100.0	5.00	0.00
Raw Umber	20	20.4	1.02	3.98
	40	40.3	2.02	2.99
	60	59.4	3.00	2.05
	80	79.6	4.02	1.03
	100	100.0	5.00	0.00
Burnt Sienna	20	20.2	1.01	4.00
	40	40.4	2.05	3.03

	60	60.6	3.03	1.97
	80	80.0	4.11	1.03
	100	100.0	5.00	0.00
Burnt Umber	20	19.9	1.00	4.02
	40	40.6	2.04	2.99
	60	59.9	3.02	2.02
	80	79.4	4.02	1.04
	100	100.0	5.00	0.00
Jenkins Green	20	20.2	1.01	3.99
	40	39.8	2.00	3.03
	60	60.0	3.02	2.01
	80	78.6	4.01	1.09
	100	100.0	5.00	0.00
Anthraquinone Blue	20	20.3	1.02	4.00
	40	40.2	2.01	2.99
	60	59.9	3.03	2.03
	80	80.2	4.05	1.00
	100	100.0	5.00	0.00
Permanent Green Light	20	19.8	1.00	4.04
	40	39.5	2.00	3.06
	60	59.6	3.02	2.05
	80	78.4	4.00	1.10
	100	100.0	5.00	0.00
Naphthol Red Medium	20	20.2	1.03	4.06
	40	40.4	2.06	3.04
	60	59.6	3.04	2.06
	80	79.5	4.00	1.03
	100	100.0	5.00	0.00
Raw Sienna	20	20.7	1.06	4.06
	40	40.1	2.05	3.06
	60	59.6	3.04	2.06
	80	79.8	4.07	1.03
	100	100.0	5.00	0.00
Paynes Gray	20	19.9	1.07	4.30
	40	39.7	2.10	3.19
	60	60.2	3.02	2.00
	80	79.3	4.05	1.06
	100	100.0	5.00	0.00

A.2. Colorimetric Data

Table A-IIa, A-IIb, A-IIc, A-IId show the colorimetric measurement values of the non-varnished and varnished paint specimens based on different specular integration. The colorimetric values were computed from the spectral reflectances of the database on Matlab. For the colorimetric calculation, the ASTM method including the ASTM weights (D65 and A) was used.

Table A-IIa. Colorimetric values in SPIN under D65 illuminant and the 1931 standard observer.

Paint Name	Concentration		Colorimetric values (non-varnished)			Colorimetric values (varnished)		
	Target [%]	Actual [%]	L*	C* _{ab}	h* _{ab}	L*	C* _{ab}	h* _{ab}
Titanium White	100	100.0	96.9	1.3	117.9	96.9	2.1	111.7
Carbon Black	5	5.0	59.9	2.5	257.0	58.9	2.4	253.8
	10	10.1	52.1	2.7	258.8	51.0	2.6	256.2
	20	20.5	42.2	2.8	263.0	40.8	2.9	261.9
	40	40.5	33.9	2.6	268.5	32.2	2.8	265.3
	60	60.1	30.0	2.0	271.2	28.2	2.3	267.0
	80	80.2	27.5	1.0	277.0	25.7	1.4	268.4
	100	100.0	26.1	0.2	312.1	23.9	0.5	270.8
Cobalt Blue	20	20.0	79.1	24.0	264.5	78.7	24.2	264.0
	40	40.2	71.1	32.8	268.7	70.2	33.4	268.8
	60	59.7	63.8	39.7	272.3	63.1	39.7	272.3
	80	80.2	55.0	46.7	277.2	54.4	46.4	277.2
	100	100.0	33.3	56.2	292.8	32.7	54.7	293.2
Chromium Oxide Green	20	19.9	75.5	14.5	147.8	74.8	15.4	146.4
	40	40.2	67.2	18.5	145.8	66.6	19.3	145.0
	60	59.5	60.3	21.7	143.7	59.8	22.3	143.1
	80	80.6	53.1	24.8	141.5	52.5	24.9	141.2
	100	100.0	44.8	28.4	137.9	44.4	28.1	138.0
Quinacridone Magenta	10	10.1	73.6	37.8	337.4	73.1	37.8	337.9
	20	20.5	67.5	44.9	338.9	66.7	45.3	339.3
	40	40.1	59.6	51.6	341.6	59.1	51.5	341.8
	60	60.3	52.6	53.8	345.2	52.0	54.3	345.7
	80	80.4	44.7	52.1	351.0	44.1	52.4	351.3
	100	100.0	29.9	27.7	5.7	----	----	----
Diarylide Yellow	10	10.3	88.3	50.2	79.3	88.1	51.4	79.7
	20	20.5	86.6	60.6	77.9	86.0	61.7	78.5
	40	40.2	83.8	73.3	76.3	83.6	74.0	76.2
	60	59.7	82.0	80.6	75.3	81.3	80.1	75.7
	80	80.2	79.9	86.9	74.3	79.2	85.5	74.8
	100	100.0	77.6	91.1	73.7	77.4	90.7	72.9
Phthalo Blue (Green Shade)	10	9.9	69.4	36.4	243.6	68.4	36.8	243.2
	20	19.8	62.7	40.9	248.1	61.9	41.3	247.7
	40	39.6	53.9	43.6	254.7	53.5	43.7	254.4
	60	59.9	46.5	42.9	261.6	46.5	43.0	261.2

	80	80.0	37.7	40.1	271.4	38.0	39.2	271.1
	100	100.0	23.5	24.8	300.2	24.2	21.8	298.7
Hansa Yellow Opaque	20	20.1	91.8	57.6	94.2	91.1	59.1	94.9
	40	39.9	90.0	71.1	92.5	89.2	71.8	92.8
	60	60.4	88.8	81.8	90.9	88.0	81.2	91.6
	80	80.5	87.6	90.2	89.9	87.1	89.4	90.1
	100	100.0	86.1	97.5	88.9	85.9	96.4	88.5
Red Oxide	20	20.3	67.1	23.8	25.2	66.4	24.4	26.2
	40	39.8	58.8	28.7	27.3	58.1	29.0	28.2
	60	60.5	51.8	32.2	29.5	51.2	32.5	30.1
	80	79.6	45.7	35.4	32.2	45.7	35.1	32.4
	100	100.0	38.7	39.1	36.7	38.9	37.9	36.1
Phthalo Green (Blue Shade)	10	10.2	74.9	37.9	185.3	74.1	38.7	185.2
	20	20.5	70.0	42.5	184.0	69.6	43.5	183.9
	40	40.2	62.2	47.0	182.1	61.8	47.6	182.1
	60	60.1	55.2	47.5	180.9	54.6	47.5	180.8
	80	79.7	46.4	42.3	181.0	45.5	42.0	180.7
	100	100.0	25.7	8.2	234.1	25.4	6.8	229.1
Pyrrole Red	10	9.9	72.3	31.8	7.5	71.4	31.6	8.2
	20	19.8	65.5	40.3	10.7	64.5	40.8	11.3
	40	40.0	57.6	50.3	15.3	57.1	50.4	15.6
	60	60.1	52.4	57.1	19.2	52.2	56.7	19.4
	80	80.2	47.6	63.4	23.8	47.3	61.8	23.9
	100	100.0	43.1	69.3	29.8	43.2	68.4	29.4
Yellow Ochre	20	20.4	84.7	26.4	79.2	83.9	27.3	79.2
	40	40.2	79.1	33.6	77.1	78.2	34.2	77.1
	60	60.2	74.2	39.1	75.4	73.6	40.1	75.3
	80	80.4	67.8	43.4	73.6	66.8	43.7	73.4
	100	100.0	58.2	45.7	71.3	57.4	45.2	70.8
Ultramarine Blue	20	19.8	71.2	33.8	276.3	70.5	33.9	275.9
	40	40.0	62.0	43.9	281.1	61.2	43.9	281.0
	60	60.0	53.5	52.8	285.9	52.9	52.3	285.9
	80	79.7	43.8	60.7	291.6	43.5	60.0	291.7
	100	100.0	22.9	25.4	299.1	23.6	26.3	299.8
Dioxazine Purple	20	19.9	54.2	42.9	304.2	52.9	43.6	304.3
	40	40.4	43.9	46.0	304.9	43.0	46.0	304.7
	60	59.9	37.0	44.3	305.3	35.8	44.0	304.8
	80	79.7	30.6	35.1	305.5	29.9	34.1	304.3
	100	100.0	23.7	2.9	1.1	23.7	1.4	2.3

Green Gold	20	20.7	85.8	55.5	105.6	85.5	57.1	105.6
	40	40.2	80.8	61.5	104.3	80.6	62.5	104.3
	60	59.8	76.0	63.4	103.5	75.2	63.5	103.5
	80	79.9	68.2	61.0	103.2	67.5	60.2	103.3
	100	100.0	49.2	42.1	112.3	48.1	39.2	112.9
Cerulean Blue	20	20.4	79.0	20.3	252.3	78.5	20.5	251.7
	40	40.2	71.8	26.2	254.4	71.0	26.3	254.0
	60	60.6	64.4	30.9	256.2	63.9	31.2	256.2
	80	80.2	56.3	35.0	258.7	55.5	35.0	259.0
	100	100.0	41.2	39.1	266.0	39.9	37.6	267.8
Titan Buff	20	19.9	92.7	6.7	94.0	92.6	7.3	92.5
	40	40.2	90.3	9.4	90.1	90.3	10.3	90.2
	60	60.4	88.2	11.7	88.7	88.0	12.4	88.5
	80	79.9	86.4	13.5	87.8	86.5	14.1	87.8
	100	100.0	83.7	16.7	86.3	83.9	17.2	86.8
Quinacridone Crimson	20	20.2	63.5	33.0	355.3	62.6	33.3	356.0
	40	40.1	54.8	38.1	359.7	54.2	38.3	360.0
	60	60.4	48.1	40.2	3.7	47.2	40.5	3.9
	80	80.5	39.9	38.4	8.9	39.2	38.7	9.0
	100	100.0	27.9	18.5	15.8	26.5	18.4	14.9
Raw Umber	20	20.4	64.6	2.9	81.5	63.9	3.3	85.1
	40	40.3	54.5	2.1	85.6	54.1	2.3	89.4
	60	59.4	45.8	1.6	83.6	45.7	1.8	89.9
	80	79.6	36.7	1.3	80.0	37.0	1.3	83.0
	100	100.0	24.5	0.4	86.8	27.2	3.4	70.0
Burnt Sienna	20	20.2	75.4	15.8	41.0	74.7	16.2	42.1
	40	40.4	66.2	20.3	40.7	65.7	20.7	41.3
	60	60.6	58.6	23.6	40.5	58.0	23.9	40.8
	80	80.0	50.4	25.9	39.8	50.2	25.8	40.1
	100	100.0	32.9	23.9	36.6	32.7	23.0	36.0
Burnt Umber	20	19.9	69.4	11.4	60.9	68.8	11.8	61.9
	40	40.6	60.0	13.4	59.1	58.8	13.5	59.8
	60	59.9	51.1	14.2	57.3	50.2	13.9	57.5
	80	79.4	41.2	13.6	54.5	40.7	12.6	53.4
	100	100.0	25.6	7.3	50.1	26.1	5.6	46.6
Jenkins Green	20	20.2	64.5	22.9	151.0	63.8	23.6	150.7
	40	39.8	54.7	23.9	150.2	53.6	24.2	150.4
	60	60.0	46.1	21.7	150.9	45.0	22.0	151.0
	80	78.6	37.7	16.5	153.2	37.1	16.2	153.2

	100	100.0	26.0	3.7	173.5	25.7	3.1	173.7
Anthraquinone Blue	20	20.3	59.5	32.3	269.2	58.9	32.4	268.8
	40	40.2	50.2	36.0	274.8	49.6	35.8	274.6
	60	59.9	42.2	37.0	280.3	41.6	36.9	280.1
	80	80.2	33.3	33.2	287.3	32.8	32.4	286.9
	100	100.0	23.1	7.6	308.6	23.6	6.1	306.7
Permanent Green Light	20	19.8	79.1	42.7	159.2	78.6	45.1	157.6
	40	39.5	72.3	52.5	158.3	71.6	54.2	157.6
	60	59.6	65.6	59.1	157.9	64.6	59.7	157.7
	80	78.4	59.5	60.7	158.2	58.6	61.0	157.9
	100	100.0	45.8	56.9	160.1	45.5	56.2	160.1
Naphthol Red Medium	20	20.2	61.8	46.3	0.8	61.1	46.9	1.4
	40	40.4	54.5	52.1	6.0	53.9	52.9	6.5
	60	59.6	48.8	54.4	11.2	48.5	54.9	11.5
	80	79.5	43.6	54.5	17.5	43.1	55.4	17.8
	100	100.0	36.2	48.3	27.8	36.9	48.6	26.4
Raw Sienna	20	20.7	83.8	20.5	76.5	83.5	21.2	77.0
	40	40.1	77.8	26.8	74.9	77.4	27.3	75.1
	60	59.6	72.2	31.9	73.5	71.6	32.1	73.4
	80	79.8	65.1	35.9	71.6	64.3	36.3	71.5
	100	100.0	50.7	37.8	68.3	50.6	37.2	67.9
Paynes Gray	20	19.9	66.9	17.1	264.8	65.7	17.2	264.1
	40	39.7	57.1	20.1	266.9	56.5	19.7	266.3
	60	60.2	47.9	21.3	269.5	47.4	20.8	268.9
	80	79.3	37.8	20.7	273.0	37.7	19.6	272.4
	100	100.0	21.0	3.8	282.2	23.3	2.7	282.7

Table A-IIb. Colorimetric values in SPEX under D65 illuminant and the 1931 standard observer.

Paint Name	Concentration		Colorimetric values (non-varnished)			Colorimetric values (varnished)		
	Target [%]	Actual [%]	L*	C* _{ab}	h* _{ab}	L*	C* _{ab}	h* _{ab}
Titanium White	100	100.0	96.6	1.4	118.3	95.5	2.3	113.4
Carbon Black	5	5.0	59.9	2.4	255.7	56.1	2.5	252.5
	10	10.1	52.1	2.7	258.6	47.6	3.0	255.6
	20	20.5	42.2	2.9	261.3	37.6	3.2	260.1
	40	40.5	34.0	2.6	266.8	27.4	3.4	264.3
	60	60.1	29.9	2.0	270.8	22.3	3.1	265.1
	80	80.2	27.5	1.1	278.9	17.2	1.9	265.8
	100	100.0	26.1	0.3	309.1	8.6	1.3	279.0
Cobalt Blue	20	20.0	78.9	23.8	264.3	76.7	24.8	263.7
	40	40.2	70.9	32.5	268.6	68.4	34.1	268.5
	60	59.7	63.5	39.4	272.3	60.5	41.4	272.2
	80	80.2	54.9	46.4	277.1	51.0	49.1	277.4
	100	100.0	32.8	55.7	292.7	23.3	68.3	296.7
Chromium Oxide Green	20	19.9	75.2	14.4	147.7	72.8	16.0	146.3
	40	40.2	67.0	18.4	145.7	64.2	20.5	144.6
	60	59.5	60.2	21.5	143.8	56.5	24.5	142.6
	80	80.6	53.0	24.7	141.5	48.9	28.3	140.6
	100	100.0	44.7	28.3	138.0	38.9	36.2	135.9
Quinacridone Magenta	10	10.1	73.3	37.4	337.5	71.1	38.9	337.8
	20	20.5	67.3	44.5	339.0	64.3	47.2	339.3
	40	40.1	59.4	51.2	341.6	56.1	54.9	341.8
	60	60.3	52.3	53.6	345.2	47.8	59.5	345.5
	80	80.4	44.5	51.9	350.9	38.7	60.1	351.1
	100	100.0	29.8	28.3	5.4	----	----	----
Diarylide Yellow	10	10.3	88.0	50.0	79.5	86.4	53.8	79.9
	20	20.5	86.2	60.3	78.0	84.3	65.1	78.9
	40	40.2	83.5	72.9	76.4	81.7	80.1	76.9
	60	59.7	81.5	80.1	75.3	79.4	88.8	76.7
	80	80.2	79.5	86.3	74.3	77.3	96.3	76.1
	100	100.0	77.1	90.4	73.6	75.6	101.9	74.4
Phthalo Blue (Green Shade)	10	9.9	69.2	36.1	243.4	67.0	37.7	242.5
	20	19.8	62.5	40.6	247.9	59.3	43.0	246.8
	40	39.6	53.7	43.3	254.5	49.8	46.7	253.5
	60	59.9	46.3	43.3	261.3	41.5	47.3	260.4

	80	80.0	37.5	40.3	271.2	30.5	46.2	271.2
	100	100.0	23.2	25.1	299.8	11.5	36.0	303.9
Hansa Yellow Opaque	20	20.1	91.3	57.2	94.3	89.8	60.8	94.9
	40	39.9	89.5	70.6	92.5	88.0	74.8	92.9
	60	60.4	88.4	81.3	91.0	86.7	86.5	91.5
	80	80.5	87.2	89.6	90.0	85.0	98.3	90.7
	100	100.0	85.7	96.8	88.9	83.9	112.0	88.7
Red Oxide	20	20.3	67.0	23.5	25.2	64.3	25.3	26.5
	40	39.8	58.2	28.4	27.2	55.1	31.3	28.8
	60	60.5	51.3	32.2	29.6	47.7	36.1	31.4
	80	79.6	45.4	35.5	32.3	41.4	40.7	34.9
	100	100.0	38.0	39.7	36.8	31.6	54.5	45.2
Phthalo Green (Blue Shade)	10	10.2	74.7	37.7	185.4	72.6	40.2	185.1
	20	20.5	69.7	42.3	184.0	67.5	45.9	183.8
	40	40.2	62.0	46.7	182.1	59.6	50.9	182.0
	60	60.1	55.0	47.3	181.0	51.3	53.6	180.7
	80	79.7	46.3	42.1	181.1	42.0	49.3	180.6
	100	100.0	25.6	8.1	235.1	11.6	14.4	221.7
Pyrrole Red	10	9.9	72.0	31.4	7.6	69.7	32.9	8.1
	20	19.8	65.1	40.0	10.7	62.6	42.1	11.3
	40	40.0	57.3	50.0	15.3	54.6	53.1	16.0
	60	60.1	52.1	56.7	19.2	49.2	60.5	20.2
	80	80.2	47.4	63.0	23.8	43.8	67.6	25.8
	100	100.0	42.8	68.8	29.6	37.8	86.5	38.3
Yellow Ochre	20	20.4	84.2	26.2	79.2	82.4	28.2	79.4
	40	40.2	78.9	33.3	77.2	76.8	36.1	77.5
	60	60.2	73.9	38.8	75.5	71.4	43.0	75.9
	80	80.4	67.7	43.0	73.7	65.6	45.8	73.9
	100	100.0	57.9	45.2	71.4	54.0	54.7	72.8
Ultramarine Blue	20	19.8	70.8	33.4	276.2	68.5	34.7	275.8
	40	40.0	61.8	43.5	281.0	58.8	45.5	281.1
	60	60.0	53.2	52.4	285.9	49.7	55.1	286.4
	80	79.7	43.6	60.3	291.6	38.6	65.4	292.9
	100	100.0	22.8	24.7	299.0	5.3	53.3	308.7
Dioxazine Purple	20	19.9	53.9	42.5	304.2	49.8	46.5	304.9
	40	40.4	43.9	45.8	304.8	39.1	50.7	305.7
	60	59.9	36.7	44.1	305.2	31.9	48.7	305.9
	80	79.7	30.5	34.9	305.5	21.4	44.8	307.1
	100	100.0	23.9	2.8	360.0	9.7	3.1	15.6

Green Gold	20	20.7	85.5	55.1	105.8	84.1	58.9	105.4
	40	40.2	80.5	61.1	104.4	79.2	65.6	103.9
	60	59.8	75.9	62.8	103.6	74.0	67.4	103.1
	80	79.9	68.1	60.7	103.4	66.0	65.7	102.6
	100	100.0	49.0	41.9	112.4	44.5	48.8	110.6
Cerulean Blue	20	20.4	78.8	20.1	252.2	76.9	20.9	251.3
	40	40.2	71.4	26.0	254.4	69.1	27.3	253.5
	60	60.6	64.2	30.6	256.1	61.7	32.4	255.8
	80	80.2	56.2	34.8	258.6	52.6	36.8	258.6
	100	100.0	41.0	38.9	266.1	34.1	42.7	267.8
Titan Buff	20	19.9	92.2	6.7	94.0	91.3	7.5	92.9
	40	40.2	89.8	9.4	90.7	88.7	10.4	90.3
	60	60.4	87.7	11.6	88.7	86.7	12.6	88.7
	80	79.9	85.9	13.4	88.0	84.9	14.4	88.1
	100	100.0	83.2	16.5	86.3	81.8	18.1	87.1
Quinacridone Crimson	20	20.2	63.2	32.8	355.3	60.6	34.7	356.1
	40	40.1	54.6	37.8	359.7	51.1	40.9	359.9
	60	60.4	47.8	39.9	3.6	43.0	45.3	4.0
	80	80.5	39.9	38.3	8.8	33.0	46.6	9.8
	100	100.0	27.7	18.4	15.7	17.7	27.5	17.8
Raw Umber	20	20.4	64.4	2.9	82.9	61.7	3.6	86.0
	40	40.3	54.2	2.1	86.0	50.9	2.7	89.5
	60	59.4	45.1	1.6	85.1	40.8	2.2	88.0
	80	79.6	34.8	1.3	80.0	29.2	2.1	83.8
	100	100.0	13.1	0.6	68.0	14.7	7.5	72.9
Burnt Sienna	20	20.2	75.1	15.6	41.4	72.6	16.9	42.5
	40	40.4	66.0	20.2	41.0	62.9	22.1	42.1
	60	60.6	58.3	23.5	40.7	54.7	26.1	41.8
	80	80.0	50.3	25.7	40.0	46.4	29.3	41.7
	100	100.0	32.7	23.5	36.6	23.2	38.7	46.2
Burnt Umber	20	19.9	69.1	11.4	61.3	66.4	12.6	62.6
	40	40.6	59.3	13.2	59.3	56.4	14.5	60.2
	60	59.9	50.9	14.0	57.5	47.4	15.4	58.0
	80	79.4	41.0	13.5	54.6	36.5	15.1	55.0
	100	100.0	25.4	7.2	49.8	15.7	10.3	50.6
Jenkins Green	20	20.2	64.2	22.9	151.0	61.0	25.7	150.5
	40	39.8	54.3	23.7	150.3	50.6	26.8	150.1
	60	60.0	46.0	21.6	150.9	41.9	24.7	150.6
	80	78.6	37.6	16.4	153.1	31.2	21.2	152.9

	100	100.0	25.9	3.8	174.4	12.0	7.3	170.4
Anthraquinone Blue	20	20.3	59.2	32.1	269.1	55.9	34.2	268.5
	40	40.2	49.9	35.6	274.6	45.7	38.6	274.7
	60	59.9	42.0	36.7	280.1	37.1	40.5	280.6
	80	80.2	33.2	33.0	287.3	25.4	39.4	288.6
	100	100.0	23.1	7.7	308.0	5.6	15.8	310.7
Permanent Green Light	20	19.8	78.6	42.3	159.1	76.6	47.0	157.6
	40	39.5	71.9	52.0	158.2	69.4	57.7	157.5
	60	59.6	65.0	58.9	157.9	62.0	65.1	157.6
	80	78.4	59.3	60.6	158.2	55.2	69.8	157.8
	100	100.0	45.7	56.9	160.1	40.1	74.4	160.3
Naphthol Red Medium	20	20.2	61.6	46.0	0.9	59.3	48.6	1.6
	40	40.4	54.2	51.8	6.0	51.3	55.7	6.7
	60	59.6	48.7	54.0	11.1	45.5	58.8	12.0
	80	79.5	43.5	53.8	17.5	40.2	59.7	18.7
	100	100.0	36.1	48.1	27.7	29.2	68.1	36.6
Raw Sienna	20	20.7	83.4	20.5	76.8	81.7	21.9	77.1
	40	40.1	77.4	26.6	75.1	75.6	28.4	75.4
	60	59.6	71.7	31.8	73.5	69.5	34.2	73.9
	80	79.8	64.7	35.7	71.7	62.7	38.3	71.9
	100	100.0	50.6	37.5	68.1	46.2	47.6	70.6
Paynes Gray	20	19.9	66.6	16.9	264.3	64.3	17.8	263.9
	40	39.7	57.0	19.8	266.7	54.3	20.4	266.1
	60	60.2	47.8	21.2	269.4	44.0	22.4	268.9
	80	79.3	37.6	20.5	272.8	33.3	22.2	272.5
	100	100.0	20.7	3.9	283.1	3.4	6.9	283.3

Table A-IIc. Colorimetric values in SPIN under illuminant A and the 1931 standard observer.

Paint Name	Concentration		Colorimetric values (non-varnished)			Colorimetric values (varnished)		
	Target [%]	Actual [%]	L*	C* _{ab}	h* _{ab}	L*	C* _{ab}	h* _{ab}
Titanium White	100	100.0	96.9	0.8	100.3	97.0	1.6	90.4
Carbon Black	5	5.0	59.7	3.0	242.4	58.7	2.9	241.3
	10	10.1	51.9	3.2	243.6	50.8	3.2	242.3
	20	20.5	42.0	3.3	246.2	40.6	3.5	245.8
	40	40.5	33.7	2.9	250.6	32.0	3.2	248.5
	60	60.1	29.9	2.1	253.9	28.0	2.6	249.4
	80	80.2	27.4	1.0	257.3	25.6	1.6	251.0
	100	100.0	26.1	0.2	304.2	23.9	0.5	252.6
Cobalt Blue	20	20.0	77.1	27.8	255.9	76.7	28.1	255.8
	40	40.2	68.5	37.6	257.8	67.5	38.3	257.8
	60	59.7	60.6	45.0	259.4	59.9	45.2	259.5
	80	80.2	51.3	51.9	262.0	50.7	51.7	262.0
	100	100.0	30.0	55.6	273.6	29.7	53.7	274.3
98145.451	20	19.9	74.7	11.8	147.9	74.1	12.4	145.9
	40	40.2	66.3	15.0	144.9	65.7	15.7	143.8
	60	59.5	59.3	17.6	141.8	58.8	18.1	140.9
	80	80.6	52.1	20.1	138.5	51.5	20.1	138.1
	100	100.0	43.8	22.9	133.2	43.4	22.6	133.4
Quinacridone Magenta	10	10.1	76.4	33.8	346.1	75.9	33.9	346.6
	20	20.5	71.0	40.2	348.3	70.2	40.7	348.7
	40	40.1	63.9	46.7	352.0	63.4	46.7	352.3
	60	60.3	57.5	50.0	356.8	57.0	50.6	357.3
	80	80.4	50.0	50.8	3.8	49.5	51.3	4.1
	100	100.0	33.1	31.9	17.0	----	----	----
Diarylide Yellow	10	10.3	91.6	56.0	76.0	91.5	57.1	76.3
	20	20.5	90.5	67.4	75.4	89.9	68.3	76.0
	40	40.2	88.5	80.9	74.7	88.3	81.7	74.7
	60	59.7	87.1	88.6	74.3	86.3	87.9	74.6
	80	80.2	85.4	95.2	73.7	84.5	93.5	74.2
	100	100.0	83.2	99.3	73.4	83.1	99.2	72.8
Phthalo Blue (Green Shade)	10	9.9	64.9	48.6	235.7	63.9	49.3	235.3
	20	19.8	57.7	54.9	237.7	56.8	55.8	237.3
	40	39.6	48.7	58.2	241.2	48.2	58.6	240.9
	60	59.9	41.5	55.7	246.0	41.6	56.0	245.6

	80	80.0	33.7	48.7	253.9	34.1	47.7	253.7
	100	100.0	23.1	21.4	283.0	23.8	18.9	279.9
Hansa Yellow Opaque	20	20.1	94.0	59.1	85.7	93.2	60.1	86.4
	40	39.9	92.6	72.8	85.0	91.8	73.2	85.5
	60	60.4	91.8	83.5	84.3	90.9	82.6	85.0
	80	80.5	90.9	91.6	83.9	90.3	90.8	84.1
	100	100.0	89.6	98.4	83.5	89.4	97.7	83.1
Red Oxide	20	20.3	69.9	28.0	32.7	69.3	28.8	33.4
	40	39.8	62.1	33.7	34.6	61.5	34.2	35.2
	60	60.5	55.5	37.7	36.6	55.0	38.1	36.9
	80	79.6	49.8	41.3	38.9	49.7	41.0	38.9
	100	100.0	43.1	45.4	42.6	43.1	44.1	42.0
Phthalo Green (Blue Shade)	10	10.2	70.9	40.9	198.6	70.1	41.7	198.4
	20	20.5	65.6	45.1	197.6	65.2	46.2	197.5
	40	40.2	57.5	48.3	196.3	57.1	48.9	196.3
	60	60.1	50.5	47.1	195.8	49.9	47.0	195.8
	80	79.7	42.2	40.2	197.3	41.4	39.9	197.0
	100	100.0	24.7	10.0	250.6	24.5	8.4	247.9
Pyrrole Red	10	9.9	75.7	36.5	17.2	74.9	36.4	17.7
	20	19.8	70.1	46.3	20.3	69.1	46.9	20.7
	40	40.0	63.4	57.6	24.7	62.9	57.7	24.9
	60	60.1	59.1	65.4	28.3	58.8	64.9	28.4
	80	80.2	55.0	72.5	32.4	54.5	70.6	32.4
	100	100.0	51.0	79.2	37.2	51.1	78.4	36.8
Yellow Ochre	20	20.4	86.6	29.5	72.1	85.9	30.4	72.1
	40	40.2	81.6	37.5	71.3	80.7	38.1	71.3
	60	60.2	77.0	43.6	70.7	76.5	44.6	70.7
	80	80.4	71.0	48.4	70.1	70.0	48.7	70.0
	100	100.0	61.5	50.9	69.2	60.8	50.5	68.9
Ultramarine Blue	20	19.8	68.9	36.8	259.7	68.2	37.1	259.5
	40	40.0	59.2	46.7	261.9	58.4	46.8	261.9
	60	60.0	50.2	54.5	264.5	49.7	53.9	264.5
	80	79.7	40.4	59.8	268.2	40.1	59.0	268.4
	100	100.0	22.4	21.4	277.7	23.1	21.8	278.2
Dioxazine Purple	20	19.9	54.0	36.3	292.4	52.8	37.0	292.6
	40	40.4	43.7	38.5	292.2	42.8	38.7	292.0
	60	59.9	36.8	36.7	292.0	35.6	36.7	291.3
	80	79.7	30.6	28.6	292.4	29.7	28.2	290.8
	100	100.0	24.0	3.3	14.6	23.9	2.0	13.2

Green Gold	20	20.7	86.7	50.1	94.4	86.4	51.3	94.4
	40	40.2	81.9	55.9	94.2	81.6	56.7	94.3
	60	59.8	77.1	58.0	94.3	76.3	58.0	94.4
	80	79.9	69.3	56.1	95.0	68.6	55.5	95.2
	100	100.0	49.4	37.4	105.9	48.3	35.0	106.7
Cerulean Blue	20	20.4	77.0	24.6	249.0	76.5	24.9	248.9
	40	40.2	69.2	31.8	249.9	68.4	32.1	249.8
	60	60.6	61.3	37.8	250.7	60.7	38.2	250.7
	80	80.2	52.6	43.0	251.9	51.8	42.9	252.3
	100	100.0	37.1	47.7	256.5	36.1	45.2	258.6
Titan Buff	20	19.9	93.0	6.7	76.7	93.0	7.4	75.0
	40	40.2	90.8	9.8	73.8	90.9	10.6	73.6
	60	60.4	88.9	12.4	73.0	88.8	13.0	72.6
	80	79.9	87.2	14.3	72.5	87.3	14.8	72.3
	100	100.0	84.8	17.9	71.7	84.9	18.3	72.0
Quinacridone Crimson	20	20.2	66.7	35.9	5.7	65.9	36.4	6.3
	40	40.1	58.8	41.9	10.1	58.3	42.2	10.4
	60	60.4	52.5	44.8	14.0	51.7	45.2	14.2
	80	80.5	44.3	43.9	18.7	43.7	44.3	18.7
	100	100.0	30.0	22.6	23.1	28.7	22.8	22.2
Raw Umber	20	20.4	64.8	3.2	71.7	64.2	3.6	73.7
	40	40.3	54.7	2.3	75.9	54.3	2.4	78.1
	60	59.4	45.9	1.8	75.1	45.8	1.9	79.3
	80	79.6	36.8	1.5	72.5	37.1	1.4	74.8
	100	100.0	24.6	0.5	80.1	27.5	4.0	63.2
Burnt Sienna	20	20.2	77.2	19.7	42.5	76.6	20.2	43.0
	40	40.4	68.5	25.1	42.7	68.0	25.6	43.0
	60	60.6	61.3	29.0	42.8	60.7	29.3	43.0
	80	80.0	53.4	31.6	42.7	53.1	31.5	42.8
	100	100.0	35.6	28.9	40.5	35.4	27.9	39.9
Burnt Umber	20	19.9	70.6	14.0	55.2	69.9	14.4	55.7
	40	40.6	61.3	16.4	54.4	60.2	16.5	54.8
	60	59.9	52.6	17.4	53.5	51.6	17.1	53.5
	80	79.4	42.6	16.8	51.8	42.1	15.6	50.9
	100	100.0	26.4	9.1	48.2	26.8	7.1	45.7
Jenkins Green	20	20.2	63.2	18.7	158.1	62.5	19.2	157.8
	40	39.8	53.3	19.5	157.0	52.2	19.8	157.2
	60	60.0	44.9	17.7	157.9	43.8	18.0	158.1
	80	78.6	36.7	13.5	161.0	36.1	13.3	161.0

	100	100.0	25.7	3.2	191.1	25.4	2.7	191.2
Anthraquinone Blue	20	20.3	56.8	37.7	250.1	56.1	38.0	249.8
	40	40.2	47.4	40.6	253.2	46.8	40.5	253.2
	60	59.9	39.6	39.8	257.2	38.9	39.8	257.0
	80	80.2	31.5	32.9	264.2	31.0	32.3	263.6
	100	100.0	23.2	5.4	297.6	23.7	4.3	292.5
Permanent Green Light	20	19.8	76.1	35.1	169.8	75.5	36.2	168.1
	40	39.5	68.8	42.6	168.6	68.0	43.6	167.9
	60	59.6	61.7	47.6	168.1	60.7	47.8	167.9
	80	78.4	55.5	48.6	168.4	54.6	48.4	168.1
	100	100.0	41.9	44.0	171.0	41.7	43.2	171.0
Naphthol Red Medium	20	20.2	66.8	48.1	13.1	66.2	48.9	13.6
	40	40.4	60.4	55.6	18.2	59.9	56.6	18.6
	60	59.6	55.2	59.6	22.7	55.0	60.3	22.9
	80	79.5	50.2	61.7	27.7	49.7	62.9	27.7
	100	100.0	41.9	57.1	34.4	42.7	58.1	32.9
Raw Sienna	20	20.7	85.4	23.3	69.7	85.1	24.0	70.0
	40	40.1	79.9	30.4	69.2	79.5	30.9	69.3
	60	59.6	74.7	36.1	68.8	74.1	36.3	68.7
	80	79.8	67.9	40.6	68.1	67.2	41.0	68.0
	100	100.0	53.7	42.7	66.8	53.6	42.1	66.5
Paynes Gray	20	19.9	65.5	19.8	253.2	64.3	20.0	252.9
	40	39.7	55.5	23.1	254.2	54.9	22.8	253.8
	60	60.2	46.2	24.3	255.6	45.8	23.8	255.2
	80	79.3	36.2	23.2	257.9	36.3	22.0	257.4
	100	100.0	20.8	3.7	267.5	23.2	2.6	267.2

Table A-IIId. Colorimetric values in SPEX under illuminant A and the 1931 standard observer.

Paint Name	Concentration		Gloss scale (non-varnished)			Gloss scale (varnished)		
	Target [%]	Actual [%]	L*	C* _{ab}	h* _{ab}	L*	C* _{ab}	h* _{ab}
Titanium White	100	100.0	96.6	0.9	100.7	95.5	1.7	93.1
Carbon Black	5	5.0	59.7	2.9	241.6	55.9	3.1	240.4
	10	10.1	51.9	3.2	243.4	47.3	3.6	241.9
	20	20.5	41.9	3.4	245.1	37.3	3.8	244.8
	40	40.5	33.8	2.9	249.6	27.1	3.9	247.7
	60	60.1	29.8	2.2	252.6	22.1	3.5	248.4
	80	80.2	27.4	1.1	259.7	17.1	2.2	249.0
	100	100.0	26.1	0.2	300.9	8.5	1.3	261.0
Cobalt Blue	20	20.0	76.9	27.5	255.9	74.5	29.0	255.5
	40	40.2	68.2	37.3	257.7	65.6	39.3	257.5
	60	59.7	60.4	44.7	259.4	57.1	47.3	259.1
	80	80.2	51.2	51.6	261.9	46.9	55.1	261.6
	100	100.0	29.5	55.1	273.6	18.4	67.8	275.9
Chromium Oxide Green	20	19.9	74.4	11.8	147.8	72.0	13.0	145.8
	40	40.2	66.1	15.0	144.8	63.2	16.6	143.3
	60	59.5	59.2	17.5	141.9	55.5	19.9	140.4
	80	80.6	52.0	20.0	138.5	47.8	23.0	137.5
	100	100.0	43.7	22.8	133.3	37.7	29.4	131.2
Quinacridone Magenta	10	10.1	76.0	33.4	346.3	73.9	34.8	346.5
	20	20.5	70.7	39.9	348.4	68.0	42.2	348.7
	40	40.1	63.7	46.4	352.1	60.7	49.3	352.2
	60	60.3	57.2	49.8	356.9	53.4	54.4	357.3
	80	80.4	49.8	50.6	3.7	45.0	57.0	4.3
	100	100.0	33.1	32.4	16.8	----	----	----
Diarylide Yellow	10	10.3	91.2	55.7	76.1	89.8	59.6	76.5
	20	20.5	90.1	67.0	75.5	88.3	71.7	76.3
	40	40.2	88.1	80.4	74.7	86.5	87.6	75.4
	60	59.7	86.5	88.1	74.2	84.5	96.3	75.6
	80	80.2	84.9	94.5	73.7	82.7	103.8	75.4
	100	100.0	82.7	98.6	73.4	81.5	109.6	74.1
Phthalo Blue (Green Shade)	10	9.9	64.9	48.2	235.6	62.4	50.7	234.6
	20	19.8	57.6	54.6	237.6	53.9	59.0	236.0
	40	39.6	48.5	57.8	241.2	43.9	64.4	238.7
	60	59.9	41.3	56.4	245.6	35.6	64.4	242.2

	80	80.0	33.4	49.1	253.6	25.0	59.4	249.2
	100	100.0	22.8	21.8	282.5	10.7	30.7	283.2
Hansa Yellow Opaque	20	20.1	93.5	58.7	85.8	91.9	61.7	86.4
	40	39.9	92.1	72.2	85.0	90.6	75.9	85.7
	60	60.4	91.4	82.9	84.4	89.6	87.3	85.1
	80	80.5	90.4	91.0	83.9	88.2	98.0	85.0
	100	100.0	89.2	97.8	83.4	87.6	110.2	83.8
Red Oxide	20	20.3	69.8	27.7	32.7	67.3	29.9	33.7
	40	39.8	61.6	33.4	34.5	58.7	36.6	35.9
	60	60.5	55.1	37.7	36.6	51.8	42.0	38.4
	80	79.6	49.4	41.4	39.0	46.0	47.0	41.6
	100	100.0	42.4	46.0	42.8	37.1	61.6	50.6
Phthalo Green (Blue Shade)	10	10.2	70.7	40.7	198.6	68.5	43.6	198.2
	20	20.5	65.4	44.9	197.6	62.8	49.0	197.1
	40	40.2	57.3	48.0	196.3	54.6	53.1	195.8
	60	60.1	50.3	47.1	196.0	46.1	54.6	194.9
	80	79.7	42.1	40.0	197.4	37.2	48.5	195.8
	100	100.0	24.6	10.0	251.1	9.6	17.0	241.8
Pyrrole Red	10	9.9	75.4	36.1	17.2	73.3	37.7	17.7
	20	19.8	69.6	46.0	20.3	67.4	48.1	20.9
	40	40.0	63.1	57.3	24.6	60.7	60.3	25.5
	60	60.1	58.8	64.9	28.2	56.3	68.6	29.6
	80	80.2	54.7	72.1	32.3	51.6	76.4	34.6
	100	100.0	50.7	78.6	37.1	47.0	97.2	45.1
Yellow Ochre	20	20.4	86.1	29.2	72.1	84.4	31.3	72.3
	40	40.2	81.3	37.2	71.4	79.4	40.1	71.7
	60	60.2	76.7	43.3	70.7	74.4	47.5	71.3
	80	80.4	70.8	48.0	70.1	68.9	50.8	70.4
	100	100.0	61.2	50.4	69.2	57.7	59.7	70.9
Ultramarine Blue	20	19.8	68.6	36.5	259.7	66.1	38.1	259.3
	40	40.0	59.0	46.3	261.8	55.7	48.7	261.6
	60	60.0	50.0	54.0	264.5	46.1	57.1	264.3
	80	79.7	40.2	59.4	268.2	34.6	64.6	268.2
	100	100.0	22.3	20.7	277.8	4.1	41.9	281.7
Dioxazine Purple	20	19.9	53.8	35.9	292.4	49.6	39.3	293.1
	40	40.4	43.6	38.4	292.1	38.8	42.5	292.9
	60	59.9	36.5	36.5	291.9	31.5	40.5	292.4
	80	79.7	30.4	28.5	292.5	21.2	36.7	293.4
	100	100.0	24.2	3.2	13.8	10.1	4.8	21.3

Green Gold	20	20.7	86.4	49.7	94.5	85.1	52.8	94.4
	40	40.2	81.5	55.5	94.3	80.2	59.2	94.2
	60	59.8	77.0	57.4	94.4	75.1	61.3	94.2
	80	79.9	69.1	55.8	95.1	67.1	60.0	94.9
	100	100.0	49.2	37.2	106.0	44.7	43.0	105.2
Cerulean Blue	20	20.4	76.8	24.4	249.0	74.7	25.4	248.7
	40	40.2	68.7	31.6	249.9	66.3	33.4	249.4
	60	60.6	61.0	37.4	250.7	58.3	39.9	250.2
	80	80.2	52.5	42.7	251.9	48.6	45.6	251.6
	100	100.0	36.9	47.4	256.6	29.4	52.7	256.9
Titan Buff	20	19.9	92.6	6.7	76.7	91.7	7.6	75.3
	40	40.2	90.4	9.8	74.2	89.3	10.8	73.7
	60	60.4	88.4	12.3	72.9	87.5	13.2	72.8
	80	79.9	86.7	14.1	72.7	85.8	15.2	72.5
	100	100.0	84.2	17.7	71.7	82.9	19.1	72.3
Quinacridone Crimson	20	20.2	66.4	35.7	5.7	64.0	37.8	6.4
	40	40.1	58.5	41.6	10.1	55.4	44.7	10.5
	60	60.4	52.2	44.5	13.9	48.1	49.5	14.8
	80	80.5	44.3	43.8	18.6	38.5	51.6	20.5
	100	100.0	29.9	22.5	23.1	21.0	32.1	26.5
Raw Umber	20	20.4	64.6	3.3	72.7	61.9	3.9	74.1
	40	40.3	54.4	2.3	76.1	51.1	2.8	77.9
	60	59.4	45.2	1.7	76.5	41.0	2.4	76.8
	80	79.6	34.9	1.4	73.2	29.3	2.3	74.4
	100	100.0	13.2	0.7	67.9	15.3	8.7	66.4
Burnt Sienna	20	20.2	76.9	19.5	42.7	74.5	21.0	43.3
	40	40.4	68.4	25.0	42.8	65.4	27.2	43.6
	60	60.6	61.0	28.9	42.9	57.6	31.8	43.9
	80	80.0	53.2	31.4	42.7	49.7	35.4	44.4
	100	100.0	35.4	28.5	40.5	27.1	44.6	50.0
Burnt Umber	20	19.9	70.2	14.0	55.4	67.7	15.3	56.2
	40	40.6	60.6	16.2	54.4	57.8	17.6	55.1
	60	59.9	52.4	17.2	53.5	49.0	18.7	54.1
	80	79.4	42.4	16.7	51.8	38.0	18.4	52.5
	100	100.0	26.2	9.1	48.0	16.8	12.6	49.9
Jenkins Green	20	20.2	62.8	18.7	158.2	59.6	20.9	157.6
	40	39.8	52.9	19.3	157.1	49.1	21.9	157.0
	60	60.0	44.8	17.7	157.8	40.5	20.2	157.8
	80	78.6	36.6	13.4	160.9	30.0	17.6	161.1

	100	100.0	25.6	3.2	192.0	11.3	6.2	186.7
Anthraquinone Blue	20	20.3	56.5	37.4	250.1	52.9	40.5	249.1
	40	40.2	47.1	40.2	253.1	42.5	44.2	252.1
	60	59.9	39.4	39.5	257.1	34.0	44.4	255.9
	80	80.2	31.4	32.7	264.2	22.9	39.7	262.3
	100	100.0	23.2	5.5	296.8	5.8	10.9	296.6
Permanent Green Light	20	19.8	75.6	34.7	169.7	73.5	37.8	168.1
	40	39.5	68.3	42.2	168.4	65.6	46.6	167.8
	60	59.6	61.1	47.4	168.0	57.8	52.6	168.0
	80	78.4	55.3	48.4	168.4	50.8	56.4	168.2
	100	100.0	41.8	44.1	171.0	35.4	60.2	171.5
Naphthol Red Medium	20	20.2	66.6	47.8	13.2	64.6	50.4	13.9
	40	40.4	60.1	55.2	18.2	57.7	59.0	19.1
	60	59.6	55.0	59.2	22.6	52.4	63.9	23.8
	80	79.5	50.0	60.9	27.6	47.4	67.0	29.2
	100	100.0	41.8	56.7	34.3	36.8	78.0	43.0
Raw Sienna	20	20.7	85.0	23.2	69.9	83.4	24.7	70.0
	40	40.1	79.5	30.2	69.4	77.8	32.1	69.7
	60	59.6	74.2	35.9	68.8	72.1	38.5	69.2
	80	79.8	67.5	40.4	68.1	65.7	43.0	68.5
	100	100.0	53.6	42.5	66.7	49.6	52.5	69.1
Paynes Gray	20	19.9	65.3	19.6	252.9	62.8	20.8	252.7
	40	39.7	55.4	22.8	254.0	52.6	23.7	253.6
	60	60.2	46.1	24.2	255.6	42.2	25.9	254.8
	80	79.3	36.1	23.0	257.8	31.5	25.2	256.8
	100	100.0	20.6	3.9	268.3	3.2	6.7	268.9

A.3. Specular Gloss

Table A-III shows the measured values of specular gloss for the artist paint material database.

Table A-III. Measurement values of specular gloss for the artist paint material database.

Paint Name	Concentration		Gloss scale (non-varnished)			Gloss scale (varnished)		
	Target [%]	Actual [%]	20°	60°	85°	20°	60°	85°
Titanium White	100	100.0	1.4	3.0	3.3	26.9	78.2	67.2
Carbon Black	5	5.0	0.4	2.1	3.2	17.6	57.4	62.0
	10	10.1	0.3	2.1	3.2	17.2	56.8	58.2
	20	20.5	0.2	2.4	2.5	9.6	38.5	45.6
	40	40.5	0.2	2.7	3.2	8.4	37.6	37.9
	60	60.1	0.2	2.9	2.8	7.4	39.6	48.0
	80	80.2	0.2	2.7	2.3	13.0	53.0	60.8
	100	100.0	0.2	2.8	2.0	18.1	62.5	65.6
Cobalt Blue	20	20.0	0.8	2.3	2.8	21.4	65.4	69.9
	40	40.2	0.6	2.2	2.6	12.3	50.1	50.7
	60	59.7	0.4	2.0	2.6	14.5	59.3	55.1
	80	80.2	0.3	2.0	2.6	22.5	59.7	59.9
	100	100.0	0.2	2.2	3.4	23.6	68.9	73.4
Chromium Oxide Green	20	19.9	0.7	2.5	3.5	18.1	60.3	58.9
	40	40.2	0.6	2.7	3.8	17.1	58.9	66.0
	60	59.5	0.4	2.9	4.0	22.6	67.2	61.5
	80	80.6	0.4	3.0	4.3	20.9	67.3	54.1
	100	100.0	0.3	3.3	5.2	28.6	75.8	70.9
Quinacridone Magenta	10	10.1	0.8	2.9	3.1	19.1	57.3	55.2
	20	20.5	0.7	3.2	3.7	17.7	58.3	64.7
	40	40.1	0.6	3.5	3.7	25.9	68.0	68.0
	60	60.3	0.6	3.8	3.4	32.5	77.7	80.0
	80	80.4	0.5	4.1	3.0	28.5	73.4	79.9
	100	100.0	0.3	3.7	2.6	----	----	----
Diarylide Yellow	10	10.3	1.2	3.3	3.1	23.3	62.3	64.1
	20	20.5	1.2	3.5	3.4	15.4	56.2	66.9
	40	40.2	1.2	4.1	3.7	22.3	66.6	56.8
	60	59.7	1.2	4.0	3.5	20.5	61.8	66.0

	80	80.2	1.1	3.7	3.1	15.6	54.4	58.2
	100	100.0	1.1	4.7	3.7	15.6	55.3	55.7
Phthalo Blue (Green Shade)	10	9.9	0.6	2.7	3.6	7.4	38.3	48.6
	20	19.8	0.4	2.8	4.1	20.8	62.9	64.3
	40	39.6	0.4	3.4	5.2	24.4	69.1	69.8
	60	59.9	0.5	5.5	6.8	31.3	74.6	76.5
	80	80.0	0.5	5.9	9.1	46.9	82.3	86.0
	100	100.0	0.6	7.4	12.8	38.5	80.9	83.1
Hansa Yellow Opaque	20	20.1	1.3	3.5	3.3	24.8	57.1	48.7
	40	39.9	1.3	3.7	3.4	10.0	41.2	43.4
	60	60.4	1.2	3.7	3.3	18.7	56.9	57.2
	80	80.5	1.2	3.5	3.2	24.5	67.3	68.1
	100	100.0	1.4	4.4	4.0	27.5	73.5	75.9
Red Oxide	20	20.3	0.7	3.3	4.2	14.9	50.1	59.3
	40	39.8	0.7	4.8	7.0	27.8	63.5	66.4
	60	60.5	0.8	6.8	10.2	26.0	62.2	66.0
	80	79.6	1.0	9.0	14.3	23.6	60.7	64.3
	100	100.0	1.3	12.1	21.5	34.0	79.6	80.7
Phthalo Green (Blue Shade)	10	10.2	0.6	2.8	3.2	12.7	47.6	53.1
	20	20.5	0.6	3.0	3.4	12.9	51.0	58.5
	40	40.2	0.5	3.2	3.5	11.7	45.2	52.2
	60	60.1	0.4	3.2	3.2	23.1	60.1	59.3
	80	79.7	0.3	3.0	2.8	14.9	47.0	54.6
	100	100.0	0.2	2.5	2.3	27.5	71.3	75.6
Pyrrole Red	10	9.9	0.8	3.0	3.4	10.0	43.6	50.0
	20	19.8	0.7	3.0	3.2	13.3	48.6	54.3
	40	40.0	0.6	3.6	3.9	13.2	49.2	55.3
	60	60.1	0.5	3.5	3.4	17.2	50.8	54.8
	80	80.2	0.5	3.5	3.1	18.7	53.6	58.0
	100	100.0	0.5	3.4	3.0	20.3	63.8	65.5
Yellow Ochre	20	20.4	1.0	3.2	3.0	19.6	54.7	62.2
	40	40.2	0.9	3.1	3.0	18.9	51.1	59.0
	60	60.2	0.8	3.0	3.0	24.3	66.8	68.5
	80	80.4	0.7	2.9	2.9	12.9	45.8	51.1
	100	100.0	0.5	2.7	2.9	30.5	71.9	72.3
Ultramarine Blue	20	19.8	0.6	2.5	3.3	22.2	59.2	64.8
	40	40.0	0.4	2.4	3.5	23.8	54.5	57.6
	60	60.0	0.3	2.1	3.3	25.3	59.7	65.5
	80	79.7	0.2	1.7	3.0	22.2	60.2	64.4

	100	100.0	0.6	2.6	4.0	20.7	67.2	80.6
Dioxazine Purple	20	19.9	0.4	2.8	3.6	18.9	56.8	62.6
	40	40.4	0.3	3.0	3.5	13.0	46.8	53.2
	60	59.9	0.3	3.2	3.3	10.8	46.6	54.6
	80	79.7	0.3	2.9	2.5	15.0	49.6	56.0
	100	100.0	0.2	2.4	2.0	19.3	64.2	64.8
Green Gold	20	20.7	1.0	2.8	2.4	9.3	42.3	49.2
	40	40.2	0.8	2.5	1.7	10.4	45.8	47.3
	60	59.8	0.7	2.2	1.5	9.4	37.2	42.9
	80	79.9	0.5	1.7	1.1	7.3	32.5	39.3
	100	100.0	0.2	1.0	1.0	13.0	48.4	54.1
Cerulean Blue	20	20.4	0.7	2.1	2.4	19.5	56.4	59.5
	40	40.2	0.6	1.6	2.1	19.2	54.6	54.9
	60	60.6	0.4	1.2	1.9	19.4	57.1	59.8
	80	80.2	0.3	0.9	1.7	18.7	55.7	61.1
	100	100.0	0.1	0.6	1.5	23.3	63.9	66.9
Titan Buff	20	19.9	1.2	3.2	2.8	12.3	45.3	50.8
	40	40.2	1.2	3.3	2.9	17.6	57.9	57.8
	60	60.4	1.2	3.2	2.9	16.9	56.4	58.3
	80	79.9	1.1	3.2	2.7	17.6	52.3	58.3
	100	100.0	1.0	3.0	2.6	45.2	82.4	83.9
Quinacridone Crimson	20	20.2	0.6	2.6	2.9	15.4	47.8	50.8
	40	40.1	0.4	2.7	2.9	23.6	60.9	62.1
	60	60.4	0.4	2.7	2.5	21.6	63.1	67.1
	80	80.5	0.3	2.6	2.3	26.6	68.6	75.1
	100	100.0	0.2	2.6	2.2	14.3	54.0	55.7
Raw Umber	20	20.4	0.5	2.9	3.8	22.9	58.3	61.5
	40	40.3	0.6	5.4	8.7	32.5	65.5	65.8
	60	59.4	1.2	10.9	19.9	41.5	74.0	79.3
	80	79.6	3.0	19.2	28.5	43.5	76.2	76.5
	100	100.0	0.4	1.9	3.1	19.2	64.8	67.0
Burnt Sienna	20	20.2	0.8	2.7	3.1	23.4	62.9	62.5
	40	40.4	0.6	2.7	3.2	23.2	64.0	63.2
	60	60.6	0.5	2.9	3.4	23.4	66.9	70.5
	80	80.0	0.4	3.4	4.1	27.1	69.8	70.8
	100	100.0	0.3	3.4	4.0	36.0	78.8	78.3
Burnt Umber	20	19.9	0.6	2.0	2.4	23.6	61.8	62.0
	40	40.6	0.4	1.6	2.3	17.6	52.4	52.8
	60	59.9	0.3	1.3	2.1	11.6	46.5	53.9

	80	79.4	0.2	1.0	1.9	13.7	48.4	54.6
	100	100.0	0.0	0.7	1.8	14.4	56.6	66.3
Jenkins Green	20	20.2	0.4	2.3	2.7	26.4	62.1	65.2
	40	39.8	0.3	2.3	2.5	15.4	54.0	60.9
	60	60.0	0.2	2.1	2.3	9.2	42.0	48.9
	80	78.6	0.2	2.1	2.1	21.2	60.1	63.9
	100	100.0	0.1	1.6	1.6	24.3	69.1	73.1
Anthraquinone Blue	20	20.3	0.4	2.5	2.7	24.7	66.5	66.6
	40	40.2	0.3	2.6	2.5	18.3	55.1	60.3
	60	59.9	0.2	2.6	2.4	12.9	51.1	54.5
	80	80.2	0.2	2.4	1.8	18.3	58.7	61.2
	100	100.0	0.1	2.0	1.5	25.6	70.4	70.6
Permanent Green Light	20	19.8	0.8	2.8	3.2	31.8	69.6	67.9
	40	39.5	0.6	2.8	3.4	23.6	63.3	61.0
	60	59.6	0.5	3.4	4.0	22.7	65.8	69.3
	80	78.4	0.4	2.6	3.2	33.4	77.1	80.2
	100	100.0	0.2	1.4	2.1	29.3	72.4	76.1
Naphthol Red Medium	20	20.2	0.6	2.7	2.9	14.9	43.1	49.5
	40	40.4	0.5	2.7	2.8	13.2	47.4	53.8
	60	59.6	0.4	2.4	2.2	16.0	52.8	54.7
	80	79.5	0.3	2.2	1.8	8.8	42.5	49.2
	100	100.0	0.2	2.6	1.6	29.0	76.9	74.9
Raw Sienna	20	20.7	1.0	3.1	3.0	19.5	53.8	55.2
	40	40.1	0.9	3.2	3.1	16.7	55.7	59.6
	60	59.6	0.8	3.2	3.0	19.9	58.4	61.0
	80	79.8	0.6	3.0	2.7	8.1	40.4	48.5
	100	100.0	0.4	2.8	2.7	33.4	75.9	76.2
Paynes Gray	20	19.9	0.5	2.0	2.4	7.6	40.1	44.8
	40	39.7	0.3	1.8	2.4	13.2	45.3	48.5
	60	60.2	0.2	1.4	2.1	13.4	51.9	53.8
	80	79.3	0.1	1.0	1.7	9.0	40.0	47.6
	100	100.0	0.0	0.6	1.4	27.4	71.8	74.1

A.4. Recipes of Secondary Mixtures

Table A-IV shows the recipes of secondary mixtures used for the evaluation of paint mixing performance.

Table A-IV. Recipes of Secondary Mixtures.

#	Target Specimen	Weight [g]				Concentration [%]			
		TiO ₂	Y*	B*	R*	TiO ₂	Y	B	R
1	B+Y (3:1) [0% TiO ₂]	0	1.25	3.75	0	0.00	25.00	75.00	0.00
2	B+Y (1:1) [0% TiO ₂]	0	2.5	2.5	0	0.00	50.00	50.00	0.00
3	B+Y (1:3) [0% TiO ₂]	0	3.74	1.3	0	0.00	74.21	25.79	0.00
4	Y+R (3:1) [0% TiO ₂]	0	3.75	0	1.25	0.00	75.00	0.00	25.00
5	Y+R (1:1) [0% TiO ₂]	0	2.5	0	2.51	0.00	49.90	0.00	50.10
6	Y+R (1:3) [0% TiO ₂]	0	1.27	0	3.75	0.00	25.30	0.00	74.70
7	R+B (3:1) [0% TiO ₂]	0	0	1.24	3.76	0.00	0.00	24.80	75.20
8	R+B (1:1) [0% TiO ₂]	0	0	2.5	2.5	0.00	0.00	50.00	50.00
9	R+B (1:3) [0% TiO ₂]	0	0	3.77	1.24	0.00	0.00	75.25	24.75
10	B+Y (1:1) [20% TiO ₂]	1.03	2.01	1.98	0	20.52	40.04	39.44	0.00
11	Y+R (1:1) [20% TiO ₂]	1	2.01	0	1.99	20.00	40.20	0.00	39.80
12	R+B (1:1) [20% TiO ₂]	1.02	0	1.99	1.99	20.40	0.00	39.80	39.80
13	B+Y (1:1) [40% TiO ₂]	2	1.5	1.5	0	40.00	30.00	30.00	0.00
14	Y+R (1:1) [40% TiO ₂]	2	1.52	0	1.54	39.53	30.04	0.00	30.43
15	R+B (1:1) [40% TiO ₂]	2.04	0	1.5	1.46	40.80	0.00	30.00	29.20
16	B+Y (1:1) [60% TiO ₂]	3.01	1.03	0.99	0	59.84	20.48	19.68	0.00
17	Y+R (1:1) [60% TiO ₂]	3.02	0.98	0	1.01	60.28	19.56	0.00	20.16
18	R+B (1:1) [60% TiO ₂]	3	0	1	1	60.00	0.00	20.00	20.00
19	B+Y (1:1) [80% TiO ₂]	4	0.52	0.54	0	79.05	10.28	10.67	0.00
20	Y+R (1:1) [80% TiO ₂]	4	0.51	0	0.49	80.00	10.20	0.00	9.80
21	R+B (1:1) [80% TiO ₂]	4.01	0	0.51	0.49	80.04	0.00	10.18	9.78
22	B+Y (3:1) [50% TiO ₂]	2.51	0.63	1.87	0	50.10	12.57	37.33	0.00
23	B+Y (1:3) [50% TiO ₂]	2.49	1.88	0.65	0	49.60	37.45	12.95	0.00
24	Y+R (3:1) [50% TiO ₂]	2.49	1.87	0	0.65	49.70	37.33	0.00	12.97
25	Y+R (1:3) [50% TiO ₂]	2.51	0.62	0	1.88	50.10	12.38	0.00	37.53
26	R+B (3:1) [50% TiO ₂]	2.49	0	0.63	1.88	49.80	0.00	12.60	37.60
27	R+B (1:3) [50% TiO ₂]	2.52	0	1.93	0.62	49.70	0.00	38.07	12.23

* Y : Hansa Yellow Opaque, B : Ultramarine Blue, R : Pyrrole Red

B. Matlab Source Codes

Handling an enormous amount of measured data all at once is a complicated process so that it might doubt accuracy when doing mathematical calculations on Matlab. That is, without revealing the complexity, a fundamental mistake in data manipulation will make the numerical dataset of our achievement unreliable. The most thinkable issue of data manipulation is on how to import a raw data from a spectrophotometer to your own Matlab source codes. Unfortunately, the situation is not simple because different instrument applications generate different format files as outputs. To simplify data manipulation, therefore, we developed the Matlab source codes applicable for an instrument application, “GretagMacbeth ProPalette (*used for GretagMacbeth ColorEye XTH*)”. The output converted from each of the raw files by a Matlab code execution is an array structure in author’s designed format. The details of the unique format developed for this research is to be introduced in Appendix C.1.

B.1. Importing Measurement Data from ProPalette to Matlab

The function, “*read_XTH*”, is to import a raw data of measurement from ProPalette into the Matlab data format. Before executing this function, you must confirm if a raw data has been stored correctly in your work space as a text file. The reason is due to multiple choices available for saving in different formats. The introduction of storing a raw data in ProPalette is the following:

[How To] Saving a measurement data as a spreadsheet in ProPalette.

Source Code (“*read_XTH*”) :

```
function xthformat = read_XTH(void)
% / xthformat = read_XTH(void) /
%
% This function is to read any ascii data files of ColorEye XTH in a current
% directory. After reading, the files is to be converted into MATLAB matrixies
% in Workspace with their shorten names without extension.
% (e.g. 'abc.txt' -> 'abc')
%
% //- How to use -//
% Step1: Make a new directory in your computer and put ascii files of
%       ProPalette into the directory.
% Step2: A current directory in MATLAB should be pointed to the directory
%       in which ProPalette datas exist.
% Step3: Run the command 'read_XTH' in Command Window.
%
% //- Structure of the data format -//
% It is a structure.
%
% ex) 'abc.txt' ascii file
%
% EDU>> abc                -> see the structure of 'abc' data
% EDU>> abc.header          -> see the header info of 'abc' data
% EDU>> abc.wave             -> see the range of wavelength of 'abc' data
% EDU>> abc.data             -> see 'all reflectances(sample)' [%]
% EDU>> abc.data(xx,:)      -> see 'xx-th reflectance' [%]
%
% //- How to plot with this data format -//
% ex) 'abc' structure
%
% EDU>> plot(abc.wave, abc.data)    <--- all data
% EDU>> plot(abc.wave, abc.data(xx,:)) <--- xx-th data
%
%
% Good luck!
% Produced by Yoshio Okumura, 2004

% Get the list of filenames in current directory.
sD = dir('');

% Check the number of the total files.
```



```

famount = size(sD(:),1);

% Initialization of a variable
count = 1;

% This loop is to pick up the filenames of the only text file, not directory names.
for i = 1:famount
    if (sD(i).isdir)
        continue
    else
        % The picked-up file is to be stocked in the cell array 'flist'.
        check1 = findstr(sD(i).name, '.txt');
        check2 = findstr(sD(i).name, '.TXT');

        if (~size(check1) & ~size(check2))
            continue
        else
            flist{count} = sD(i).name;
            count = count + 1;
        end
    end
end

% *****
%
% // Initial Value Settings //
%
% *****

% The wavelength range of a XTH data
waveRange = 360:10:750;
% The number of the column lines for a single dataset ('Name' + 'Spec' + 'Ref')
colNum = 34;
% The area of the trial name and spec of a column dataset
nsArea = 1:2;
% The area of the reflectance factor of a column dataset
dataArea = 3:34;

% *****
%
% // Read each of dataset //
%
% *****

```

```

for j = 1:(count-1)
    % -----
    % -- Data Input --
    % -----

    [textStrings] = textread(flist{j}, '%s');

    numData = (size(textStrings,1) - 4) / 9;    % Number of measured data %
    dataArray = zeros(numData,40);             % Allocate the array space for measured data %
    dataLabel = cell(0);                       % Initialize Cell Array %

    for xx = 1:numData
        yy = (xx-1) * 9 + 4;
        dataLabel{xx} = textStrings{yy};
        data1 = sscanf(textStrings{yy+2}, '%f,%f,%f,%f,%f,%f');
        data2 = sscanf(textStrings{yy+3}, '%f,%f,%f,%f,%f,%f');
        data3 = sscanf(textStrings{yy+4}, '%f,%f,%f,%f,%f,%f');
        data4 = sscanf(textStrings{yy+5}, '%f,%f,%f,%f,%f,%f');
        data5 = sscanf(textStrings{yy+6}, '%f,%f,%f,%f,%f,%f');
        data6 = sscanf(textStrings{yy+7}, '%f,%f,%f,%f,%f,%f');
        data7 = sscanf(textStrings{yy+8}, '%f,%f,%f,%f');
        dataArray(xx,:) = [data1',data2',data3',data4',data5',data6',data7'];
    end

    % -----
    % -- Data Output --
    % -----

    % Construction of the structure of header and data array
    dataXTH.header = {textStrings{1};textStrings{2};textStrings{3}};
    dataXTH.label = dataLabel;
    dataXTH.wave = waveRange;
    dataXTH.data = dataArray ./ 100.0;

    % Transfer the structure onto Work space
    [wsName, trash] = strtok(flist{j}, './');

    % If wsName is numerical, change into 'z' + wsName
    numornot = regexp(wsName, '^d');

    if (numornot)
        wsName = strcat('z', wsName);
    end

```

```

else
    wsName = wsName;
end

% stock a matrix name
mlist{j} = wsName;

% transfer a matrix to Workspace
assignin('base',wsName,dataXTH);
end

% return the list of filenames and matrix names
xthformat.count = count - 1;
xthformat.flist = flist;
xthformat.mlist = mlist;

```

C. Data Formats of Artist Material Database

The achievement of this paint research including spectral dataset (e.g. reflectance, unit K and unit S) was recorded as Matlab data files. For this purpose, we prepared two data formats, each of which was designed for storing reflectance factors and a series of unit k and unit s .

C.1. Reflectance Factors in Matlab

The Matlab data format of reflectance factors is a cell structure containing “header”, “label”, “wave”, and “data”. The data format can be generated by executing the functions, “*read_XTH*” that is introduced in the previous chapter. The details of each component is summarized in Table C-I.

Table C-I. Data formats of the measured reflectance factors for the database.

Component	Data type	Contents	Example
header	cell array	name of an instrument application name of a raw data file (imported)	‘propalette’
label	cell array	name of each of the paint names	‘TiO2’
wave	double array	sampling wavelengths of an instrument	[360, 370, ..., 750]
data	double array	reflectance factor (raw vector)	[0.92, 0.93, ..., 0.86]

Now, let a “*VarnishSPIN*” be a data structure of reflectance factors that follows the format. And then, we assume that the number, ‘7’, indicates the address of “Carbon Black 40%” as an example. When you try to access to the name of and the reflectance factor (vector) of “Carbon Black 40%” in the structure, you are required to type in command window or write in your source code on Matlab:

[*Commands*]

VarnishSPIN.label{7} : paint name (or measurement title)
(e.g. 'CarbonBlack_40')

VarnishSPIN.data(7,:) : reflectance factor (raw vector)
(e.g. [0.0346, 0.0355, ..., 0.0601])

Another possible occasion to access the structure is the case of finding a target reflectance factor by searching a specific paint name. The case is that, you do not know the address of a target paint except paint's name and its concentration. In such a case, you can use some keywords in String type on Matlab to get the address of the target. With regard to the previous example, the keywords can be "CarbonBlack" (no space) and "40". Hence, you can obtain the address by using an example Matlab code as the following:

[*Commands*]

```
key1 = 'CarbonBlack';  
key2 = '40';  
map1 = cellfun('isempty',regexp(VarnishSPIN.label,key1));  
map2 = cellfun('isempty',regexp(VarnishSPIN.label,key2));  
map3 = map1 + map2;  
address = find(map3 == 0);      % the address of 'Carbon Black 40%' %
```

For all the paint tint ladders, the measured reflectance factors by a GretagMacbeth ColorEye based on varnish application and specular integration are stored with the following names. All the data files are the achievements of this paint project.

Table C-II. Data files of the measured reflectance factors for the database.

Filename	Target specimen	Specular integration
GOLDEN_non-varnish_SPIN.mat	non-varnished	SPIN
GOLDEN_non-varnish_SPEX.mat	non-varnished	SPEX
GOLDEN_varnish_SPIN.mat	varnished	SPIN
GOLDEN_varnish_SPEX.mat	varnished	SPEX

C.2. Unit k and Unit s in Matlab

The Matlab data format of unit k and unit s that were obtained by global optimization is a cell structure containing “colKandS”, “whKandS”, “K1andK2”, “stKoS”, “stRm”, “de00”, “RMS”, and “wave”. We assume for explanation that, “*KSDataset*” is a cell structure stored with following the data format as an example. The details of each component in this format is summarized in Table C-III,

Table C-III. Data formats of the optimized paint mixing model coefficients for the database.

Component	Data type	Contents*	How to access*
colKandS	cell array	optimized unit k for i th paint optimized unit s for i th paint	KSDataset.colKandS{ i }(1,:) KSDataset.colKandS{ i }(2,:)
whKandS	cell array	unit k for Titanium White unit s for Titanium White (=1)	KSDataset.whKandS{ i }(1,:) KSDataset.whKandS{ i }(2,:)
K1andK2	double array	optimized K_1 optimized K_2	KSDataset.K1andK2(1) KSDataset.K1andK2(2)
stKoS	cell array	measured K/S for i th paint estimated K/S for i th paint	KSDataset.stKoS{ i }.org KSDataset.stKoS{ i }.est
stRm	cell array	measured R_m for i th paint estimated R_m for i th paint	KSDataset.stRm{ i }.org KSDataset.stRm{ i }.est
de00	cell array	estimation errors in CIEDE2000 for i th paint	KSDataset.de00{ i }
RMS	cell array	estimation errors in R_m RMS for i th paint	KSDataset.RMS{ i }
wave	double array	sampling wavelengths of an instrument	KSDataset.wave

* ‘ i ’ indicates the indexing number of a target paint.

where the indexing number of each paint is listed in Table C-IV.

Table C-IV. The indexing number of each paint for the database.

Index	Paint name	Index	Paint name
1	Carbon Black	14	Green Gold
2	Cobalt Blue	15	Cerulean Blue
3	Chromium Oxide Green	16	Titan Buff
4	Quinacridone Magenta	17	Quinacridone Crimson
5	Diarylide Yellow	18	Raw Umber
6	Phthalo Blue (Green Shade)	19	Burnt Sienna
7	Hansa Yellow Opaque	20	Burnt Umber
8	Red Oxide	21	Jenkins Green
9	Phthalo Green (Blue Shade)	22	Anthraquinone Blue
10	Pyrrole Red	23	Permanent Green Light
11	Yellow Ochre	24	Naphthol Red Medium
12	Ultramarine Blue	25	Raw Sienna
13	Dioxazine Purple	26	Payness Gray

Table C-V shows the list of the data files that store a series of unit k and unit s for all the paint set. The optimized unit ks and unit ss result from several types of global optimization.

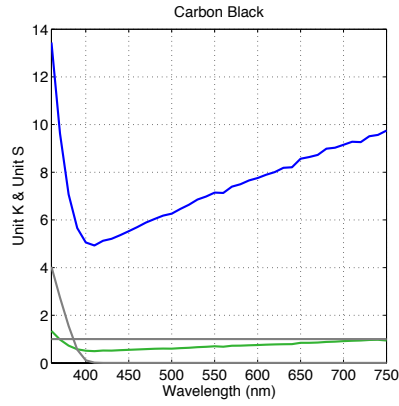
Table C-V. Data files of the optimized unit ks and unit ss for the database.

Filename	Optimization method	Minimizing criterion
goptlnOUT.mat	non-negative least square	R_m RMS
goptlinMahOUT.mat*	non-negative least square	R_m RMS
goptOUT1.mat	non-linear optimization	R_m RMS
goptOUT2.mat	non-linear optimization	R_m weighted RMS by the diagonal of Matrix R
goptOUT3.mat	non-linear optimization	R_m weighted RMS by Viggiano's

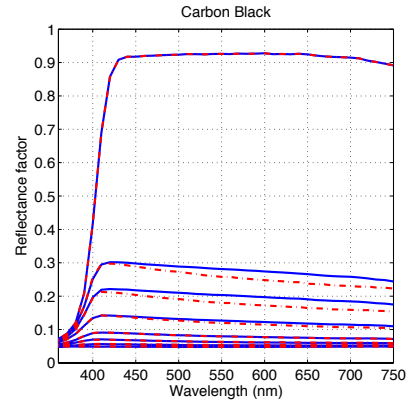
* The file resulted from the non-negative least-square method using only tint, maximum chroma, and massstone.

D. Plots of the Spectral Database of Artist Paint Materials

The optimized values of the unit k s and unit s s for all the acrylic paints based on the two-constant Kubelka-Munk theory are plotted in this section. The information of the plots can be considered to be useful to confirm the absorption and scattering properties of a paint. Each plot of unit k , unit s , and estimation performance for a single paint is shown for the case of a global optimization by a non-negative least square method. The original data of the plots is stored in a Matlab M-file, “*goptlnOUT.mat*”.

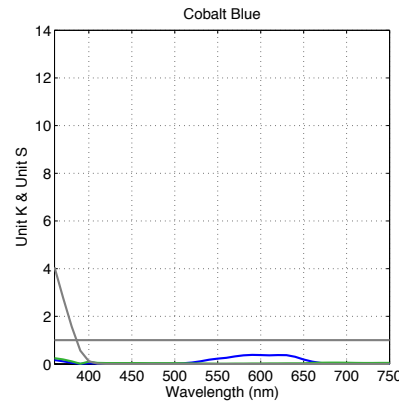


† blue line: unit k , green line: unit s

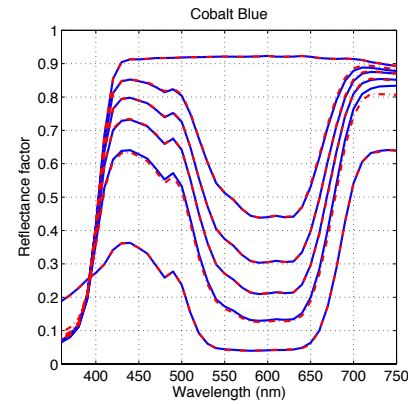


†† blue line: measurement, red line: estimation

(a) Carbon Black

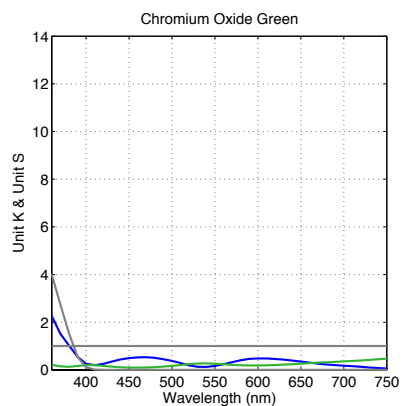


† blue line: unit k , green line: unit s

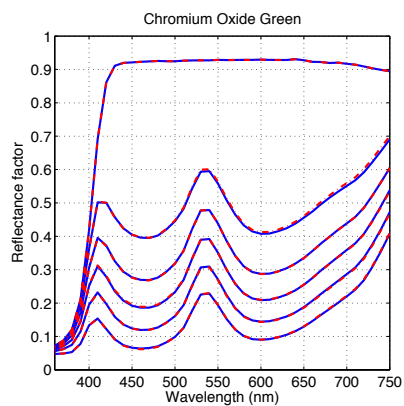


†† blue line: measurement, red line: estimation

(b) Cobalt Blue

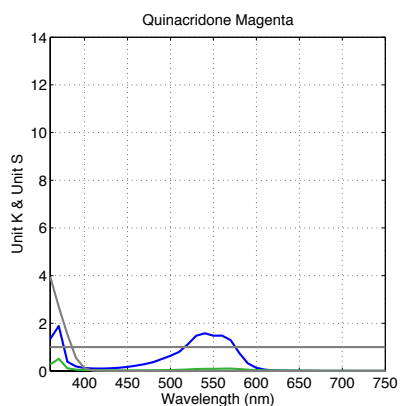


† blue line: unit k, green line: unit s

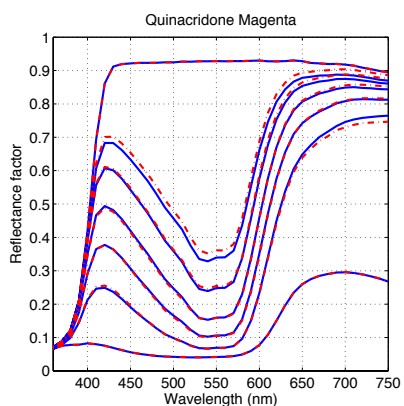


†† blue line: measurement, red line: estimation

(c) Chromium Oxide Green

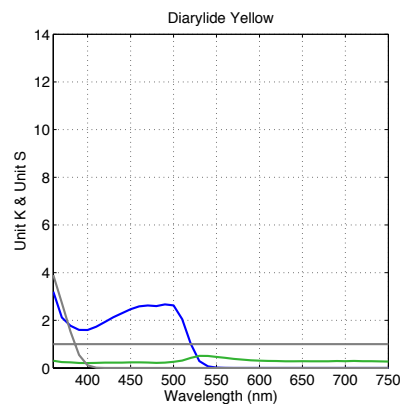


† blue line: unit k, green line: unit s

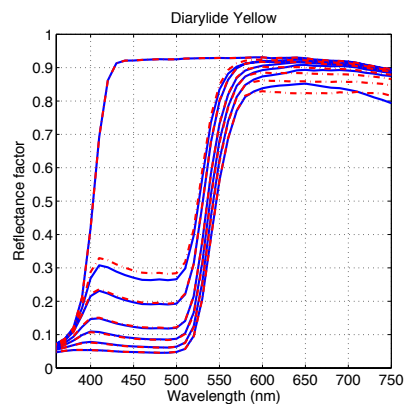


†† blue line: measurement, red line: estimation

(d) Quinacridone Magenta

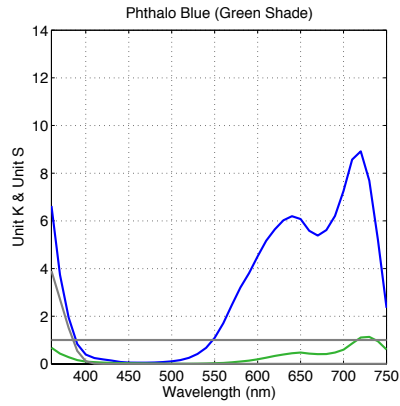


† blue line: unit k, green line: unit s

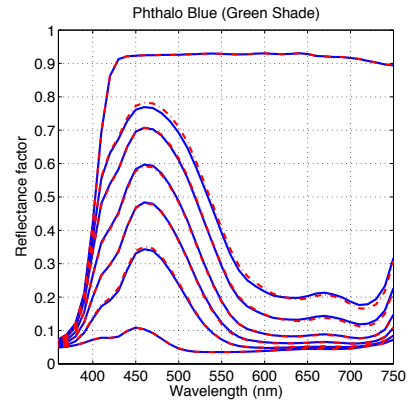


†† blue line: measurement, red line: estimation

(e) Diarylide Yellow

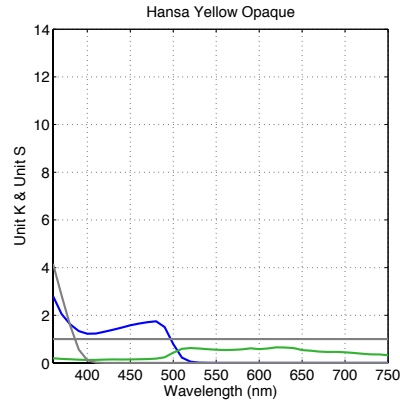


$\dagger$ blue line: unit k, green line: unit s

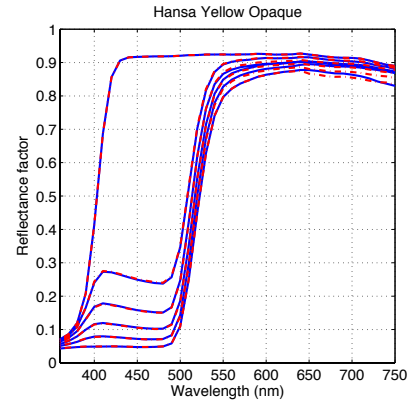


$\dagger\dagger$ blue line: measurement, red line: estimation

(f) Phthalo Blue (Green Shade)

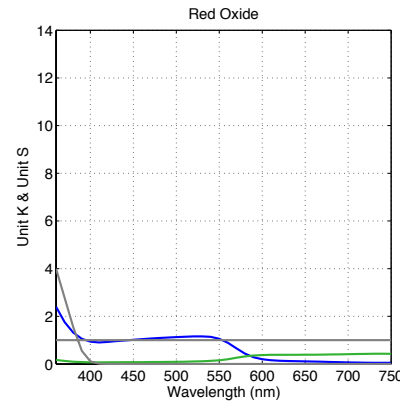


$\dagger$ blue line: unit k, green line: unit s

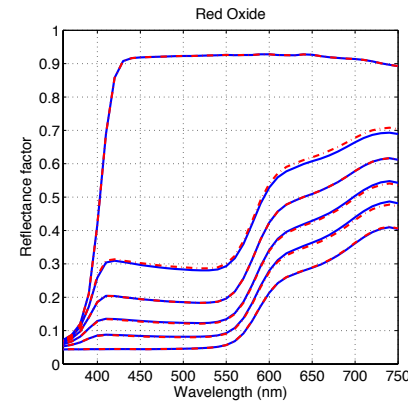


$\dagger\dagger$ blue line: measurement, red line: estimation

(g) Hansa Yellow Opaque

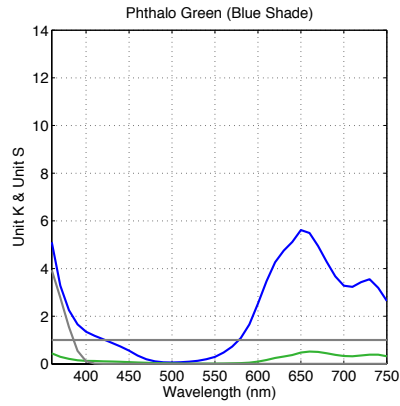


$\dagger$ blue line: unit k, green line: unit s

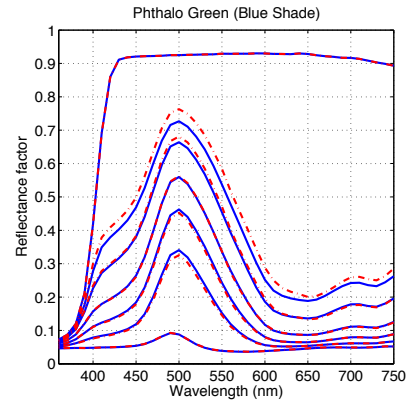


$\dagger\dagger$ blue line: measurement, red line: estimation

(h) Red Oxide

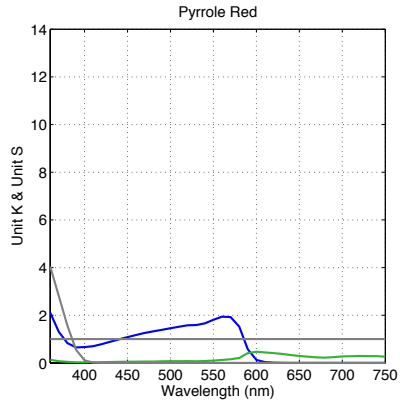


† blue line: unit k, green line: unit s

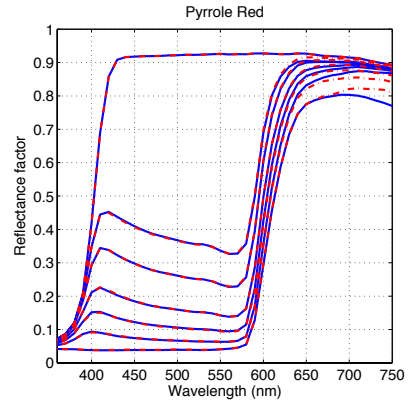


†† blue line: measurement, red line: estimation

(i) Phthalo Green (Blue Shade)

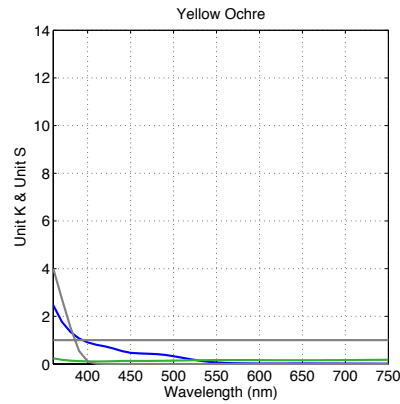


† blue line: unit k, green line: unit s

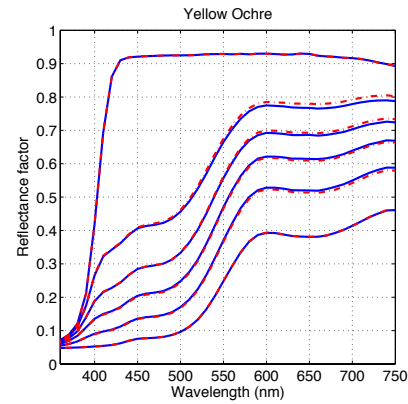


†† blue line: measurement, red line: estimation

(j) Pyrrole Red

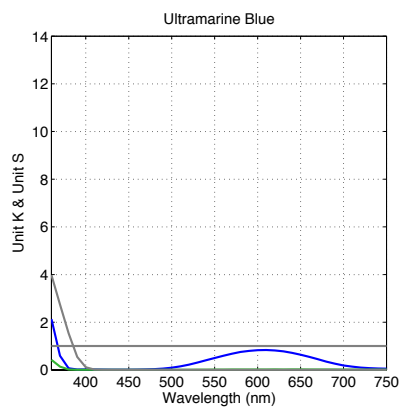


† blue line: unit k, green line: unit s

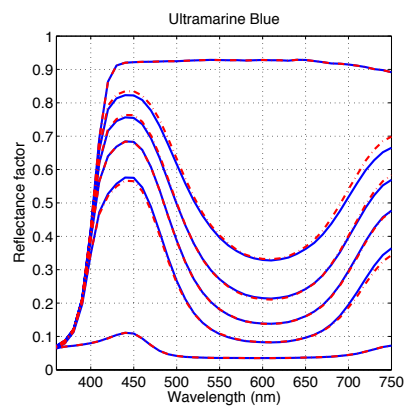


†† blue line: measurement, red line: estimation

(k) Yellow Ochre

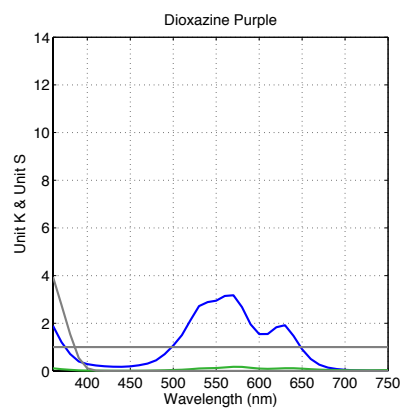


† blue line: unit k, green line: unit s

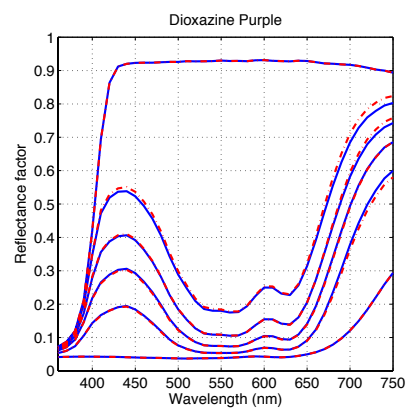


†† blue line: measurement, red line: estimation

(l) Ultramarine Blue

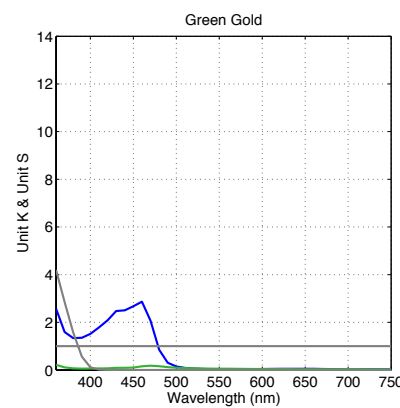


† blue line: unit k, green line: unit s

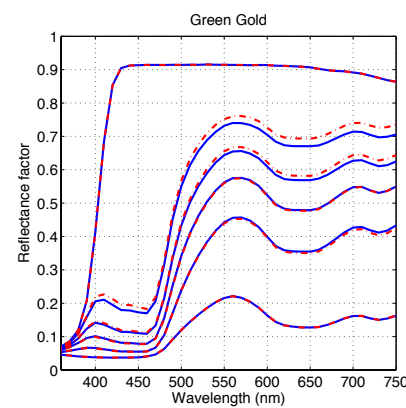


†† blue line: measurement, red line: estimation

(m) Dioxazine Purple

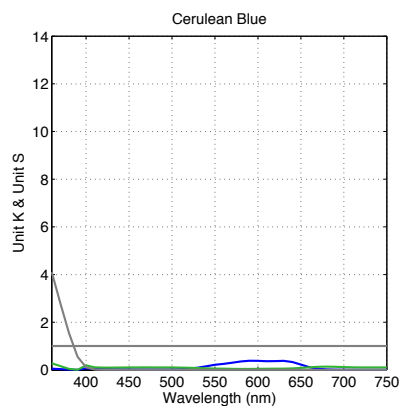


† blue line: unit k, green line: unit s

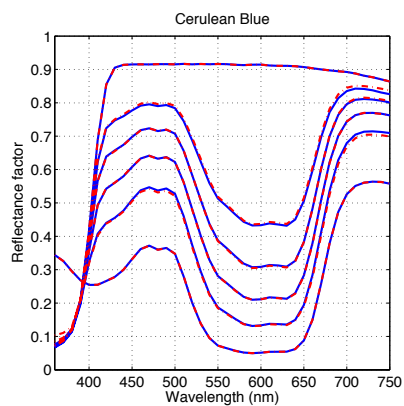


†† blue line: measurement, red line: estimation

(n) Green Gold

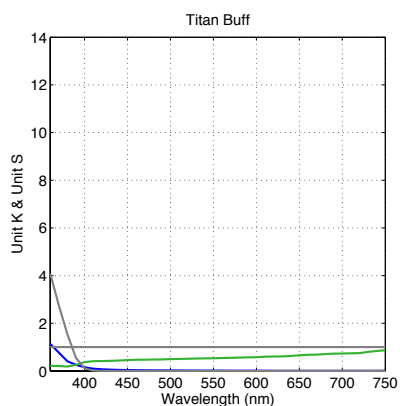


† blue line: unit k, green line: unit s

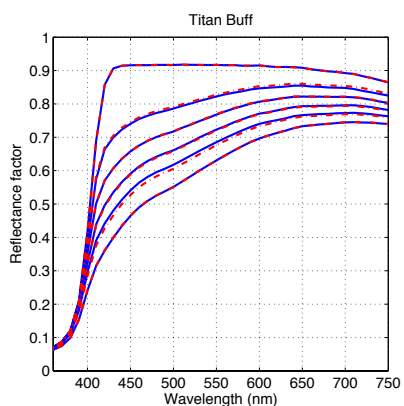


†† blue line: measurement, red line: estimation

(o) Cerulean Blue

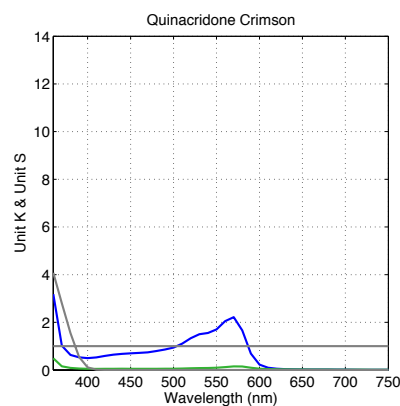


† blue line: unit k, green line: unit s

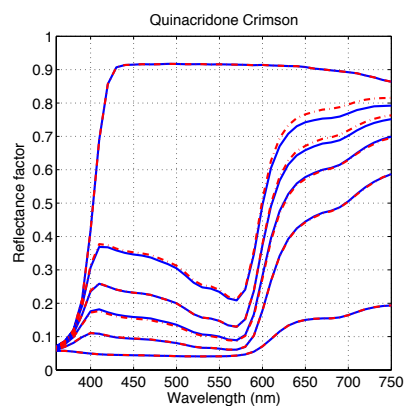


†† blue line: measurement, red line: estimation

(p) Titan Buff

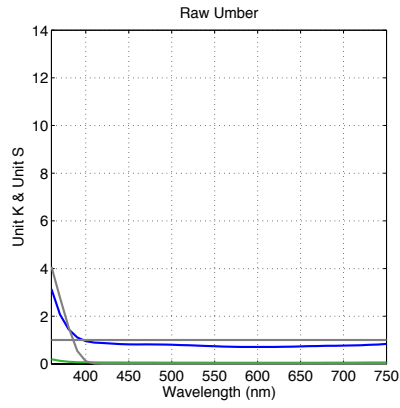


† blue line: unit k, green line: unit s

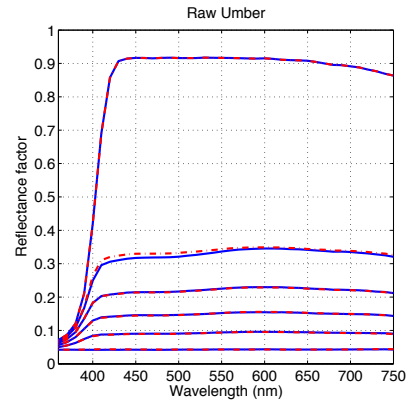


†† blue line: measurement, red line: estimation

(q) Quinacridone Crimson

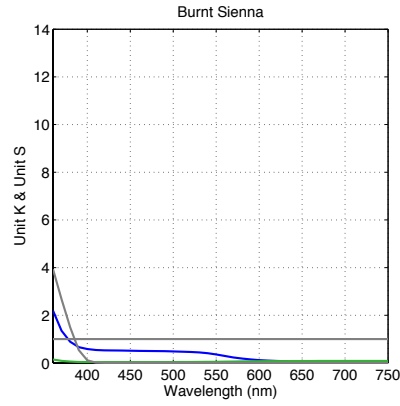


† blue line: unit k, green line: unit s

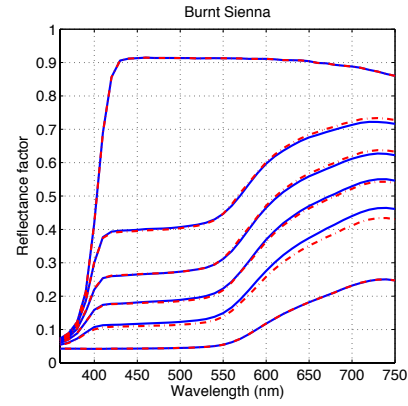


†† blue line: measurement, red line: estimation

(r) Raw Umber

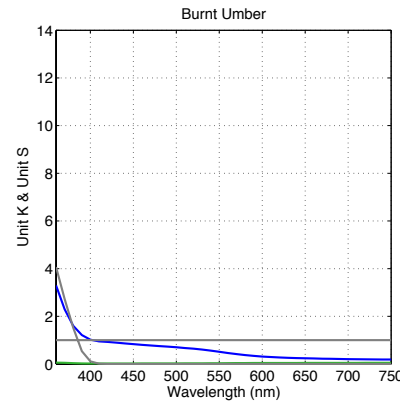


† blue line: unit k, green line: unit s

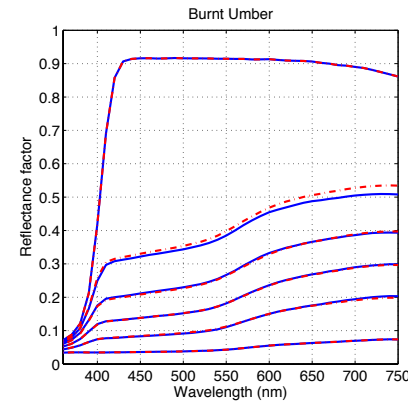


†† blue line: measurement, red line: estimation

(s) Burnt Sienna

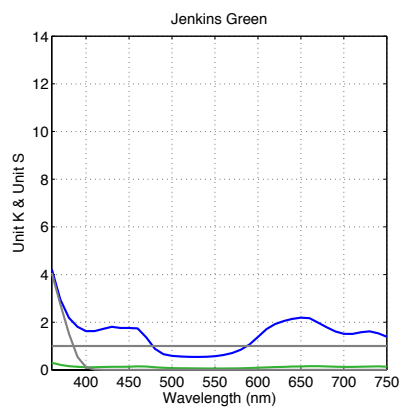


† blue line: unit k, green line: unit s

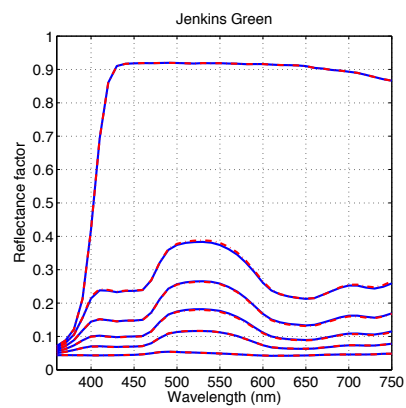


†† blue line: measurement, red line: estimation

(t) Burnt Umber

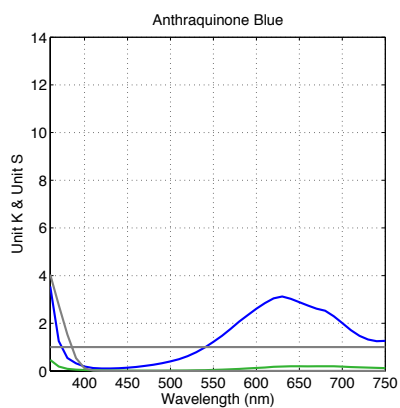


† blue line: unit k, green line: unit s

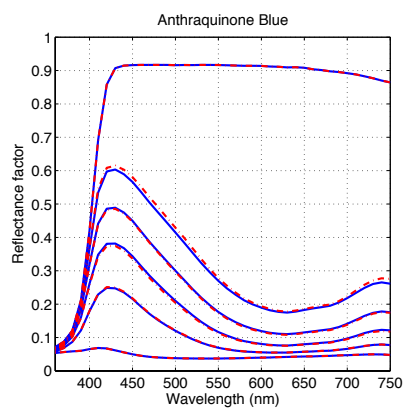


†† blue line: measurement, red line: estimation

(u) Jenkins Green

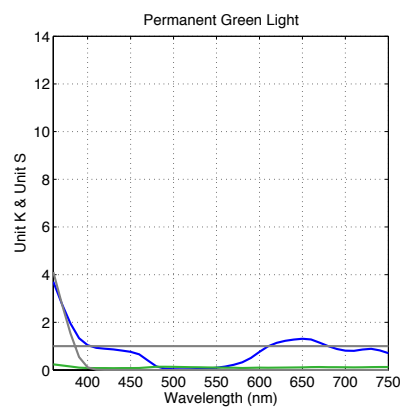


† blue line: unit k, green line: unit s

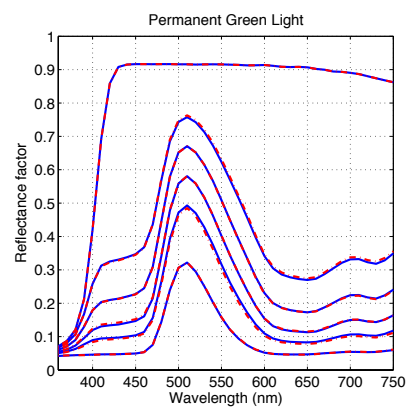


†† blue line: measurement, red line: estimation

(v) Anthraquinone Blue

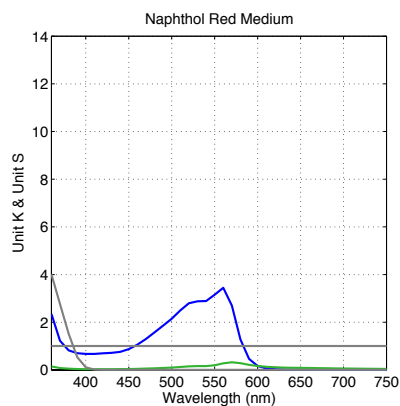


† blue line: unit k, green line: unit s

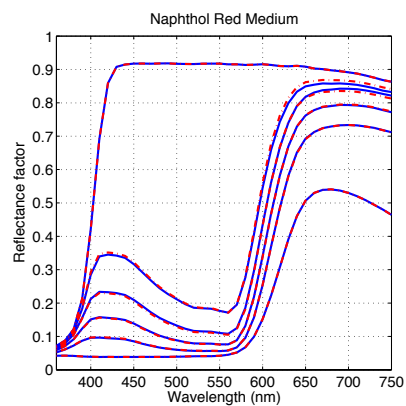


†† blue line: measurement, red line: estimation

(w) Permanent Green Light

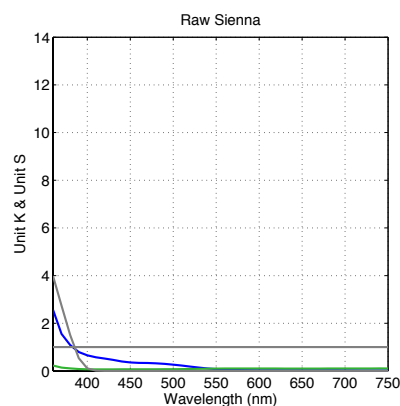


† blue line: unit k, green line: unit s

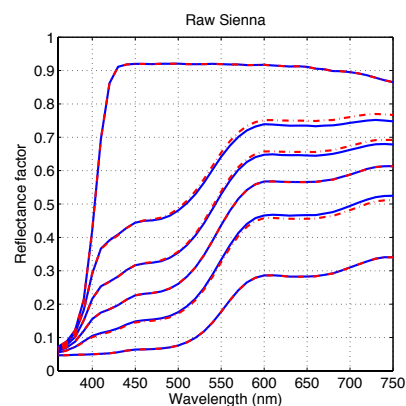


†† blue line: measurement, red line: estimation

(x) Naphthol Red Medium

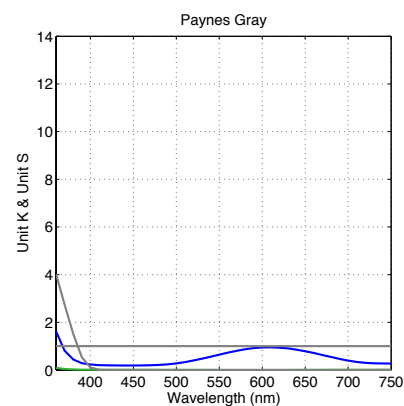


† blue line: unit k, green line: unit s

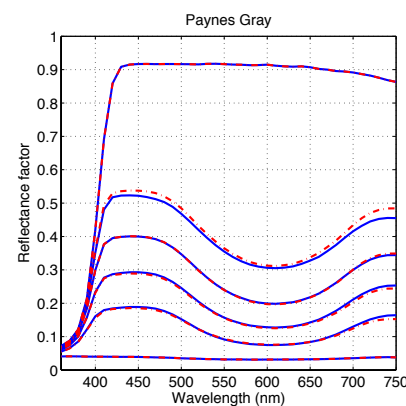


†† blue line: measurement, red line: estimation

(y) Raw Sienna



† blue line: unit k, green line: unit s



†† blue line: measurement, red line: estimation

(z) Paynes Gray

E. Spreadsheets of Paint Database Performances

The details of the paint database performances, which were evaluated after getting the optimal paint mixing model coefficients, are shown in this section. The target optimization methods are:

- (1) Local optimization (least square method for each paint)*
- (2) Global optimization (least square method for the entire paints)*
- (3) Global optimization (least square method for the entire paints, only using tint, max chroma, and masstone)*
- (4) Global optimization (minimizing R_m RMS)*
- (5) Global optimization (minimizing R_m weighted RMS by the diagonal of Matrix R)*
- (6) Global optimization (minimizing R_m weighted RMS by Viggiano's weight)*

Table E-I. Spectral characterization performances by local optimization.

#	Paint name	RMS (Reflectance)		CIEDE2000		K1	K2
		mean	max	mean	max		
1	Carbon Black	0.003	0.010	0.77	2.05	0.026	0.551
2	Cobalt Blue	0.004	0.013	0.15	0.71	0.040	0.599
3	Chromium Oxide Green	0.002	0.006	0.19	0.40	0.031	0.601
4	Quinacridone Magenta	0.006	0.017	0.45	0.94	0.034	0.623
5	Diarylide Yellow	0.007	0.028	0.20	0.49	0.032	0.610
6	Phthalo Blue (Green Shade)	0.003	0.011	0.35	0.98	0.031	0.601
7	Hansa Yellow Opaque	0.003	0.007	0.11	0.24	0.029	0.597
8	Red Oxide	0.003	0.009	0.20	0.48	0.033	0.614
9	Phthalo Green (Blue Shade)	0.007	0.023	0.71	1.50	0.035	0.643
10	Pyrrole Red	0.004	0.013	0.19	0.44	0.030	0.600
11	Yellow Ochre	0.004	0.010	0.19	0.39	0.031	0.608
12	Ultramarine Blue	0.004	0.013	0.18	0.41	0.035	0.650
13	Dioxazine Purple	0.005	0.012	0.21	0.46	0.031	0.613
14	Green Gold	0.006	0.017	0.31	0.62	0.034	0.650
15	Cerulean Blue	0.003	0.007	0.14	0.29	0.046	0.616
16	Titan Buff	0.003	0.009	0.14	0.50	0.035	0.620
17	Quinacridone Crimson	0.005	0.016	0.29	0.66	0.031	0.609
18	Raw Umber	0.002	0.009	0.29	1.10	0.030	0.604
19	Burnt Sienna	0.006	0.016	0.42	1.39	0.031	0.617
20	Burnt Umber	0.004	0.016	0.26	0.79	0.030	0.599
21	Jenkins Green	0.001	0.004	0.18	0.42	0.030	0.605
22	Anthraquinone Blue	0.003	0.010	0.32	0.84	0.030	0.604
23	Permanent Green Light	0.003	0.007	0.29	0.84	0.031	0.616
24	Naphthol Red Medium	0.003	0.009	0.19	0.48	0.030	0.600
25	Raw Sienna	0.005	0.011	0.27	0.61	0.031	0.608
26	Paynes Gray	0.003	0.012	0.23	0.60	0.031	0.650

Table E-II. Spectral characterization performances by global optimization (least square method).

#	Paint name	RMS (Reflectance)		CIEDE2000		K1	K2
		mean	max	mean	max		
1	Carbon Black	0.007	0.023	1.01	3.24	0.031	0.650
2	Cobalt Blue	0.010	0.036	0.15	0.30		
3	Chromium Oxide Green	0.002	0.006	0.24	0.72		
4	Quinacridone Magenta	0.006	0.017	0.40	1.11		
5	Diarylide Yellow	0.003	0.014	0.20	0.77		
6	Phthalo Blue (Green Shade)	0.004	0.009	0.31	0.71		
7	Hansa Yellow Opaque	0.003	0.006	0.11	0.22		
8	Red Oxide	0.003	0.012	0.18	0.61		
9	Phthalo Green (Blue Shade)	0.007	0.030	0.54	1.88		
10	Pyrrole Red	0.004	0.009	0.25	0.70		
11	Yellow Ochre	0.004	0.010	0.18	0.45		
12	Ultramarine Blue	0.005	0.015	0.21	0.53		
13	Dioxazine Purple	0.005	0.013	0.15	0.48		
14	Green Gold	0.006	0.018	0.23	0.64		
15	Cerulean Blue	0.013	0.058	0.13	0.33		
16	Titan Buff	0.004	0.008	0.18	0.31		
17	Quinacridone Crimson	0.005	0.014	0.27	0.63		
18	Raw Umber	0.002	0.006	0.29	1.10		
19	Burnt Sienna	0.007	0.014	0.43	0.79		
20	Burnt Umber	0.003	0.012	0.29	0.65		
21	Jenkins Green	0.001	0.002	0.17	0.28		
22	Anthraquinone Blue	0.003	0.007	0.31	0.55		
23	Permanent Green Light	0.003	0.008	0.33	0.73		
24	Naphthol Red Medium	0.003	0.008	0.21	0.45		
25	Raw Sienna	0.005	0.012	0.28	0.70		
26	Paynes Gray	0.004	0.014	0.28	0.54		

Table E-III. Spectral characterization performances by global optimization (least square - max chroma).

#	Paint name	RMS (Reflectance)		CIEDE2000		K1	K2
		mean	max	mean	max		
1	Carbon Black	0.007	0.021	1.18	3.60	0.000	0.599
2	Cobalt Blue	0.008	0.022	0.22	0.92		
3	Chromium Oxide Green	0.002	0.007	0.18	0.41		
4	Quinacridone Magenta	0.008	0.026	0.54	1.87		
5	Diarylide Yellow	0.012	0.025	0.39	0.81		
6	Phthalo Blue (Green Shade)	0.004	0.014	0.40	1.29		
7	Hansa Yellow Opaque	0.003	0.009	0.13	0.32		
8	Red Oxide	0.005	0.012	0.44	1.03		
9	Phthalo Green (Blue Shade)	0.009	0.039	0.72	2.36		
10	Pyrrole Red	0.005	0.017	0.22	0.65		
11	Yellow Ochre	0.004	0.011	0.22	0.58		
12	Ultramarine Blue	0.010	0.025	0.31	0.91		
13	Dioxazine Purple	0.006	0.016	0.26	0.65		
14	Green Gold	0.007	0.024	0.28	0.94		
15	Cerulean Blue	0.011	0.041	0.19	0.52		
16	Titan Buff	0.003	0.009	0.14	0.52		
17	Quinacridone Crimson	0.006	0.023	0.43	1.36		
18	Raw Umber	0.004	0.009	0.62	1.41		
19	Burnt Sienna	0.013	0.030	0.74	1.75		
20	Burnt Umber	0.004	0.020	0.26	1.01		
21	Jenkins Green	0.002	0.007	0.20	0.72		
22	Anthraquinone Blue	0.004	0.018	0.45	1.72		
23	Permanent Green Light	0.009	0.025	0.57	1.34		
24	Naphthol Red Medium	0.003	0.010	0.21	0.53		
25	Raw Sienna	0.005	0.017	0.31	1.02		
26	Paynes Gray	0.006	0.024	0.33	1.15		

Table E-IV. Spectral characterization performances by global optimization (R_m RMS).

#	Paint name	RMS (Reflectance)		CIEDE2000		K1	K2
		mean	max	mean	max		
1	Carbon Black	0.006	0.021	0.98	3.11	0.031	0.606
2	Cobalt Blue	0.005	0.015	0.17	0.78		
3	Chromium Oxide Green	0.002	0.006	0.21	0.40		
4	Quinacridone Magenta	0.006	0.019	0.41	1.17		
5	Diarylide Yellow	0.010	0.019	0.33	0.81		
6	Phthalo Blue (Green Shade)	0.003	0.011	0.36	0.94		
7	Hansa Yellow Opaque	0.004	0.009	0.11	0.19		
8	Red Oxide	0.003	0.010	0.21	0.51		
9	Phthalo Green (Blue Shade)	0.007	0.029	0.66	1.78		
10	Pyrrole Red	0.006	0.012	0.21	0.38		
11	Yellow Ochre	0.004	0.010	0.19	0.38		
12	Ultramarine Blue	0.005	0.015	0.19	0.51		
13	Dioxazine Purple	0.005	0.013	0.19	0.59		
14	Green Gold	0.006	0.020	0.25	0.71		
15	Cerulean Blue	0.004	0.008	0.15	0.36		
16	Titan Buff	0.003	0.009	0.15	0.49		
17	Quinacridone Crimson	0.005	0.016	0.29	0.69		
18	Raw Umber	0.002	0.008	0.34	0.95		
19	Burnt Sienna	0.006	0.018	0.47	1.55		
20	Burnt Umber	0.003	0.015	0.28	0.72		
21	Jenkins Green	0.001	0.003	0.18	0.41		
22	Anthraquinone Blue	0.003	0.009	0.33	0.86		
23	Permanent Green Light	0.003	0.007	0.30	0.83		
24	Naphthol Red Medium	0.003	0.009	0.20	0.43		
25	Raw Sienna	0.005	0.012	0.28	0.70		
26	Paynes Gray	0.004	0.014	0.27	0.67		

Table E-V. Spectral characterization performances by global optimization (R_m wRMS1).

#	Paint name	RMS (Reflectance)		CIEDE2000		K1	K2
		mean	max	mean	max		
1	Carbon Black	0.006	0.021	0.98	3.16	0.031	0.605
2	Cobalt Blue	0.011	0.037	0.19	0.72		
3	Chromium Oxide Green	0.002	0.005	0.24	0.64		
4	Quinacridone Magenta	0.007	0.019	0.44	1.30		
5	Diarylide Yellow	0.022	0.065	0.48	0.95		
6	Phthalo Blue (Green Shade)	0.004	0.010	0.39	0.90		
7	Hansa Yellow Opaque	0.007	0.015	0.12	0.20		
8	Red Oxide	0.004	0.011	0.22	0.59		
9	Phthalo Green (Blue Shade)	0.008	0.032	0.65	1.98		
10	Pyrrole Red	0.011	0.029	0.23	0.54		
11	Yellow Ochre	0.004	0.009	0.19	0.40		
12	Ultramarine Blue	0.005	0.016	0.27	0.49		
13	Dioxazine Purple	0.005	0.014	0.22	0.59		
14	Green Gold	0.007	0.018	0.27	0.67		
15	Cerulean Blue	0.013	0.057	0.17	0.34		
16	Titan Buff	0.003	0.008	0.15	0.41		
17	Quinacridone Crimson	0.005	0.014	0.30	0.65		
18	Raw Umber	0.002	0.008	0.32	1.12		
19	Burnt Sienna	0.006	0.013	0.44	1.07		
20	Burnt Umber	0.004	0.013	0.33	0.66		
21	Jenkins Green	0.001	0.003	0.20	0.41		
22	Anthraquinone Blue	0.003	0.009	0.36	0.75		
23	Permanent Green Light	0.003	0.008	0.32	0.69		
24	Naphthol Red Medium	0.003	0.008	0.22	0.45		
25	Raw Sienna	0.005	0.011	0.27	0.62		
26	Paynes Gray	0.004	0.016	0.32	0.59		

Table E-VI. Spectral characterization performances by global optimization (R_m wRMS2).

#	Paint name	RMS (Reflectance)		CIEDE2000		K1	K2
		mean	max	mean	max		
1	Carbon Black	0.006	0.022	0.99	3.19	0.031	0.610
2	Cobalt Blue	0.009	0.032	0.15	0.72		
3	Chromium Oxide Green	0.002	0.005	0.19	0.46		
4	Quinacridone Magenta	0.007	0.018	0.39	0.99		
5	Diarylide Yellow	0.019	0.058	0.25	0.65		
6	Phthalo Blue (Green Shade)	0.004	0.010	0.34	0.85		
7	Hansa Yellow Opaque	0.007	0.013	0.11	0.19		
8	Red Oxide	0.004	0.010	0.19	0.52		
9	Phthalo Green (Blue Shade)	0.007	0.029	0.62	1.79		
10	Pyrrole Red	0.010	0.027	0.18	0.41		
11	Yellow Ochre	0.004	0.011	0.17	0.47		
12	Ultramarine Blue	0.005	0.015	0.17	0.52		
13	Dioxazine Purple	0.005	0.014	0.17	0.47		
14	Green Gold	0.006	0.019	0.25	0.68		
15	Cerulean Blue	0.012	0.052	0.14	0.36		
16	Titan Buff	0.003	0.009	0.14	0.50		
17	Quinacridone Crimson	0.005	0.015	0.28	0.62		
18	Raw Umber	0.002	0.008	0.28	1.11		
19	Burnt Sienna	0.006	0.013	0.40	1.17		
20	Burnt Umber	0.004	0.015	0.24	0.76		
21	Jenkins Green	0.001	0.004	0.16	0.48		
22	Anthraquinone Blue	0.003	0.010	0.34	1.01		
23	Permanent Green Light	0.003	0.006	0.28	0.80		
24	Naphthol Red Medium	0.003	0.008	0.19	0.43		
25	Raw Sienna	0.005	0.012	0.27	0.71		
26	Paynes Gray	0.004	0.016	0.25	0.67		

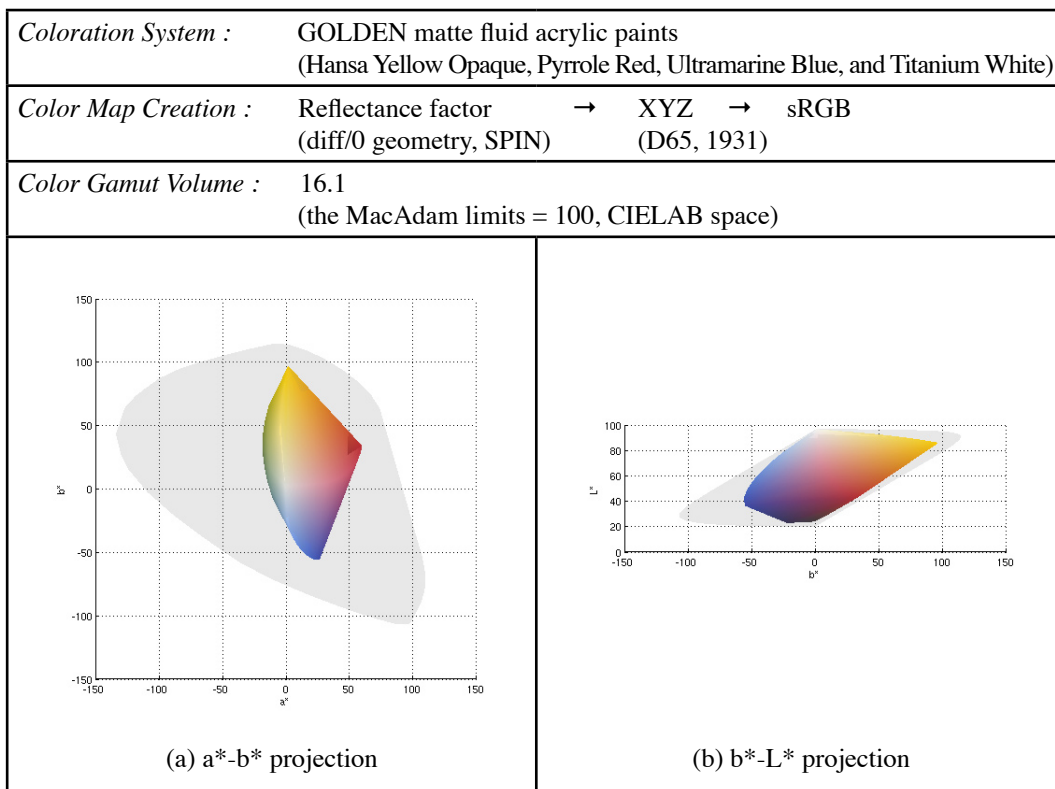
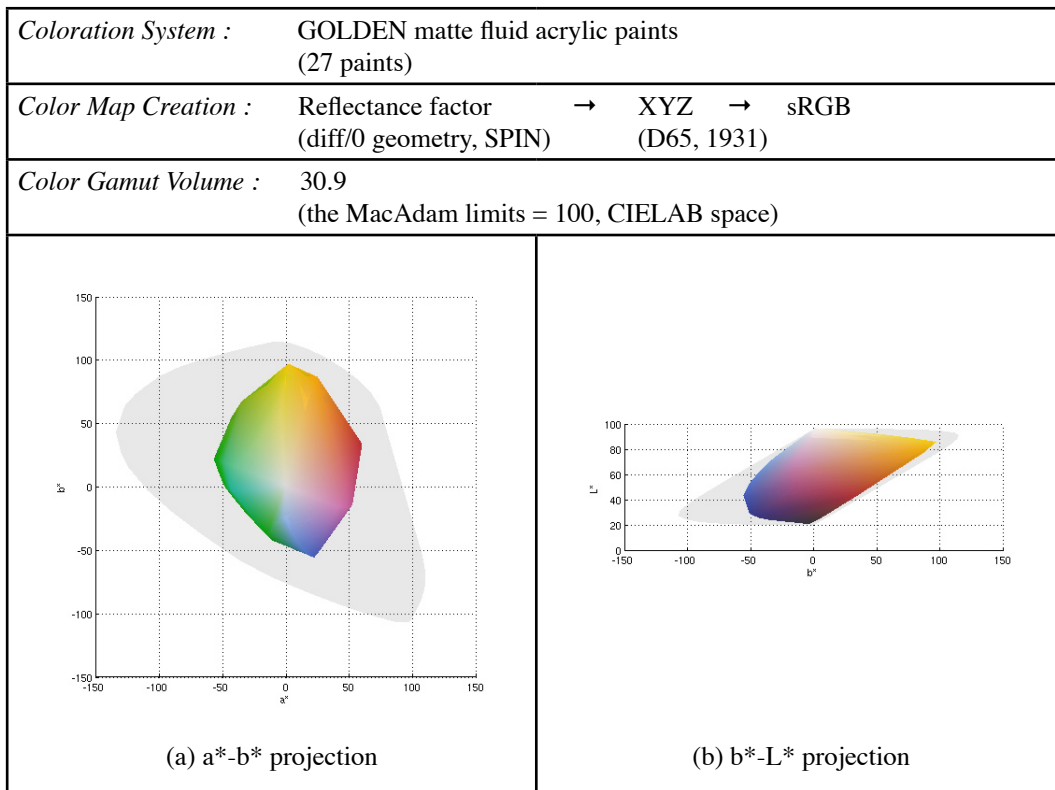
F. Color Gamut Collection for Various Coloration Systems

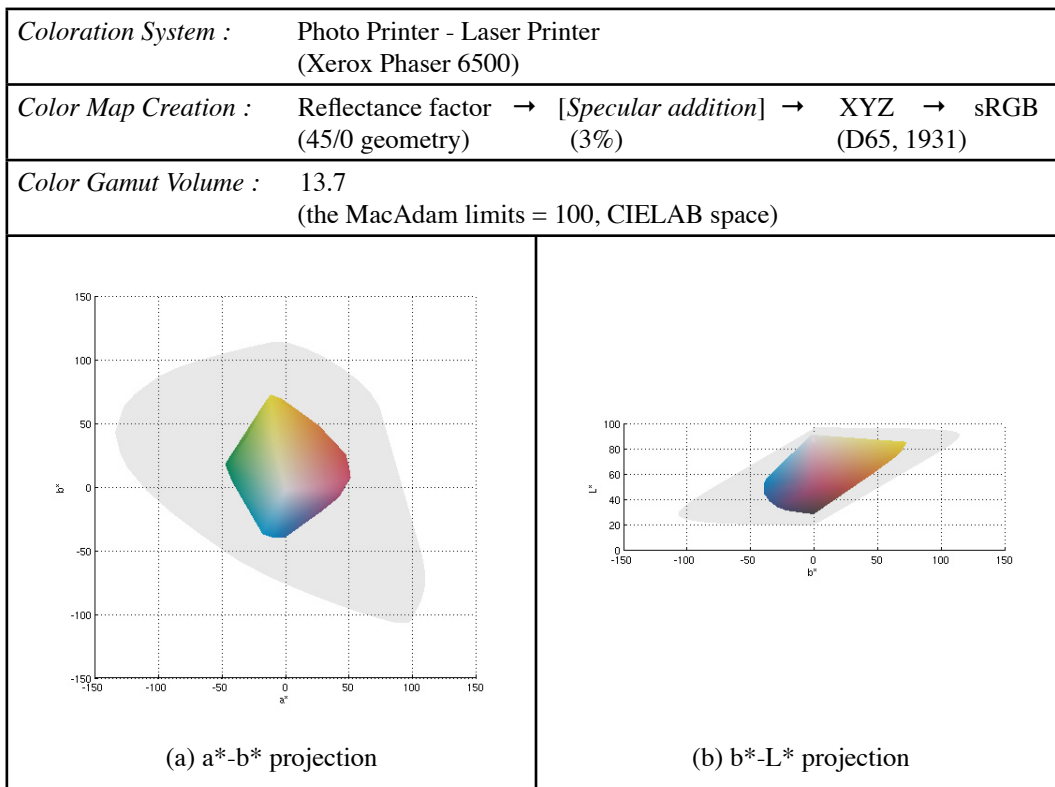
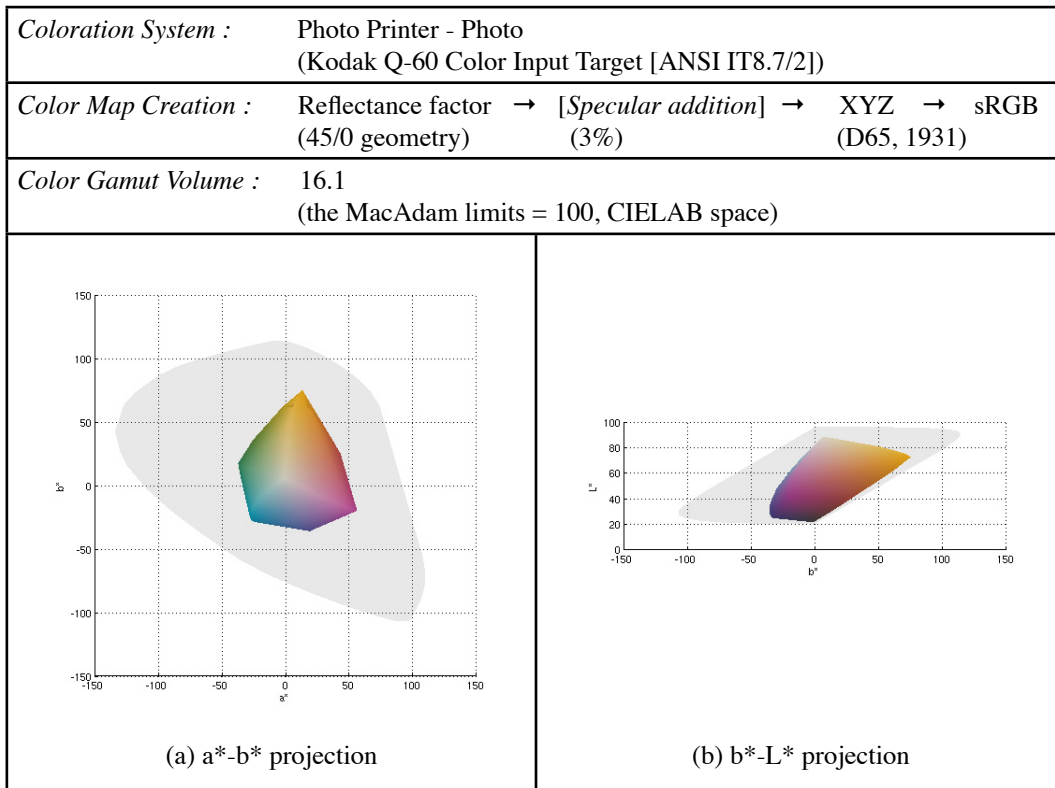
This section exhibits the color gamuts rendered for various coloration systems including photo printer and color chart. Before rendering color gamuts, spectrophotometric and colorimetric data had been collected from various resources. Unfortunately, some of the measurements were obtained under different measurement geometries or different illuminants. To keep constancy of measurement condition, we converted any measurement values into CIELAB values under D65 illuminant and the 1931 standard observer with adding a 3% specular reflection first. And then, CIELAB values were converted to sRGB digits in order to create color map for displaying. The important spectral and colorimetric data manipulations, which were used for color gamut rendering, are listed in the following.

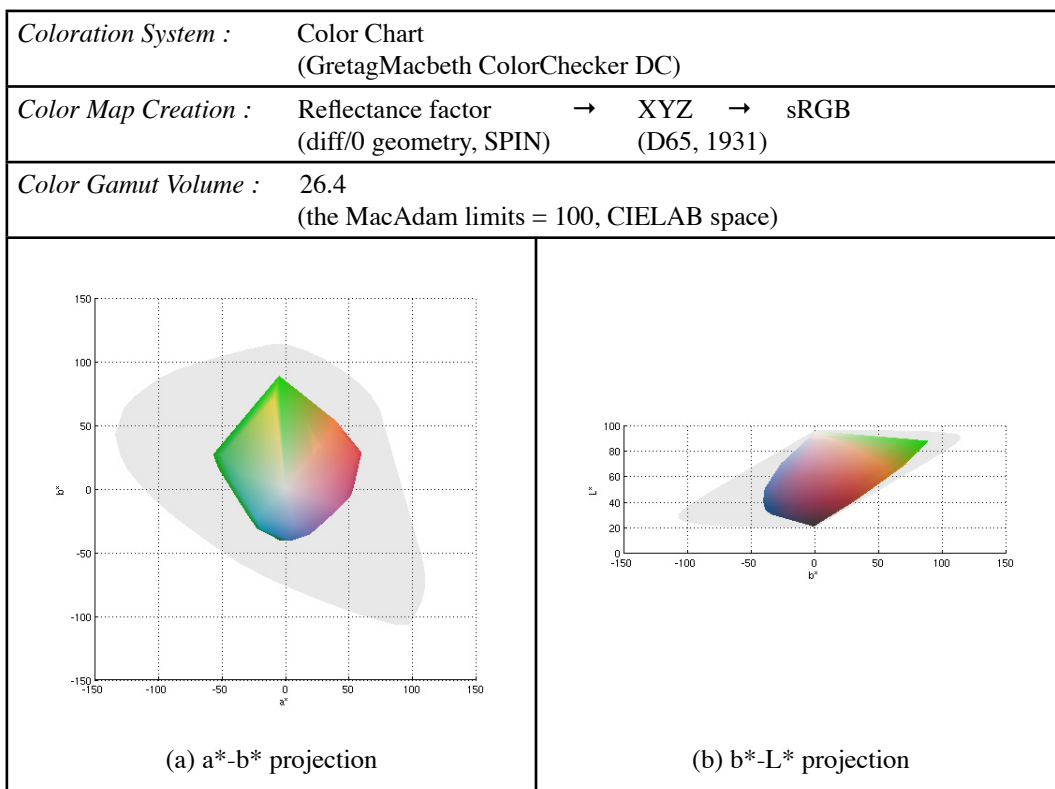
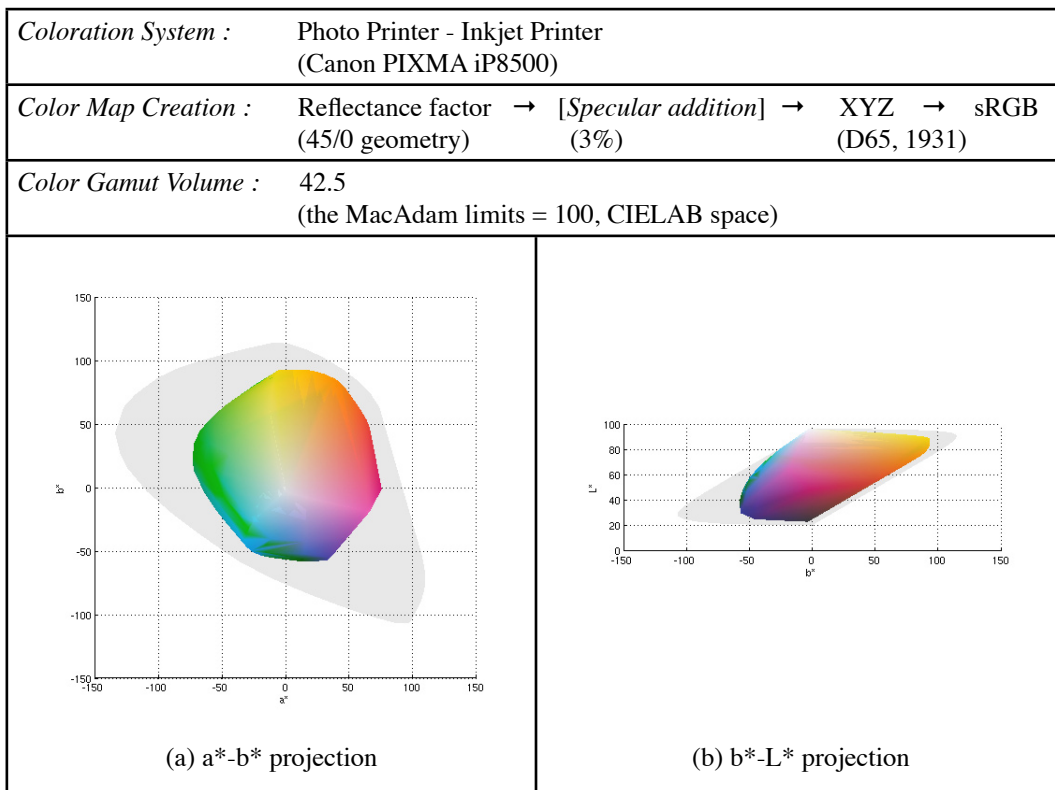
- For 45/0 geometry measurement, we simply added a 3% flat spectral reflectance into the original measured reflectance curves over all wavelengths in order to simulate measurements by diff/0 geometry in the SPIN mode.
- For XYZ values under illuminant C, the von Kries transformation was used to convert XYZ values (illuminant C) into XYZ values (illuminant D65) values in order to simulate measurements under illuminant D65. The following is the transformation equation, where $(X_{w,D65}, Y_{w,D65}, Z_{w,D65})$ and $(X_{w,C}, Y_{w,C}, Z_{w,C})$ are the white points of illuminant D65 and illuminant C, respectively.

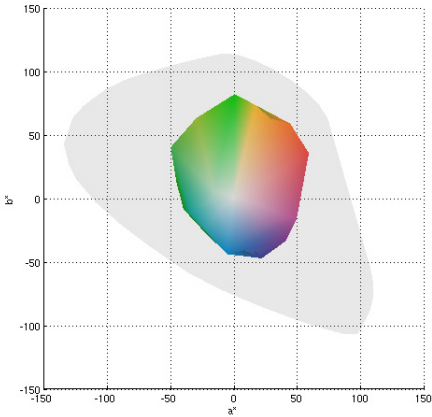
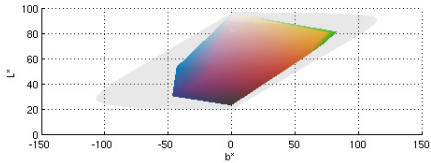
$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{illuminant D65}} = \begin{pmatrix} X_{w,D65}/X_{w,C} & 0 & 0 \\ 0 & Y_{w,D65}/Y_{w,C} & 0 \\ 0 & 0 & Z_{w,D65}/Z_{w,C} \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{illuminant C}}$$

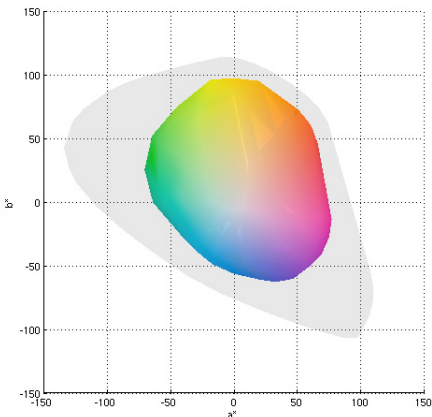
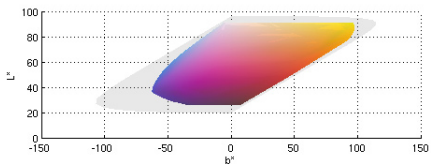
- For XYZ values without specular reflection, ΔXYZ , which was calculated by multiplying a 0.3 flat reflectance factor over all wavelengths with the ASTM weights for illuminant D65 and the 1931 standard observer and summing them, added into the original XYZ values in order to simulate a 3% specular intrusion into the colorimetric values.







<i>Coloration System :</i>	Color Chart (GretagMacbeth ColorChecker SG)		
<i>Color Map Creation :</i>	Reflectance factor (diff/0 geometry, SPIN)	→ XYZ (D65, 1931)	→ sRGB
<i>Color Gamut Volume :</i>	23.0 (the MacAdam limits = 100, CIELAB space)		
 (a) a*-b* projection		 (b) b*-L* projection	

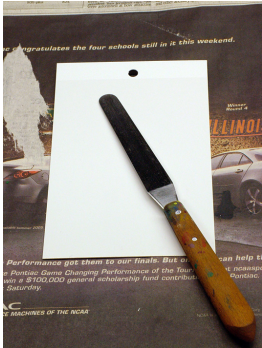
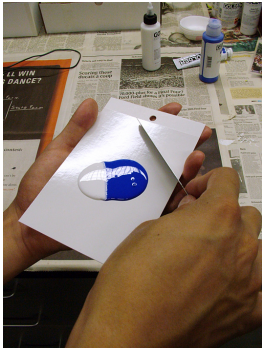
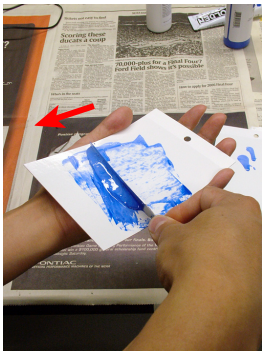
<i>Coloration System :</i>	Real surface colors (Pointer)		
<i>Color Map Creation :</i>	CIELAB → XYZ (C, 1931)	→ XYZ (C, 1931)	→ [Specular addition] → sRGB (D65, 1931) (3%)
<i>Color Gamut Volume :</i>	45.7 (the MacAdam limits = 100, CIELAB space)		
 (a) a*-b* projection		 (b) b*-L* projection	

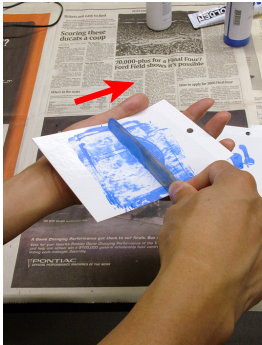
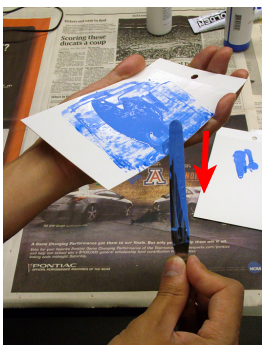
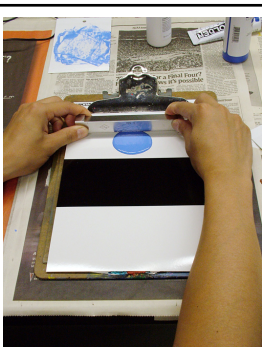
G. Tips for Paint Sample Preparation

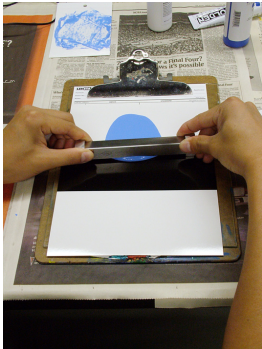
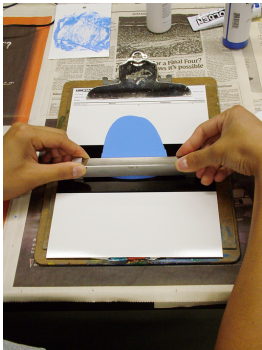
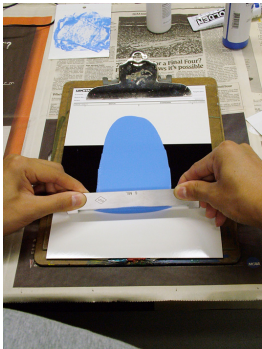
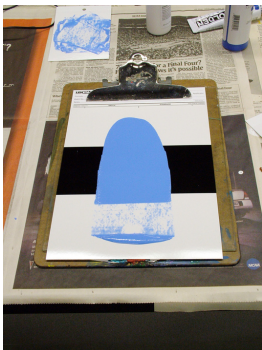
G.1. Paint Mixing and Paint Application

The expertise of a paint mixing technique determines the final quality of the paint drawdowns. An insufficient mixing can cause a marble pattern to appear on a paint surface leading to a poor paint characterization. To prevent this possible issue from occurring, we need to take appropriate mixing steps. This section is to introduce an instructive procedure of paint drawdown preparation. The basic procedure follows the way which technical staff of GOLDEN Artist Colors, Inc. have instructed us for this research.

Table G-I. Instruction on paint drawdown preparation.

Step	Instruction	Photo
1	Prepare two paint mixing tools. (1) A flat palette knife (2) A laminated chart A flat palette knife having a long blade is easy-to-use for paint mixing.	
2	Extract two paints (or single paint) from the containers onto the center of the chart and weigh them going to being 5g totally. (In the right photo, 3g Cobalt Blue and 2g Titanium White.) For GOLDEN matte fluid acrylic paints, you do not need to thin them with water because they are already fluid. A 5g paint portion is appropriate for covering the whole area of a drawdown paper.	
3	With your left hand, a laminated chart is hold. And, a knife blade is adjusted to be perpendicular to the chart with your right fingers.	
4	Scrape a paint portion toward your body a couple of times. At this time, it would be much easier if the chart is tilted that makes the far side of a chart lower. Just reminder, a paint portion is mixed only during this scraping. And then, you are expected to check whether some pigments are left at the far side of a chart or not. If some are left, you need to scrape them hard a little bit.	

5	A paint portion which has been approached to your near side is to be slid back to the center of a chart with a palette knife. This step is to repeat Step 4 and Step 5 until it is completely mixed. Go back to Step 4 if it is necessary.	
6	Some paint portion on the blade can be removed by scraping it with the edge of a chart. This technique is necessary for perfect mixing and efficient paint use. If you still find some paint portion unmixed on a palette knife, go back to Step 4 after removing it from the blade onto a chart.	
7	When mixing is completed, a mixed paint portion is put down at the top of a drawdown paper by using the tip of a palette knife. And then, level off a paint portion to make a wide painted area in application.	
8	Your both hands should gently hold both side of the metal bar as shown in the right photo. Too much holding force will prevent a drawdown bar from a smooth slide.	

9	Move a drawdown bar toward your body with pushing it onto the substrate.	
10	Keep moving a drawdown bar at 6 cm/s.	
11	At the end of a drawdown paper, you need to do half-filp at your near side of a metal bar. Do not release, keep pushing it.	
12	Scrape a paint portion on a drawdown bar on the level quickly. Finally, a paint application onto a drawdown paper is done.	

G.2. Coating Defects in Varnishing

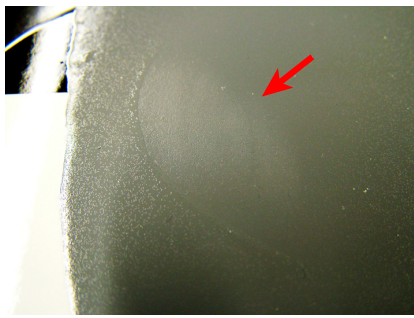
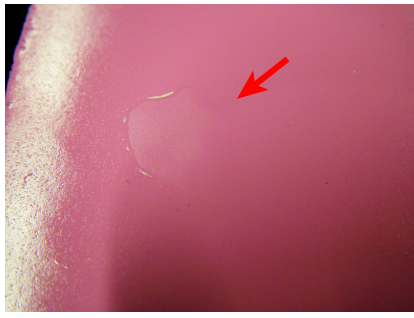
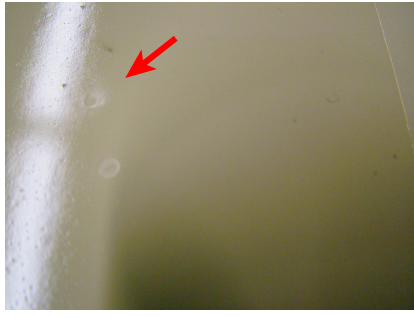
In a varnishing process, we observed coating defects on the surfaces of some varnished-paint specimens. The surface defects can be considered to result from inhomogeneous surface forces in the coating [Gutoff 1995]. Inhomogeneous surface forces are usually generated by several internal and external factors: the compatibility of chemical materials, surface contamination by dusts, temperature fluctuations in solvent evaporation, and so on. Paint analysts will possibly meet difficulty in measuring defective surfaces. Table F-II shows the list of some surface defects observed in our paint samples. To remove the defects, understanding chemical processes on coating and some treatments are necessary. Gutoff introduced several solutions for the surface defects in his reference.

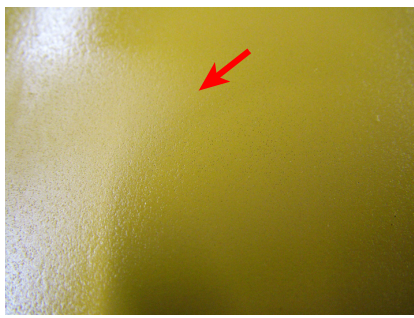
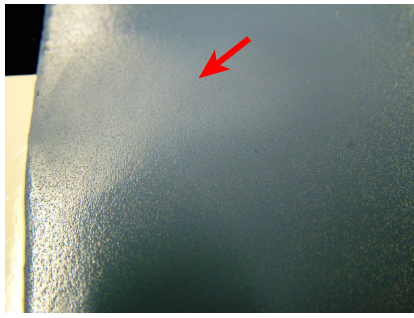
The first treatment which color scientists can take is to change the recipe of a varnish solution. For example, using a varnish at high concentration is expected to raise viscosity so as to stabilize dispersed varnish resins in liquid more effectively. That is, a high-viscous solution will reduce the fluctuation of varnish resins that cause fat edges and craters. From this point, paint operators are required to determine the recipe of a varnish solution based on their aiming goals (e.g. gloss level and surface quality).

The second treatment to choose possibly is to control room temperature in order to slow drying speed. The approach of lowering temperature can help self-recovery by which craters on a paint surface will disappear naturally in evaporation.

The last treatment is to prepare for a clean varnishing room. Without such a clean environment, paint researchers must apply a varnish onto paint drawdowns as soon as after the specimens are completely dried. Taking a lot of drying time more than the necessary raises the possibility for contaminating the surface with dusts and particles in the air.

Table G-II. Coating defects appeared on varnished paint surfaces.

Paint Name	Defect	Photo
Carbon Black (10%)	Fat edge	
Cobalt Blue (80%)	Craters	
Quinacridone Magenta (60%)	Fat edge	
Titan Buff (100%)	Craters	

Raw Sienna (80%)	Fat edge	
Paynes Gray (40%)	Fat edge	
Titanium White (100%)	Fat edge	

The varnished drawdowns of Cobalt Blue have the largest number of defective craters on the surfaces than those of the other paints. Actually, we felt difficulty in varnishing Cobalt Blue more than the other paints with good varnish quality. To confirm how the recipe of a varnish solution affects the creation of craters for the masstone of Cobalt Blue, therefore, we prepared two different recipes of the varnish solutions: (1) 2 varnish : 1 mineral spirit, (2) varnish only. And then, we observed the surface state of each of varnish coatings after they were completely dried. This approach so far is

equivalent to the first treatment that has been introduced in the previous page. The result was that, even though we could see the decrease of craters in a specimen coated by a 100% varnish solution; however, the effects of the treatment were not maximized. This fact implies that craters appeared on the drawdown surfaces of Cobalt Blue do not result from the fluctuation of varnish resins.

With regard to the second treatment and the third treatment, we did not attempt them actually because of using a limited environment. Therefore, whether these treatments can work or not is open to discussion.